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abb8bef · 18 minutes ago

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Experiment 2:- Docker Installation, Configuration, and Running Images

Step 1: Pull Image

```
bash docker --version
```

```
PS C:\Users\acer\OneDrive\Desktop\Vagrent> wsl
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrent$ docker --version
Docker version 28.4.0, build d8eb465
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrent$ |
```

Step 2: Run Container with Port Mapping

```
bash docker run -d -p 8080:80 nginx
```

```

acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrent$ sudo docker run -d -p 8080:80 nginx
[sudo] password for acer:
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
a158ce1337b952374b010619b948ffd0568366936eee5a7facb39836cbe641cd
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrent$ |

```

Step 3: Verify Running Containers

bash docker ps

```

acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrent$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
NAMES
a158ce1337b9   nginx    "/docker-entrypoint...." 2 minutes ago  Up 2 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
gracious_elion
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrent$ |

```

Step 4: Stop Container

bash docker stop <container_id>

```
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrant$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
a158ce1337b9	nginx	"/docker-entrypoint..."	5 minutes ago	Exited (0)	About a minute ago	gracious_eli

```
on
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrant$ |
```

Step 5: Remove Container

```
bash docker rm <container_id>
```

```
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrant$ sudo docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
nginx	latest	4af177a024eb	2 weeks ago	161MB

```
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrant$ |
```

Step 5: Remove Image

```
bash docker rmi nginx
```

```
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrant$ sudo docker rm a15
a15
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Vagrant$ |
```

Result Docker images were successfully pulled, containers executed, and lifecycle commands performed.



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abb8bef · 19 minutes ago



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Lab Experiment 3: Deploying NGINX Using Different Base Images and Comparing Image Layers



Aim

To deploy NGINX using three different approaches (official image, Ubuntu-based image, Alpine-based image), compare image si:

<hr>

<h4 align="center"> Pre-requisite </h4>

- Docker installed and running (using `sudo` if required)
- Basic knowledge of `docker run`, `Dockerfile`, and port mapping
- WSL / Linux terminal (the lab uses Ubuntu in WSL)

> **Directory overview**

> The experiment directory contains Dockerfiles, a custom `html` folder, and a `test\_lab` subfolder.

>

> ![Experiment directory](./Images/image.png)

<hr>

Step-by-Step Procedure

****Step 1 - Pull the Official NGINX Image****

Because the user is not yet in the ``docker`` group, ``sudo`` is required. The official ``nginx:latest`` image is pulled from Docker Hub.

```
```bash
```

```
sudo docker pull nginx:latest
```

```
sudo docker images
```

```

acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ # Add your user to the docker group
sudo usermod -aG docker $USER

Apply the group changes immediately
newgrp docker

Verify you're in the docker group
groups
[sudo] password for acer:
usermod: group 'docker' does not exist
newgrp: group 'docker' does not exist
acer adm cdrom sudo dip plugdev users
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ docker pull nginx:latest
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.51/images/create?fromImage=docker.io%2Flibrary%2Fnginx&tag=latest": dial unix /var/run/docker.sock: connect: permission denied
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ # Use sudo for all docker commands in this session
sudo docker pull nginx:latest
sudo docker images
latest: Pulling from library/nginx
0c8d55a45c0d: Pull complete
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 5cdef4ac3335 2 days ago 161MB
nginx <none> 4af177a024eb 3 weeks ago 161MB
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$

```

## Step 2 – Run the Official NGINX Container

Start a container from the official image, mapping port **8080** on the host to port **80** inside the container.

```

sudo docker run -d --name nginx-official -p 8080:80 nginx
curl http://localhost:8080

```



(The first `curl` attempt may fail with "connection reset" while the container is still starting; repeating the `curl` after a few seconds returns the welcome page.)

```

acer@DESKTOP-DOEGC13: /m X + v
sudo docker images
latest: Pulling from library/nginx
0c8d55a45c0d: Pull complete
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 5cdef4ac3335 2 days ago 161MB
nginx <none> 4af177a024eb 3 weeks ago 161MB
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ docker run -d --name nginx-offi
cial -p 8080:80 nginx
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.soc
k/_ping": dial unix /var/run/docker.sock: connect: permission denied

Run 'docker run --help' for more information
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ # Already done: sudo docker pul
l nginx:latest

sudo docker run -d --name nginx-official -p 8080:80 nginx
curl http://localhost:8080

Check image
sudo docker images nginx
0e98675fa86ed1693ec64c0d9f9338f6e81351ae7b4e03e3799a80599b83fa60
curl: (56) Recv failure: Connection reset by peer
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx latest 5cdef4ac3335 2 days ago 161MB
nginx <none> 4af177a024eb 3 weeks ago 161MB
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ |

```

### Step 3 – Create a Dockerfile for Ubuntu-Based NGINX

Use `nano` to write a `Dockerfile.ubuntu` that installs NGINX on top of Ubuntu 22.04.



**FROM** ubuntu:22.04

**RUN** apt-get update && \  
apt-get install -y nginx && \  
apt-get clean && \  
rm -rf /var/lib/apt/lists/\*

**EXPOSE** 80

**CMD** ["nginx", "-g", "daemon off;"]

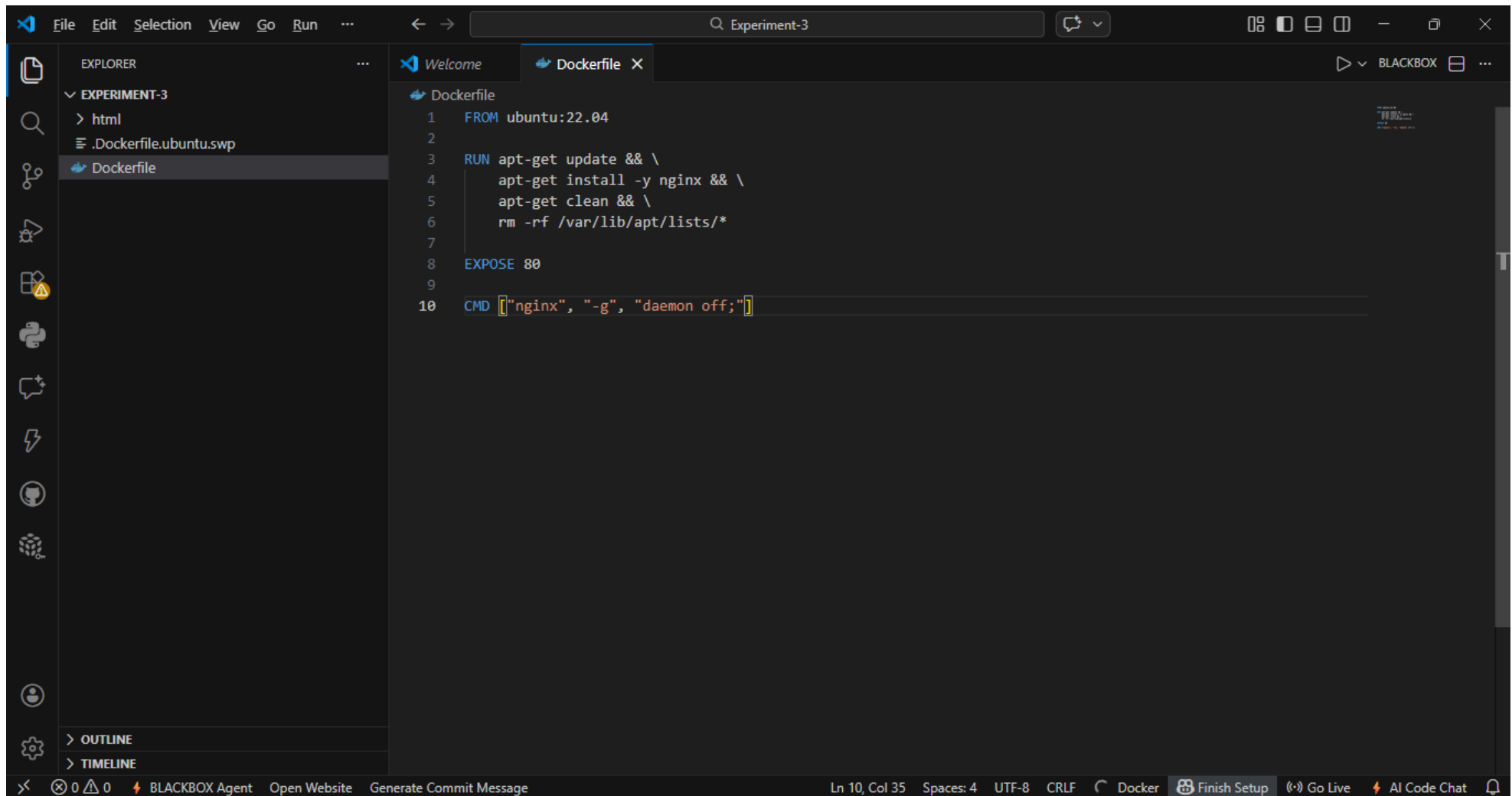


acer@DESKTOP-DOEGC13: /m

GNU nano 7.2 Dockerfile.ubuntu

[ New File ]

<b>^G</b> Help	<b>^O</b> Write Out	<b>^W</b> Where Is	<b>^K</b> Cut	<b>^T</b> Execute	<b>^C</b> Location	<b>M-U</b> Undo	<b>M-A</b> Set Mark	<b>M-]</b> To Bracket
<b>^X</b> Exit	<b>^R</b> Read File	<b>^_\</b> Replace	<b>^U</b> Paste	<b>^J</b> Justify	<b>^/_</b> Go To Line	<b>M-E</b> Redo	<b>M-6</b> Copy	<b>^O</b> Where Was



The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project named 'EXPERIMENT-3' with a subdirectory 'html' and a file named 'Dockerfile'. The main editor area displays the content of the 'Dockerfile' file. The Dockerfile contains the following instructions:

```
1 FROM ubuntu:22.04
2
3 RUN apt-get update && \
4 apt-get install -y nginx && \
5 apt-get clean && \
6 rm -rf /var/lib/apt/lists/*
7
8 EXPOSE 80
9
10 CMD ["nginx", "-g", "daemon off;"]
```

The status bar at the bottom indicates the current position is Line 10, Column 35, with 4 spaces, UTF-8 encoding, and CRLF line endings. It also shows the Docker extension is active and offers options like 'Finish Setup', 'Go Live', and 'AI Code Chat'.

## Step 4 – Build and Run the Ubuntu-Based NGINX Image

```
sudo docker build -t nginx-ubuntu -f Dockerfile.ubuntu .
sudo docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu
```



```
curl http://localhost:8081
sudo docker images nginx-ubuntu
```



```
GNU nano 7.2 Dockerfile.ubuntu *
Build image
sudo docker build -t nginx-ubuntu -f Dockerfile.ubuntu .

Run container
sudo docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu

Verify
curl http://localhost:8081

Check image size
sudo docker images nginx-ubuntu
|
```

```

acer@DESKTOP-DOEGC13: /m X + v
=> => extracting sha256:6f4ebca3e823b18dac366f72e537b1772bc3522a5c7ae299d6491fb17378410e 1.8s
=> [2/2] RUN apt-get update && apt-get install -y nginx && apt-get clean && rm -rf /var/lib/apt/lists/* 85.4s
=> exporting to image 0.5s
=> => exporting layers 0.4s
=> => writing image sha256:40bcda3f26ed36aff7f7a897d6b782b8b082a42b8f8576993ade94a77616c8f4 0.0s
=> => naming to docker.io/library/nginx-ubuntu 0.0s
c02bb8078de362fb7b4ac715838bb40b3abb53cb261fecc2ed61f5ccfac431d7
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
 body {
 width: 35em;
 margin: 0 auto;
 font-family: Tahoma, Verdana, Arial, sans-serif;
 }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
nginx.org.

Commercial support is available at
nginx.com.</p>

<p>Thank you for using nginx.</p>
</body>
</html>
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx-ubuntu latest 40bcda3f26ed 2 seconds ago 134MB
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ |

```

## Step 5 – Create a Dockerfile for Alpine-Based NGINX

Write a Dockerfile.alpine using the lightweight Alpine Linux base.

```

FROM alpine:latest
RUN apk add --no-cache nginx

```



EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

```

margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif;
}
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
nginx.org.

Commercial support is available at
nginx.com.</p>

<p>Thank you for using nginx.</p>
</body>
</html>
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx-ubuntu latest 40bcda3f26ed 2 seconds ago 134MB
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ # Create Dockerfile
nano Dockerfile.alpine
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ # Build image
sudo docker build -t nginx-alpine -f Dockerfile.alpine .

Run container
sudo docker run -d --name nginx-alpine -p 8082:80 nginx-alpine

Verify
curl http://localhost:8082

Check image size
sudo docker images nginx-alpine
[+] Building 0.0s (0/1)

```

docker:default

*(scene before build)*

## Step 6 – Build and Run the Alpine-Based NGINX Image



```

sudo docker build -t nginx-alpine -f Dockerfile.alpine .
sudo docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
curl http://localhost:8082
sudo docker images nginx-alpine

```

```

acer@DESKTOP-DOEGC13: /m X + v
</html>
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx-alpine latest 2fe6bc635982 1 second ago 10.4MB
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ # Compare all NGINX images
sudo docker images | grep nginx

Check layers
sudo docker history nginx
sudo docker history nginx-ubuntu
sudo docker history nginx-alpine
nginx-alpine latest 2fe6bc635982 15 seconds ago 10.4MB
nginx-ubuntu latest 40bcda3f26ed About a minute ago 134MB
nginx latest 5cdef4ac3335 2 days ago 161MB
nginx <none> 4af177a024eb 3 weeks ago 161MB
IMAGE CREATED CREATED BY SIZE COMMENT
5cdef4ac3335 2 days ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0
<missing> 2 days ago STOPSIGNAL SIGQUIT 0B buildkit.dockerfile.v0
<missing> 2 days ago EXPOSE map[80/tcp:{}] 0B buildkit.dockerfile.v0
<missing> 2 days ago ENTRYPOINT ["/docker-entrypoint.sh"] 0B buildkit.dockerfile.v0
<missing> 2 days ago COPY 30-tune-worker-processes.sh /docker-ent... 4.62kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 20-envsubst-on-templates.sh /docker-ent... 3.02kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 15-local-resolvers.envsh /docker-entryp... 389B buildkit.dockerfile.v0
<missing> 2 days ago COPY 10-listen-on-ipv6-by-default.sh /docker... 2.12kB buildkit.dockerfile.v0
<missing> 2 days ago COPY docker-entrypoint.sh / # buildkit 1.62kB buildkit.dockerfile.v0
<missing> 2 days ago RUN /bin/sh -c set -x && groupadd --syst... 82.2MB buildkit.dockerfile.v0
<missing> 2 days ago ENV DYNPKG_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV PKG_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV ACME_VERSION=0.3.1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_VERSION=0.9.5 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NGINX_VERSION=1.29.5 0B buildkit.dockerfile.v0
<missing> 2 days ago LABEL maintainer=NGINX Docker Maintainers <d... 0B buildkit.dockerfile.v0
<missing> 5 days ago # debian.sh --arch 'amd64' out/ 'trixie' '@1... 78.6MB debuerreotype 0.17
IMAGE CREATED CREATED BY SIZE COMMENT
40bcda3f26ed About a minute ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0

```

## Step 7 – Inspect Image Layers with `docker history`

Compare the layers of the three images to understand why sizes differ.

```
sudo docker history nginx
sudo docker history nginx-ubuntu
sudo docker history nginx-alpine
```



- **Official NGINX** – many intermediate layers (pre-optimised Debian base) → **161 MB**
- **NGINX-Ubuntu** – full Ubuntu filesystem plus NGINX installation → **134 MB**
- **NGINX-Alpine** – minimal Alpine rootfs + NGINX package → **10.4 MB**

```

acer@DESKTOP-DOEGC13: /m X + v
5cdef4ac3335 2 days ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0
<missing> 2 days ago STOPSIGNAL SIGQUIT 0B buildkit.dockerfile.v0
<missing> 2 days ago EXPOSE map[80/tcp:{}] 0B buildkit.dockerfile.v0
<missing> 2 days ago ENTRYPOINT ["/docker-entrypoint.sh"] 0B buildkit.dockerfile.v0
<missing> 2 days ago COPY 30-tune-worker-processes.sh /docker-ent... 4.62kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 20-envsubst-on-templates.sh /docker-ent... 3.02kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 15-local-resolvers.envsh /docker-entryp... 389B buildkit.dockerfile.v0
<missing> 2 days ago COPY 10-listen-on-ipv6-by-default.sh /docker... 2.12kB buildkit.dockerfile.v0
<missing> 2 days ago COPY docker-entrypoint.sh / # buildkit 1.62kB buildkit.dockerfile.v0
<missing> 2 days ago RUN /bin/sh -c set -x && groupadd --syst... 82.2MB buildkit.dockerfile.v0
<missing> 2 days ago ENV DYNPKG_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV PKG_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV ACME_VERSION=0.3.1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_VERSION=0.9.5 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NGINX_VERSION=1.29.5 0B buildkit.dockerfile.v0
<missing> 2 days ago LABEL maintainer=NGINX Docker Maintainers <d... 0B buildkit.dockerfile.v0
<missing> 5 days ago # debian.sh --arch 'amd64' out/ 'trixie' '@1... 78.6MB debuerreotype 0.17
IMAGE
40bcda3f26ed About a minute ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0
<missing> About a minute ago EXPOSE &[{{8 0} {8 0}}] 0xc000d16840} 0B buildkit.dockerfile.v0
<missing> About a minute ago RUN /bin/sh -c apt-get update && apt-get... 55.7MB buildkit.dockerfile.v0
<missing> 4 weeks ago /bin/sh -c #(nop) CMD ["/bin/bash"] 0B
<missing> 4 weeks ago /bin/sh -c #(nop) ADD file:b499000226bd9a7c5... 77.9MB
<missing> 4 weeks ago /bin/sh -c #(nop) LABEL org.opencontainers.... 0B
<missing> 4 weeks ago /bin/sh -c #(nop) LABEL org.opencontainers.... 0B
<missing> 4 weeks ago /bin/sh -c #(nop) ARG LAUNCHPAD_BUILD_ARCH 0B
<missing> 4 weeks ago /bin/sh -c #(nop) ARG RELEASE 0B
IMAGE
2fe6bc635982 15 seconds ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0
<missing> 15 seconds ago EXPOSE &[{{5 0} {5 0}}] 0xc000853540} 0B buildkit.dockerfile.v0
<missing> 15 seconds ago RUN /bin/sh -c apk add --no-cache nginx # bu... 2MB buildkit.dockerfile.v0
<missing> 10 days ago CMD ["/bin/sh"] 0B buildkit.dockerfile.v0
<missing> 10 days ago ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /... 8.44MB buildkit.dockerfile.v0
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ |

```

```

acer@DESKTOP-DOEGC13: /m X + v
</html>
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx-alpine latest 2fe6bc635982 1 second ago 10.4MB
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3$ # Compare all NGINX images
sudo docker images | grep nginx

Check layers
sudo docker history nginx
sudo docker history nginx-ubuntu
sudo docker history nginx-alpine
nginx-alpine latest 2fe6bc635982 15 seconds ago 10.4MB
nginx-ubuntu latest 40bcda3f26ed About a minute ago 134MB
nginx latest 5cdef4ac3335 2 days ago 161MB
nginx <none> 4af177a024eb 3 weeks ago 161MB
IMAGE CREATED CREATED BY SIZE COMMENT
5cdef4ac3335 2 days ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0
<missing> 2 days ago STOPSIGNAL SIGQUIT 0B buildkit.dockerfile.v0
<missing> 2 days ago EXPOSE map[80/tcp:{}] 0B buildkit.dockerfile.v0
<missing> 2 days ago ENTRYPOINT ["/docker-entrypoint.sh"] 0B buildkit.dockerfile.v0
<missing> 2 days ago COPY 30-tune-worker-processes.sh /docker-ent... 4.62kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 20-envsubst-on-templates.sh /docker-ent... 3.02kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 15-local-resolvers.envsh /docker-entryp... 389B buildkit.dockerfile.v0
<missing> 2 days ago COPY 10-listen-on-ipv6-by-default.sh /docker... 2.12kB buildkit.dockerfile.v0
<missing> 2 days ago COPY docker-entrypoint.sh / # buildkit 1.62kB buildkit.dockerfile.v0
<missing> 2 days ago RUN /bin/sh -c set -x && groupadd --syst... 82.2MB buildkit.dockerfile.v0
<missing> 2 days ago ENV DYNPKG_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV PKG_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV ACME_VERSION=0.3.1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_RELEASE=1~trixie 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_VERSION=0.9.5 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NGINX_VERSION=1.29.5 0B buildkit.dockerfile.v0
<missing> 2 days ago LABEL maintainer=NGINX Docker Maintainers <d... 0B buildkit.dockerfile.v0
<missing> 5 days ago # debian.sh --arch 'amd64' out/ 'trixie' '@1... 78.6MB debuerreotype 0.17
IMAGE CREATED CREATED BY SIZE COMMENT
40bcda3f26ed About a minute ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0

```

## Step 8 – Serve a Custom HTML Page (Volume Mount)

Create a simple `index.html` and mount it into the default NGINX web root.

```

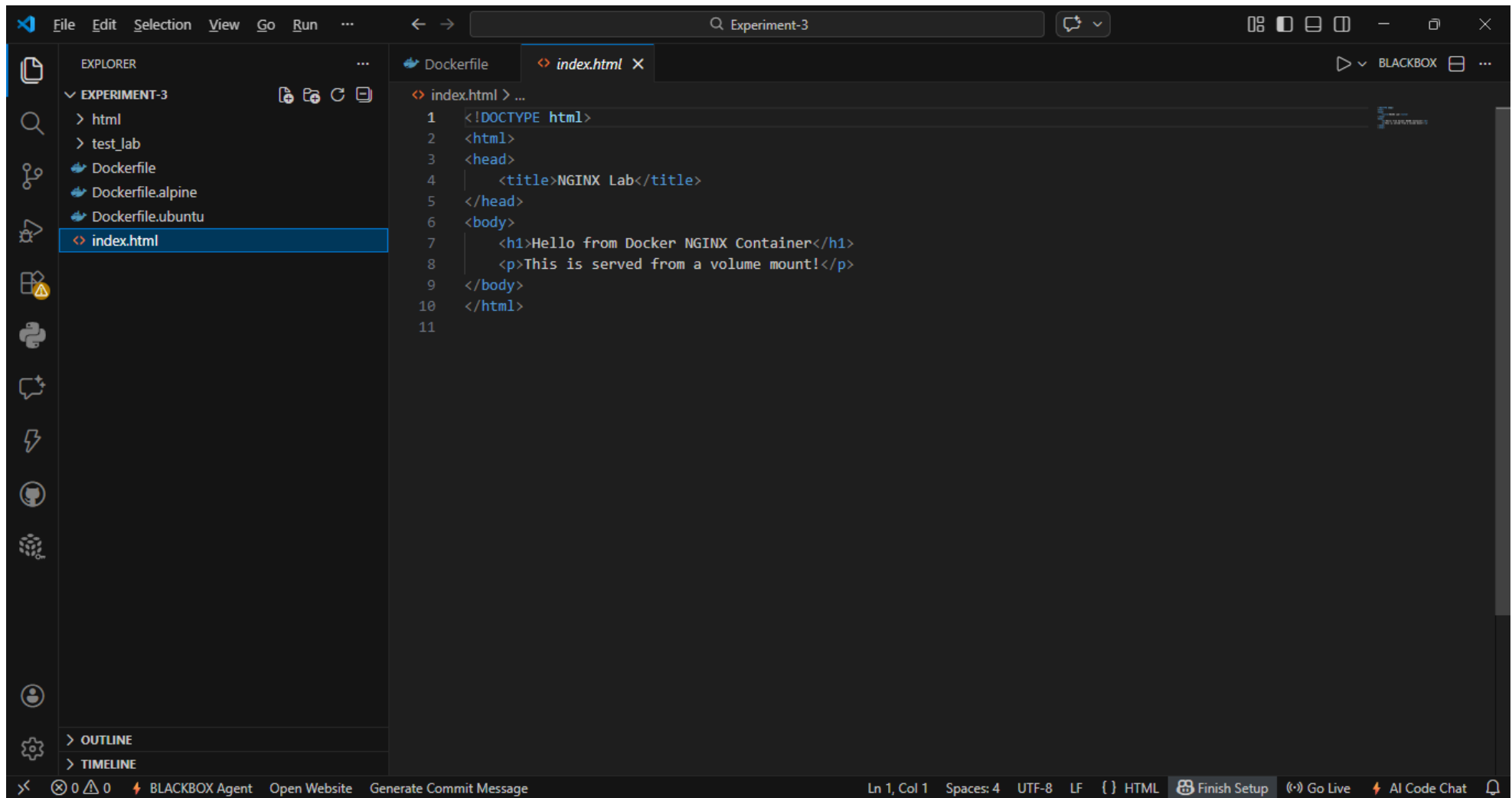
mkdir html
echo "<h1>Hello from Docker NGINX</h1>" > html/index.html

```



```
sudo docker run -d -p 8083:80 -v $(pwd)/html:/usr/share/nginx/html nginx
```





The screenshot shows a VS Code editor interface with a dark theme. The Explorer sidebar on the left displays the file structure of a project named 'EXPERIMENT-3'. The files listed are 'html', 'test\_lab', 'Dockerfile', 'Dockerfile.alpine', 'Dockerfile.ubuntu', and 'index.html'. The 'index.html' file is selected and highlighted. The main editor area shows the content of 'index.html', which is an HTML document. The code is as follows:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>NGINX Lab</title>
5 </head>
6 <body>
7 <h1>Hello from Docker NGINX Container</h1>
8 <p>This is served from a volume mount!</p>
9 </body>
10 </html>
11
```

The status bar at the bottom of the editor shows the following information: 'Ln 1, Col 1', 'Spaces: 4', 'UTF-8', 'LF', '{ } HTML', 'Finish Setup', 'Go Live', 'AI Code Chat', and a bell icon. The bottom of the sidebar shows the 'OUTLINE' and 'TIMELINE' sections.

```

acer@DESKTOP-DOEGC13: ~/ | X + v
<missing> 2 days ago RUN /bin/sh -c set -x && apkArch="$(cat ... 49.4MB buildkit.dockerfile.v0
<missing> 2 days ago ENV ACME_VERSION=0.3.1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_RELEASE=1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NJS_VERSION=0.9.5 0B buildkit.dockerfile.v0
<missing> 2 days ago CMD ["nginx" "-g" "daemon off;"] 0B buildkit.dockerfile.v0
<missing> 2 days ago STOPSIGNAL SIGQUIT 0B buildkit.dockerfile.v0
<missing> 2 days ago EXPOSE map[80/tcp:{}] 0B buildkit.dockerfile.v0
<missing> 2 days ago ENTRYPOINT ["/docker-entrypoint.sh"] 0B buildkit.dockerfile.v0
<missing> 2 days ago COPY 30-tune-worker-processes.sh /docker-ent... 4.62kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 20-envsubst-on-templates.sh /docker-ent... 3.02kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 15-local-resolvers.envsh /docker-entryp... 389B buildkit.dockerfile.v0
<missing> 2 days ago COPY 10-listen-on-ipv6-by-default.sh /docker... 2.12kB buildkit.dockerfile.v0
<missing> 2 days ago COPY docker-entrypoint.sh / # buildkit 1.62kB buildkit.dockerfile.v0
<missing> 2 days ago RUN /bin/sh -c set -x && addgroup -g 101... 4.25MB buildkit.dockerfile.v0
<missing> 2 days ago ENV DYNPKG_RELEASE=1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV PKG_RELEASE=1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NGINX_VERSION=1.29.5 0B buildkit.dockerfile.v0
<missing> 2 days ago LABEL maintainer=NGINX Docker Maintainers <d... 0B buildkit.dockerfile.v0
<missing> 10 days ago CMD ["/bin/sh"] 0B buildkit.dockerfile.v0
<missing> 10 days ago ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /... 8.44MB buildkit.dockerfile.v0
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
46e47b8ec5ba nginx-health "/docker-entrypoint..." 19 seconds ago Up 18 seconds (healthy) 0.0.0.0:8085->80/tcp, [::]:8085->80/tcp nginx-fi
d7e17173edb2 nginx-alpine "nginx -g 'daemon of..." 18 minutes ago Up 18 minutes 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp nginx-al
c02bb8078de3 nginx-ubuntu "nginx -g 'daemon of..." 20 minutes ago Up 20 minutes 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp nginx-ub
0e98675fa86e nginx "/docker-entrypoint..." 25 minutes ago Up 25 minutes 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp nginx-of
ficial
acer@DESKTOP-DOEGC13:~/nginx_lab$ sudo docker images | grep nginx
nginx-health latest 35e46240351a About a minute ago 62.1MB
nginx-alpine latest 2fe6bc635982 19 minutes ago 10.4MB
nginx-ubuntu latest 40bcda3f26ed 21 minutes ago 134MB
nginx latest 5cdef4ac3335 2 days ago 161MB
acer@DESKTOP-DOEGC13:~/nginx_lab$ |

```

## Step 9 – Add a Health Check to the Alpine Image (Extension)

Create a `Dockerfile.health` that includes the `HEALTHCHECK` instruction for Docker to monitor the container.

```

FROM nginx:alpine
COPY index.html /usr/share/nginx/html/

```



```
HEALTHCHECK --interval=5s --timeout=3s --start-period=5s --retries=3 CMD wget -q --spider http://localhost/ || exit 1
```

Build the image and run the container on port **8085**:

```
cd ~/nginx_lab
sudo docker build -t nginx-health -f Dockerfile.health .
sudo docker run -d -p 8085:80 --name nginx-final nginx-health
sleep 5
curl http://localhost:8085
```



```

acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/OneDrive/Desktop/Cloud Lab/Containerization-and-Devops-Lab-/lab/Experiment-3/test_lab$ # Create a simple Dockerfile with health check
cat > Dockerfile.health <<'EOF'
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/
HEALTHCHECK --interval=5s --timeout=3s --start-period=5s --retries=3 CMD wget -q --spider http://localhost/ || exit 1
EOF

Build and run
sudo docker build -t nginx-health -f Dockerfile.health .
sudo docker run -d -p 8084:80 --name nginx-healthy nginx-health

Wait for health check
sleep 5

Test
curl http://localhost:8084
[+] Building 16.4s (7/7) FINISHED
=> [internal] load build definition from Dockerfile.health 0.1s
=> => transferring dockerfile: 221B 0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine 6.4s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [internal] load build context 0.1s
=> => transferring context: 65B 0.0s
=> [1/2] FROM docker.io/library/nginx:alpine@sha256:5878d06ae4c83d73285438255f705bb3f9a736f41cd24876ed25bb33faf76c7d 9.4s
=> => resolve docker.io/library/nginx:alpine@sha256:5878d06ae4c83d73285438255f705bb3f9a736f41cd24876ed25bb33faf76c7d 0.1s
=> => sha256:bca5d04786e112d958f100a66f8257b2aeefc14b64d81e405c3c44acff2fb000 1.86MB / 1.86MB 1.5s
=> => sha256:5878d06ae4c83d73285438255f705bb3f9a736f41cd24876ed25bb33faf76c7d 10.33kB / 10.33kB 0.0s
=> => sha256:c032460d1fd73978317479ba23c37bcb57d93156cab122eb3c54b8e4bdc292fa 2.50kB / 2.50kB 0.0s
=> => sha256:3e2c181db1b0985ce357c7aaf48ac615f30f392cd15d5c5ba34c4faa1f4f39a2 626B / 626B 1.5s
=> => sha256:6b7b6c7061b76cdb8601e18722d12ae3232f0ddcfald2983754abcc6ce0a8a83 954B / 954B 1.9s
=> => sha256:b76de378d57272a1dd9091a05dd548a3639dfb792ebdbf95d06704d2950afdea 11.65kB / 11.65kB 0.0s
=> => extracting sha256:bca5d04786e112d958f100a66f8257b2aeefc14b64d81e405c3c44acff2fb000 0.2s
=> => sha256:955a8478f9aceb66cbf2f579fa3c24e1af278d1fa3ffd3043d6260e21d2f7416 1.21kB / 1.21kB 2.6s

```

```
acer@DESKTOP-DOEGC13: ~/ | X + v
ls -la
/home/acer/nginx_lab
total 16
drwxr-xr-x 2 acer acer 4096 Feb 7 09:31 .
drwxr-x--- 6 acer acer 4096 Feb 7 09:23 ..
-rw-r--r-- 1 acer acer 175 Feb 7 09:31 Dockerfile.health
-rw-r--r-- 1 acer acer 184 Feb 7 09:23 index.html
acer@DESKTOP-DOEGC13:~/nginx_lab$ # Create HTML file
cat > index.html <<'EOF'
<!DOCTYPE html>
<html>
<head>
 <title>NGINX Lab - Final</title>
</head>
<body>
 <h1>Lab 3 Complete!</h1>
 <p>NGINX deployed with Alpine base image</p>
 <p>Image size: ~10MB</p>
</body>
</html>
EOF
acer@DESKTOP-DOEGC13:~/nginx_lab$ # Build the image
sudo docker build -t nginx-health -f Dockerfile.health .

Run it
sudo docker run -d -p 8085:80 --name nginx-final nginx-health

Test
sleep 2
curl http://localhost:8085
[+] Building 2.7s (1/2)
=> [internal] load build definition from Dockerfile.health
=> => transferring dockerfile: 221B
=> [internal] load metadata for docker.io/library/nginx:alpine
docker:default
0.0s
0.0s
2.7s
```

## Step10 – Final Summary of All NGINX Images

List all containers and images to observe the size differences.

```
sudo docker ps
sudo docker images | grep nginx
```



```

acer@DESKTOP-DOEGC13: ~/ | X + v
<missing> 2 days ago EXPOSE map[80/tcp:{}] 0B buildkit.dockerfile.v0
<missing> 2 days ago ENTRYPOINT ["/docker-entrypoint.sh"] 0B buildkit.dockerfile.v0
<missing> 2 days ago COPY 30-tune-worker-processes.sh /docker-ent... 4.62kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 20-envsubst-on-templates.sh /docker-ent... 3.02kB buildkit.dockerfile.v0
<missing> 2 days ago COPY 15-local-resolvers.envsh /docker-entryp... 389B buildkit.dockerfile.v0
<missing> 2 days ago COPY 10-listen-on-ipv6-by-default.sh /docker... 2.12kB buildkit.dockerfile.v0
<missing> 2 days ago COPY docker-entrypoint.sh / # buildkit 1.62kB buildkit.dockerfile.v0
<missing> 2 days ago RUN /bin/sh -c set -x && addgroup -g 101... 4.25MB buildkit.dockerfile.v0
<missing> 2 days ago ENV DYNPKG_RELEASE=1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV PKG_RELEASE=1 0B buildkit.dockerfile.v0
<missing> 2 days ago ENV NGINX_VERSION=1.29.5 0B buildkit.dockerfile.v0
<missing> 2 days ago LABEL maintainer=NGINX Docker Maintainers <d... 0B buildkit.dockerfile.v0
<missing> 10 days ago CMD ["/bin/sh"] 0B buildkit.dockerfile.v0
<missing> 10 days ago ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /... 8.44MB buildkit.dockerfile.v0
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
46e47b8ec5ba nginx-health "/docker-entrypoint..." 19 seconds ago Up 18 seconds (healthy) 0.0.0.0:8085->80/tcp, [::]:8085->80/tcp nginx-fi
nal
d7e17173edb2 nginx-alpine "nginx -g 'daemon of..." 18 minutes ago Up 18 minutes 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp nginx-al
pine
c02bb8078de3 nginx-ubuntu "nginx -g 'daemon of..." 20 minutes ago Up 20 minutes 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp nginx-ub
untu
0e98675fa86e nginx "/docker-entrypoint..." 25 minutes ago Up 25 minutes 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp nginx-of
ficial
acer@DESKTOP-DOEGC13:~/nginx_lab$ sudo docker images | grep nginx
nginx-health latest 35e46240351a About a minute ago 62.1MB
nginx-alpine latest 2fe6bc635982 19 minutes ago 10.4MB
nginx-ubuntu latest 40bcd3f26ed 21 minutes ago 134MB
nginx latest 5cdef4ac3335 2 days ago 161MB
acer@DESKTOP-DOEGC13:~/nginx_lab$ # You already created:
mkdir html
echo "<h1>Hello from Docker NGINX</h1>" > html/index.html

And ran:
docker run -d -p 8083:80 -v $(pwd)/html:/usr/share/nginx/html nginx

```

The final image sizes are:

Image	Size
nginx (official)	161 MB
nginx-ubuntu	134 MB

Image	Size
nginx-alpine	10.4 MB
nginx-health	62.1 MB

## Key Observations

- **Official NGINX** is pre-optimised and ready for production, but still larger than an Alpine-based build.
- **Ubuntu-based** image includes a complete operating system, leading to the largest size and longest build time. It is useful for debugging or when additional system tools are needed.
- **Alpine-based** image is **~10 MB**, contains only the bare minimum, starts almost instantly, and has a smaller security surface – ideal for microservices and CI/CD pipelines.
- The **health check** adds a small layer that allows Docker to monitor the container's state, improving reliability in orchestrated environments.

## Result

Successfully deployed NGINX from three different base images, inspected their layers, understood the trade-offs between size, flexibility, and security, and demonstrated a real-world scenario with a custom HTML page and a health check.





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varun124477 Add Experiments 11 &amp; 12, update Experiments 6 &amp; 7, and update homepage ✓

abb8bef · 19 minutes ago

History

Preview

Code

Blame

Raw



The README I previously wrote used placeholder image names like `1.png` , `2.png` , etc. Your actual screenshots are stored in a folder named `Image` (singular) and have longer, descriptive file names – that's why they don't show up.

Below is the complete corrected README. It uses the **exact filenames** you provided, with spaces encoded as `%20` so they work in Markdown. Place this `README.md` inside your `Experiment-4` folder, and the images will display correctly on GitHub.

## ## Lab Experiment 4: Docker Essentials – Dockerfile, .dockerignore, Tagging & Publishing

### ### Aim

To containerise a Python Flask application, understand Dockerfile instructions, ``.dockerignore``, image tagging, layer caching

### ### Pre-requisites

- Docker installed (used via WSL/Ubuntu)
- Python/Flask basics
- Terminal access

> **Project structure**

> All files reside in ``my-flask-app/``.

> **Folder structure** ([./Image/image.png](#))



---

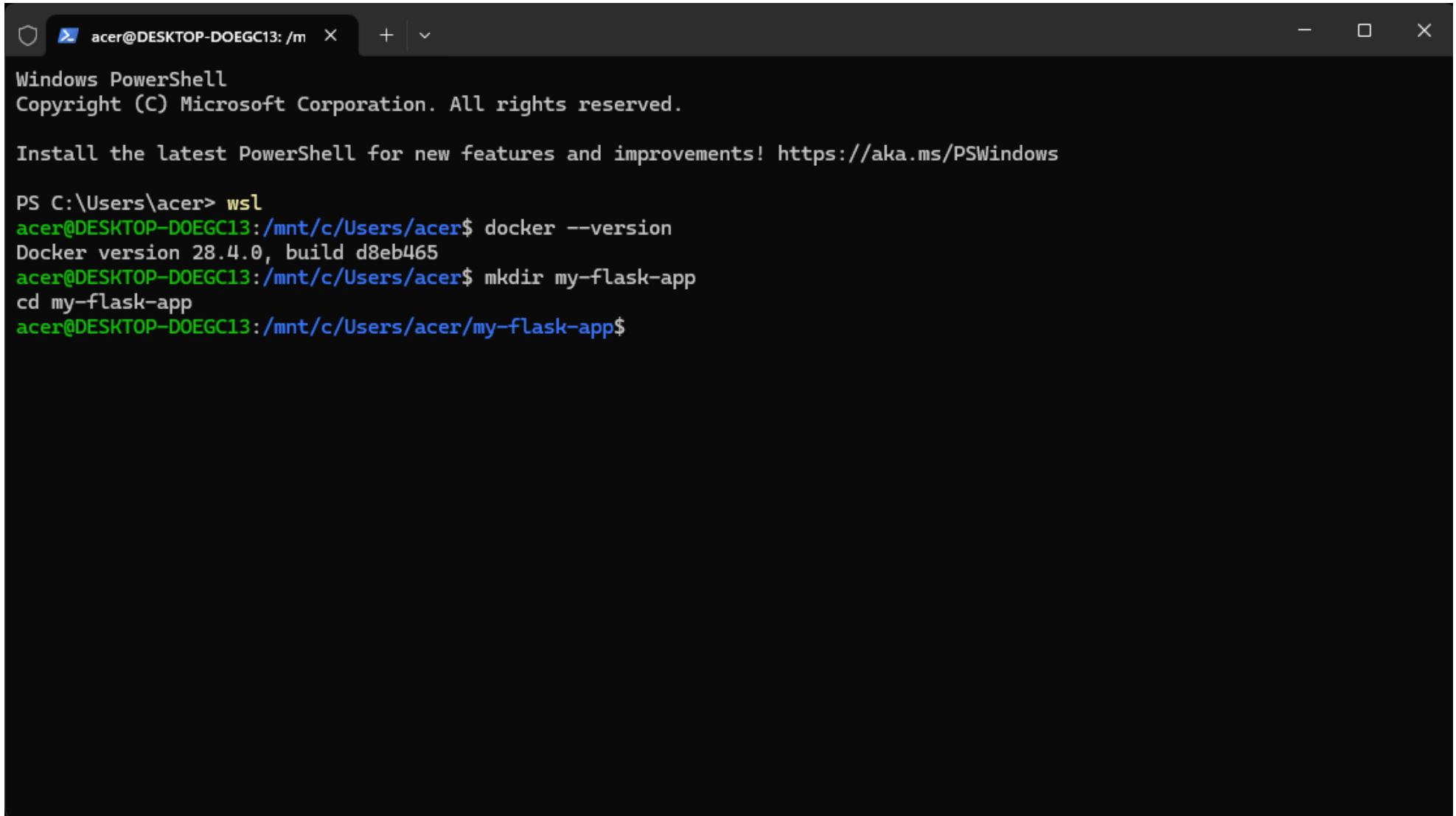
### ### Step-by-Step Procedure

**\*\*Step 1 - Create the project folder\*\***

```
```bash
```

```
mkdir my-flask-app
```

```
cd my-flask-app
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\acer> wsl
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker --version
Docker version 28.4.0, build d8eb465
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ mkdir my-flask-app
cd my-flask-app
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$
```

Step 2 – Write the Flask application (app.py)

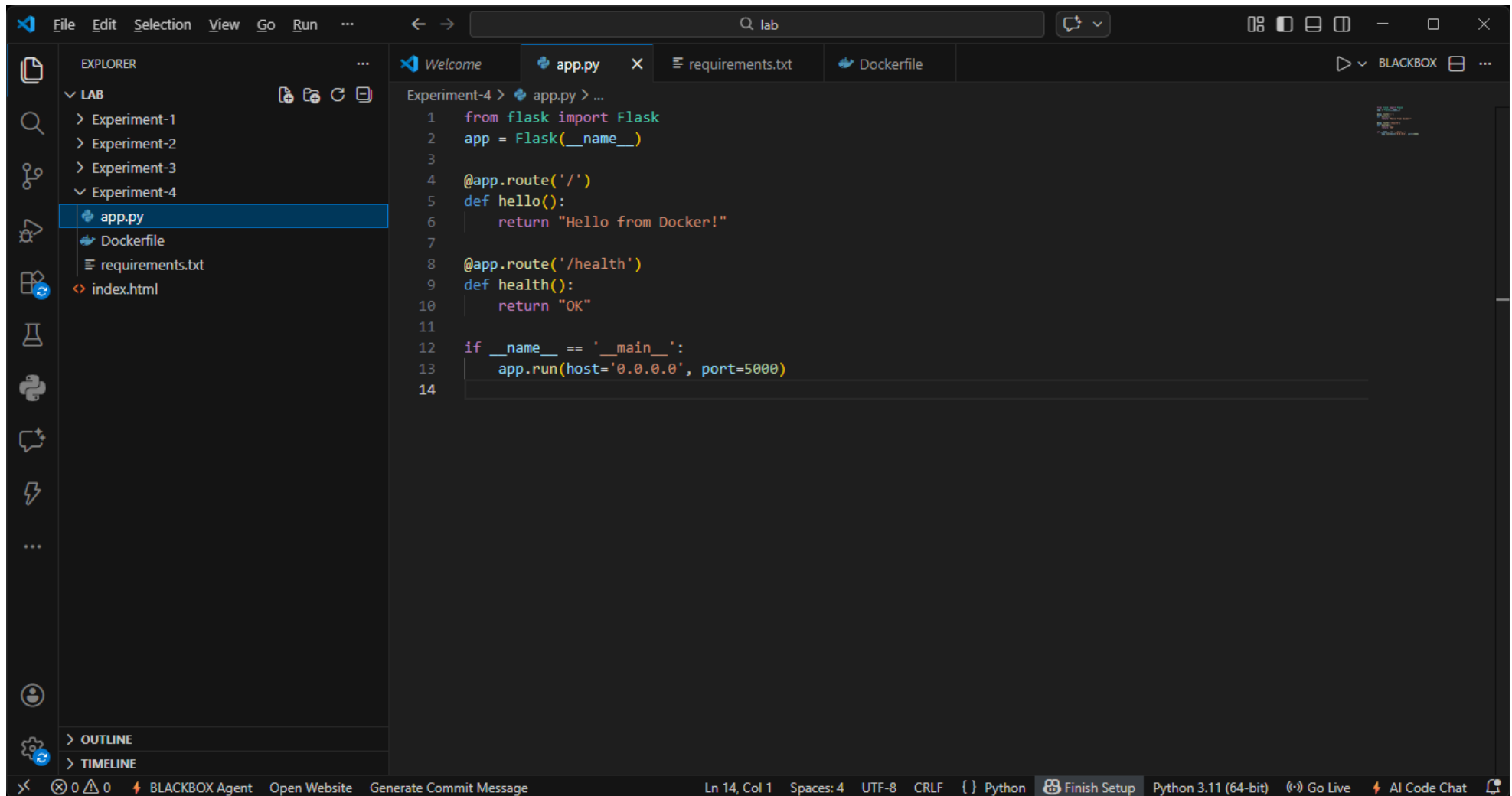
```
from flask import Flask
app = Flask(__name__)

@app.route('/')

```



```
def hello():  
    return "Hello from Docker!"  
  
@app.route('/health')  
def health():  
    return "OK"  
  
if __name__ == '__main__':  
    app.run(host='0.0.0.0', port=5000)
```



The screenshot shows a VS Code editor window with the following details:

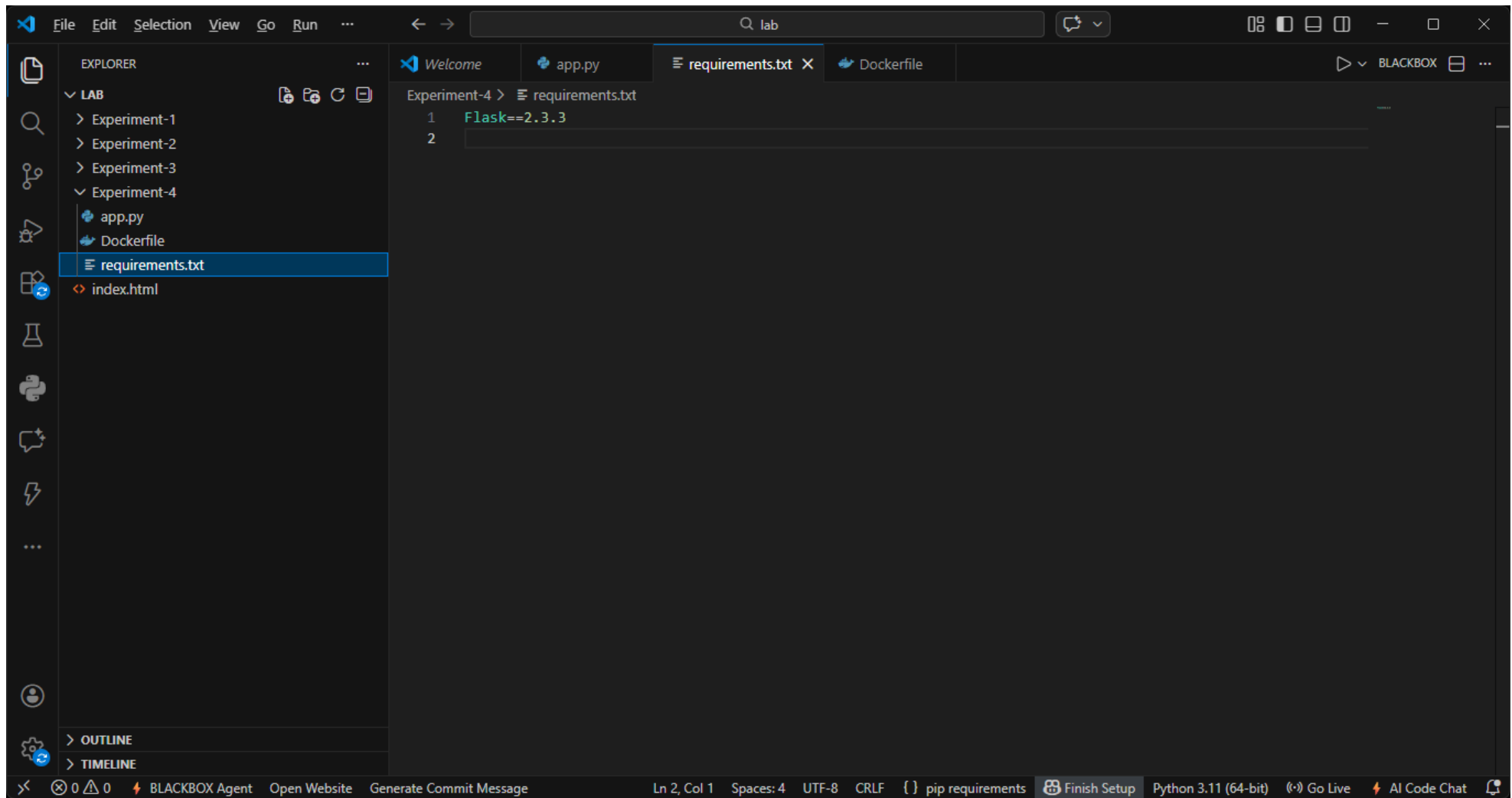
- Explorer Panel:** Shows a project structure with a 'LAB' folder containing 'Experiment-1' through 'Experiment-4'. Inside 'Experiment-4', there are files 'app.py', 'Dockerfile', 'requirements.txt', and 'index.html'. 'app.py' is selected.
- Editor Panel:** Displays the code for 'app.py'. The code is a Flask application with two routes: a root route ('/') that returns 'Hello from Docker!' and a health route ('/health') that returns 'OK'. The application is configured to run on host '0.0.0.0' and port 5000.
- Code in app.py:**

```
1 from flask import Flask
2 app = Flask(__name__)
3
4 @app.route('/')
5 def hello():
6     return "Hello from Docker!"
7
8 @app.route('/health')
9 def health():
10    return "OK"
11
12 if __name__ == '__main__':
13    app.run(host='0.0.0.0', port=5000)
14
```
- Status Bar:** Shows 'Ln 14, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', '{ }', 'Python', 'Finish Setup', 'Python 3.11 (64-bit)', 'Go Live', and 'AI Code Chat'.

Step 3 – requirements.txt

Flask==2.3.3





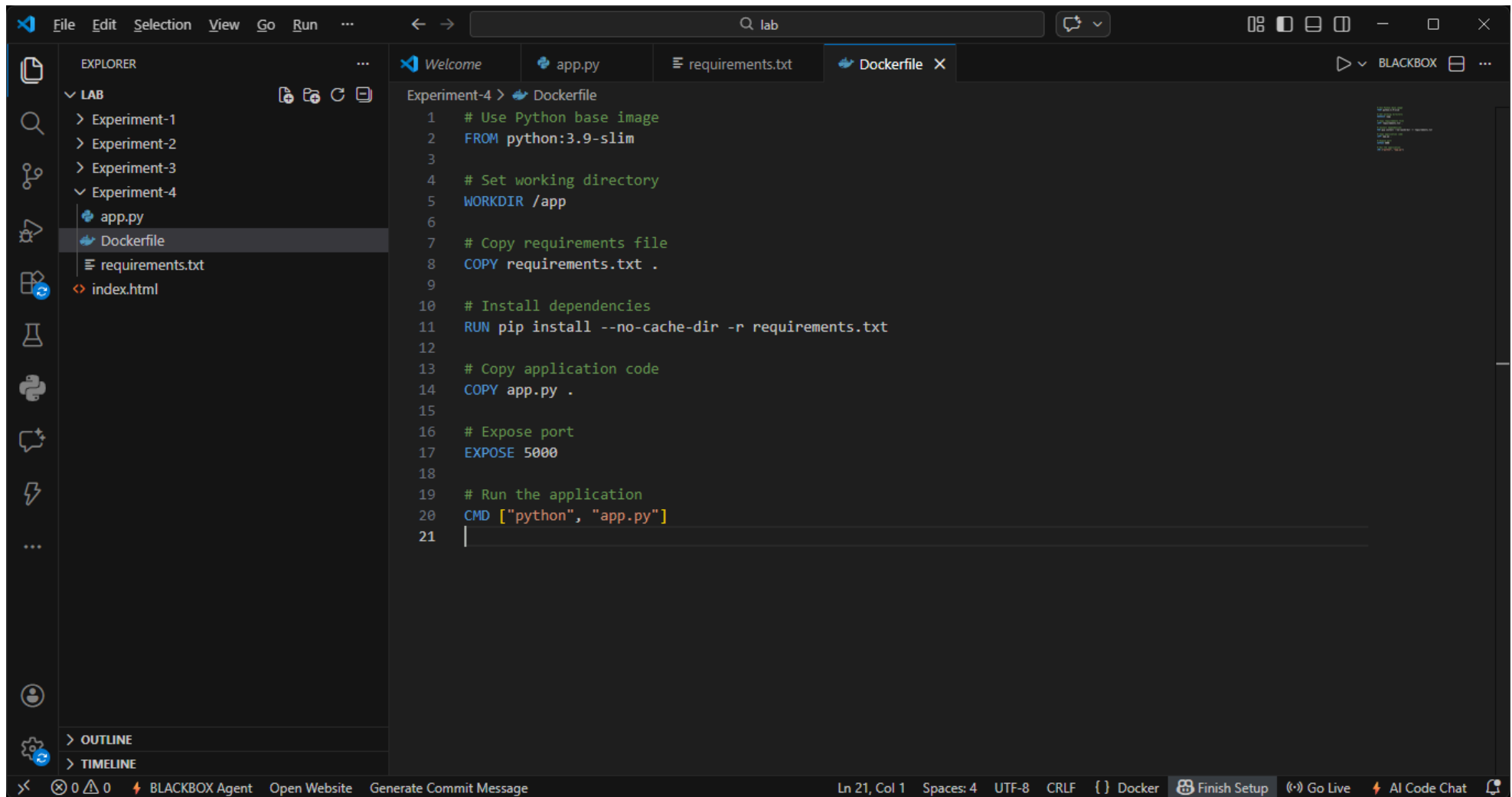
Step 4 – Create the Dockerfile

```
FROM python:3.9-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
```



EXPOSE 5000

CMD ["python", "app.py"]



The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project structure with 'LAB' containing 'Experiment-1' through 'Experiment-4'. Under 'Experiment-4', there are files 'app.py', 'Dockerfile', 'requirements.txt', and 'index.html'. The 'Dockerfile' is selected and its content is displayed in the main editor. The Dockerfile contains the following instructions:

```
1 # Use Python base image
2 FROM python:3.9-slim
3
4 # Set working directory
5 WORKDIR /app
6
7 # Copy requirements file
8 COPY requirements.txt .
9
10 # Install dependencies
11 RUN pip install --no-cache-dir -r requirements.txt
12
13 # Copy application code
14 COPY app.py .
15
16 # Expose port
17 EXPOSE 5000
18
19 # Run the application
20 CMD ["python", "app.py"]
21
```

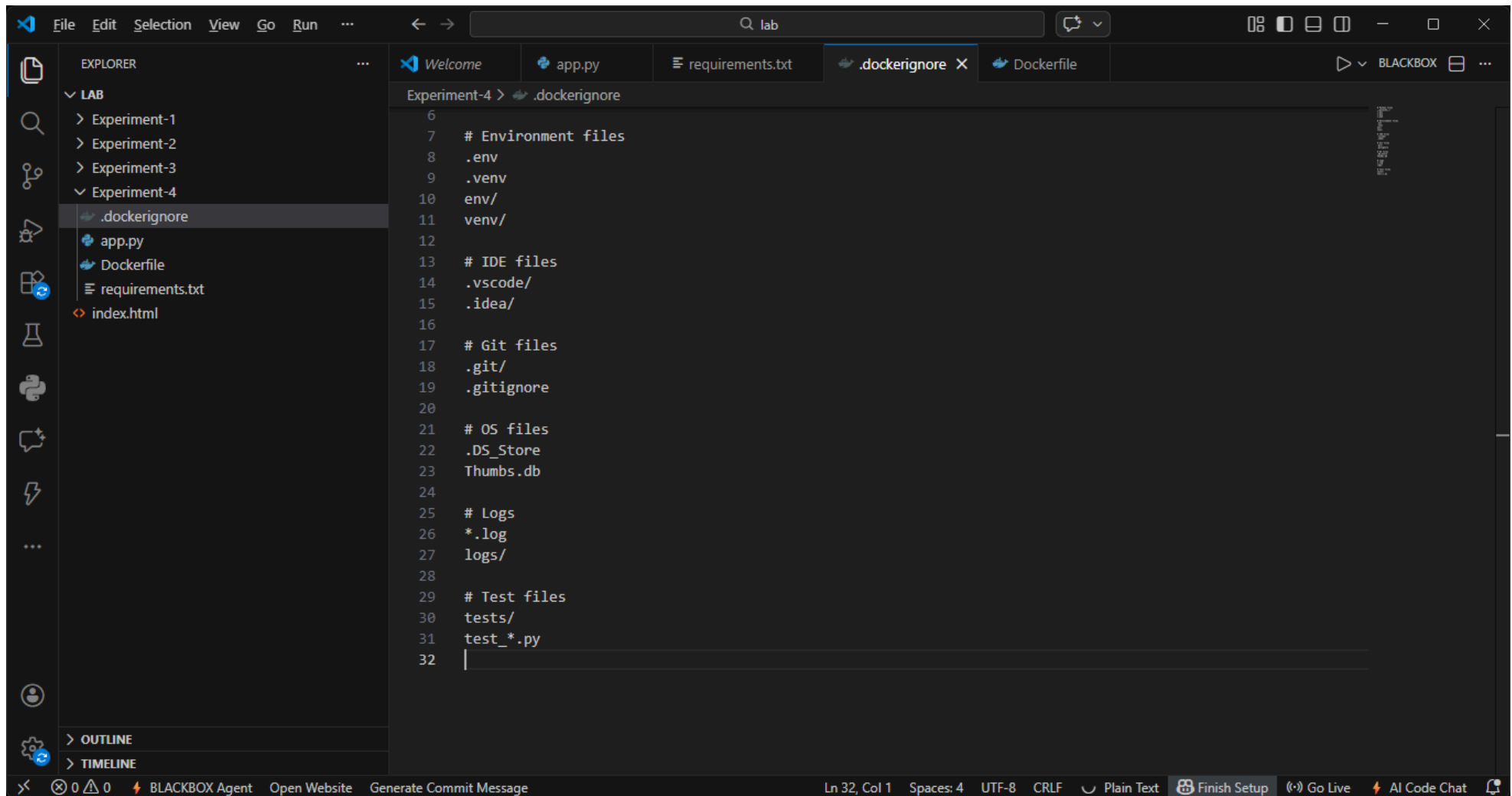
The status bar at the bottom indicates the current line is 21, column 1, with 4 spaces, UTF-8 encoding, and CRLF line endings. It also shows the Docker extension is active and offers options like 'Finish Setup', 'Go Live', and 'AI Code Chat'.

Step 5 – .dockerignore

```
__pycache__/
*.pyc
```



```
.env  
.venv  
env/  
venv/  
.vscode/  
.idea/  
.git/  
.gitignore  
.DS_Store  
Thumbs.db  
*.log  
logs/  
tests/  
test_*.py
```



The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left displays the file structure of a project named 'LAB', which includes folders 'Experiment-1', 'Experiment-2', 'Experiment-3', and 'Experiment-4'. The 'Experiment-4' folder is expanded, showing files like '.dockerignore', 'app.py', 'Dockerfile', 'requirements.txt', and 'index.html'. The main editor window is open to the '.dockerignore' file, showing the following content:

```
6
7 # Environment files
8 .env
9 .venv
10 env/
11 venv/
12
13 # IDE files
14 .vscode/
15 .idea/
16
17 # Git files
18 .git/
19 .gitignore
20
21 # OS files
22 .DS_Store
23 Thumbs.db
24
25 # Logs
26 *.log
27 logs/
28
29 # Test files
30 tests/
31 test_*.py
32 |
```

The status bar at the bottom indicates the current position is 'Ln 32, Col 1' and lists various settings like 'Spaces: 4', 'UTF-8', 'CRLF', and 'Plain Text'. It also includes buttons for 'Finish Setup', 'Go Live', and 'AI Code Chat'.

Step 6 – First build attempt (credential error)

`docker build` fails because the Docker config inside WSL points to `desktop.exe`.

```
docker build -t my-flask-app .
```




```

acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ # Build image from Dockerfile
docker build -t my-flask-app .

# Check built images
docker images
[+] Building 1.7s (2/2) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default
=> => transferring dockerfile: 381B                                              0.1s
=> ERROR [internal] load metadata for docker.io/library/python:3.9-slim          0.1s
-----
> [internal] load metadata for docker.io/library/python:3.9-slim:              1.5s
-----
Dockerfile:2
-----
1 | # Use Python base image
2 | >>> FROM python:3.9-slim
3 |
4 | # Set working directory
-----
ERROR: failed to build: failed to solve: error getting credentials - err: exec: "docker-credential-desktop.exe": executable file not found in $PATH, out: ``

IMAGE ID DISK USAGE CONTENT SIZE EXTRA
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ |

```

Step 7 – Fix the credential issue

Remove the broken Docker config and restart WSL.

```

rm -rf ~/.docker
exit
wsl --shutdown # in PowerShell

```



```
wsl
```

```
cd /mnt/c/Users/acer/my-flask-app
```

```

acer@DESKTOP-DOEGC13: /m  X + v
4 | # Set working directory
-----
ERROR: failed to build: failed to solve: error getting credentials - err: exec: "docker-credential-desktop.exe": executable file not found in $PATH, out: ``
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ cat ~/.docker/config.json
{
  "auths": {},
  "credsStore": "desktop.exe"
}
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ rm -rf ~/.docker
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ ls -a ~
. .aws .bash_history .bashrc .local .profile nginx_lab
.. .azure .bash_logout .cache .motd_shown .sudo_as_admin_successful snap
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ exit
logout
PS C:\Users\acer> wsl --shutdown
PS C:\Users\acer> wsl
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ cd /mnt/c/Users/acer/my-flask-app
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker images
Info → U In Use

IMAGE ID DISK USAGE CONTENT SIZE EXTRA
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker run -d -p 5000:5000 --name flask-container my-flask-app
docker ps
Unable to find image 'my-flask-app:latest' locally
docker: error getting credentials - err: exec: "docker-credential-desktop.exe": executable file not found in $PATH, out:
``

Run 'docker run --help' for more information
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ |

```

```

-----
> [internal] load metadata for docker.io/library/python:3.9-slim:
-----
Dockerfile:2
-----
1 | # Use Python base image
2 | >>> FROM python:3.9-slim
3 |
4 | # Set working directory
-----
ERROR: failed to build: failed to solve: error getting credentials - err: exec: "docker-credential-desktop.exe": executable file not found in $PATH, out: ``
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ cat ~/.docker/config.json
{
  "auths": {},
  "credsStore": "desktop.exe"
}
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ rm -rf ~/.docker
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ ls -a ~
.  .aws  .bash_history  .bashrc  .local  .profile  nginx_lab
.. .azure  .bash_logout  .cache  .motd_shown  .sudo_as_admin_successful  snap
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ exit
logout
PS C:\Users\acer> wsl --shutdown
PS C:\Users\acer> wsl
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ cd /mnt/c/Users/acer/my-flask-app
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker images

IMAGE ID DISK USAGE CONTENT SIZE EXTRA
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ |

```

Step 8 – Rebuild successfully

Now the build completes and creates the image.

```

docker build --no-cache -t my-flask-app .
docker images

```



```

acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ rm -rf ~/.docker
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ ls -a ~
.  .aws  .bash_history  .bashrc  .local  .profile  nginx_lab
.. .azure  .bash_logout  .cache  .motd_shown  .sudo_as_admin_successful  snap
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ exit
logout
PS C:\Users\acer> wsl --shutdown
PS C:\Users\acer> wsl
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ cd /mnt/c/Users/acer/my-flask-app
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker images

IMAGE ID          DISK USAGE  CONTENT SIZE  EXTRA
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker run -d -p 5000:5000 --name flask-container my-flask-app
docker ps
Unable to find image 'my-flask-app:latest' locally
docker: error getting credentials - err: exec: "docker-credential-desktop.exe": executable file not found in $PATH, out:
``

Run 'docker run --help' for more information
CONTAINER ID  IMAGE  COMMAND  CREATED  STATUS  PORTS  NAMES
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ printf '{ "auths": {} }\n' > ~/.docker/config.json
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ cat ~/.docker/config.json
{ "auths": {} }
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ export DOCKER_CONFIG=$HOME/.docker
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker build --no-cache -t my-flask-app .
[+] Building 0.3s (1/2)
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 381B 0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 0.1s

```

```

{ "auths": {} }
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ export DOCKER_CONFIG=$HOME/.docker
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker build --no-cache -t my-flask-app .
[+] Building 24.0s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 381B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 4.4s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 304B                                    0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf 7.5s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf 0.1s
=> => sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B 0.7s
=> => sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB 2.8s
=> => sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB 1.5s
=> => sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB 4.5s
=> => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 1.6s
=> => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.1s
=> => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 1.0s
=> => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 0.0s
=> [internal] load build context                                  0.2s
=> => transferring context: 335B                                    0.1s
=> [2/5] WORKDIR /app                                           0.3s
=> [3/5] COPY requirements.txt .                                0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt     9.6s
=> [5/5] COPY app.py .                                          0.1s
=> exporting to image                                           1.5s
=> => exporting layers                                             0.9s
=> => exporting manifest sha256:d09fd28e91b7acfd4ad0566247436cc58f4311487ba1cfc64313ca06712b303d 0.0s
=> => exporting config sha256:1c62832ccb3a30f70a49d4e3f3ea6beb851a92d87c9c9e7b11e0e0b685f82b87 0.0s
=> => exporting attestation manifest sha256:012f8076b5f9e4d60d1717860860850a59527829c1c0a27e471c8e476f340bf8 0.1s

```

```

acer@DESKTOP-DOEGC13: /m  X + v
=> => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 1.6s
=> => extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 0.1s
=> => extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 1.0s
=> => extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 0.0s
=> [internal] load build context 0.2s
=> => transferring context: 335B 0.1s
=> [2/5] WORKDIR /app 0.3s
=> [3/5] COPY requirements.txt . 0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt 9.6s
=> [5/5] COPY app.py . 0.1s
=> exporting to image 1.5s
=> => exporting layers 0.9s
=> => exporting manifest sha256:d09fd28e91b7acfd4ad0566247436cc58f4311487ba1cfc64313ca06712b303d 0.0s
=> => exporting config sha256:1c62832ccb3a30f70a49d4e3f3ea6beb851a92d87c9c9e7b11e0e0b685f82b87 0.0s
=> => exporting attestation manifest sha256:012f8076b5f9e4d60d1717860860850a59527829c1c0a27e471c8e476f340bf8 0.1s
=> => exporting manifest list sha256:3c0db181cf5050e44954e4045809c51ed177f53e236527567514f6d24af1c3a7 0.0s
=> => naming to docker.io/library/my-flask-app:latest 0.0s
=> => unpacking to docker.io/library/my-flask-app:latest 0.4s
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker images
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
my-flask-app:latest 3c0db181cf50 200MB 48.8MB
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker run -d -p 5000:5000 --name flask-container my-flask-app
docker ps
ccb04efd9461e7f08753e3f175e064d8c2b11cc969a346404797ed2fa2ba630b
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
ccb04efd9461 my-flask-app "python app.py" 2 seconds ago Up Less than a second 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp flask-container
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ |

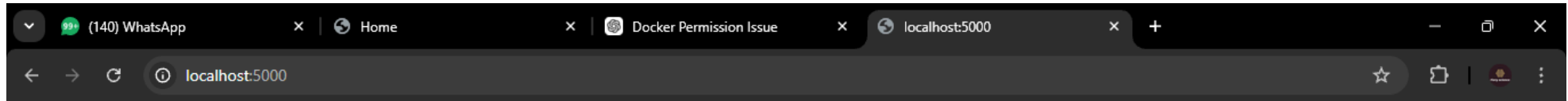
```

Step 9 – Run the container and test

```
docker run -d -p 5000:5000 --name flask-container my-flask-app
```



Open `http://localhost:5000` – you should see “Hello from Docker!”.



Hello from Docker!

Step 10 – Make a code change and rebuild (demonstrating layer caching)

Edit `app.py` , then rebuild. Notice how unchanged layers (like `pip install`) are cached.

```
nano app.py
docker build -t my-flask-app .
```




```
docker rm -f flask-container  
docker run -d -p 5000:5000 --name flask-container my-flask-app
```

```

acer@DESKTOP-DOEGC13: /m  X  +  v
=> => unpacking to docker.io/library/my-flask-app:latest 0.4s
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker images
INFO → U In Use
IMAGE          ID          DISK USAGE  CONTENT SIZE  EXTRA
my-flask-app:latest 3c0db181cf50 200MB       48.8MB
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker run -d -p 5000:5000 --name flask-container my-flask-app
docker ps
ccb04efd9461e7f08753e3f175e064d8c2b11cc969a346404797ed2fa2ba630b
CONTAINER ID  IMAGE          COMMAND          CREATED          STATUS          PORTS
              NAMES
ccb04efd9461  my-flask-app  "python app.py"  2 seconds ago   Up Less than a second  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  flask-container
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ nano app.py
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ nano app.py
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ nano app.py
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker build -t my-flask-app .
[+] Building 3.4s (10/10) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 381B 0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 2.5s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 304B 0.1s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf 0.1s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf 0.1s
=> [internal] load build context 0.1s
=> => transferring context: 336B 0.0s
=> CACHED [2/5] WORKDIR /app 0.0s
=> CACHED [3/5] COPY requirements.txt . 0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY app.py . 0.1s

```

```

acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ nano app.py
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ nano app.py
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker build -t my-flask-app .
[+] Building 3.4s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 381B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 2.5s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 304B                                    0.1s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf 0.1s
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf 0.1s
=> [internal] load build context                                  0.1s
=> => transferring context: 336B                                    0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                            0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY app.py .                                           0.1s
=> exporting to image                                             0.4s
=> => exporting layers                                             0.1s
=> => exporting manifest sha256:747dee26c647bb6d513cba6ef931616e6ecc883b903dca59fe3f0a40f3e0e7cb 0.0s
=> => exporting config sha256:702f891527308f9e411aa4b8a13eb023f5e32b9a6910f29c0e4212c13ab4d74c 0.0s
=> => exporting attestation manifest sha256:e52bbf5cbf1975ed6ee20b5b4219b36f23ff30912e009ffe1a4034eb71b232c0 0.0s
=> => exporting manifest list sha256:edfa383ffc748f52a67cf2e3841ee536c1e44f962f8da4912a10802c8c861fe6 0.0s
=> => naming to docker.io/library/my-flask-app:latest            0.0s
=> => unpacking to docker.io/library/my-flask-app:latest         0.1s
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker rm -f flask-container
flask-container
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker run -d -p 5000:5000 --name flask-container my-flask-app
950eb6a43e058b1b917945952156429739b4987a68318c22dd84f97055a8cdc6
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ |

```

Step 11 – Tagging for versioning and registry

```

docker tag my-flask-app:latest username/my-flask-app:1.0
docker tag my-flask-app:latest username/my-flask-app:latest

```



Step 12 – Publish to Docker Hub

```
docker login
docker push username/my-flask-app:1.0
docker push username/my-flask-app:latest
```



Step 13 – Multi-stage build (optional)

A multi-stage Dockerfile reduces the final image size by separating build and runtime stages.

```
# STAGE 1: Builder
FROM python:3.9-slim AS builder
WORKDIR /app
COPY requirements.txt .
RUN pip install --user -r requirements.txt

# STAGE 2: Runtime
FROM python:3.9-slim
WORKDIR /app
COPY --from=builder /root/.local /root/.local
COPY app.py .
ENV PATH=/root/.local/bin:$PATH
EXPOSE 5000
CMD ["python", "app.py"]
```



Build and compare sizes:

```
docker build -f Dockerfile.multistage -t my-flask-app:multistage .
docker images | grep my-flask-app
```



Key Concepts

Term	Description
Dockerfile	Declarative instructions to build an image
.dockerignore	Prevents sending unnecessary files to the build context
Layer caching	Docker reuses unchanged layers for faster rebuilds
Tag	A label attached to an image (e.g., <code>v1.0</code> , <code>latest</code>)
Multi-stage build	Uses multiple <code>FROM</code> statements to keep production images lean
Registry	Storage for images (Docker Hub, AWS ECR, etc.)

Result

A complete Flask application was containerised. The Docker credential issue was resolved, the image was built and run successfully, layer caching was observed, and the concepts of tagging and multi-stage builds were understood. The app served correctly on `http://localhost:5000` .



Once you save this as ``README.md`` inside ``Experiment-4``, all images will reference the exact file names in your ``Image`` folder. If you later rename the folder to ``Images``, just update all ``.`/Image/`` prefixes to ``.`/Images/``.



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Experiment 5: Docker Volumes, Environment Variables, Monitoring & Networks

Aim

To master Docker volumes for persistent data, configure containers with environment variables, monitor resource usage and l

Pre-requisites

- Docker installed (WSL / Ubuntu environment)
- Basic Linux command line
- Familiarity with ``docker run``, ``docker ps``, ``docker exec``

> ****Experiment Directory****

> All screenshots are stored inside the ``Image`` folder. Paths are case-sensitive.

> ![Local folder](./Image/image.png)

Part 1 – Docker Volumes: Persistent Data Storage

****Step 1.1 – The Ephemeral Nature of Container Data****

First, create a file inside a temporary container and observe that the data is lost once the container is removed.

```
```bash
docker run -it --name test-container ubuntu /bin/bash
Inside container:
mkdir /data
echo "Hello World" > /data/message.txt
cat /data/message.txt # Shows "Hello World"
exit
docker start test-container
docker exec test-container cat /data/message.txt # File still present (container was not removed)
docker rm -f test-container
docker run -it --name test-container ubuntu /bin/bash
cat /data/message.txt # "No such file or directory" - data is lost
```

```
acer@DESKTOP-DOEGC13: /m X + v
root@626ccbce9652:/# mkdir /data
echo "Hello World" > /data/message.txt
cat /data/message.txt
Hello World
root@626ccbce9652:/# exit
exit
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker start test-container
docker exec test-container cat /data/message.txt
test-container
Hello World
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker run -it --name test-container ubuntu /bin/bash
docker: Error response from daemon: Conflict. The container name "/test-container" is already in use by container "626ccbce9652615899b50eb20f5350665dda67b25d90e1fc311cd1140e9e5fdc". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker rm -f test-container
test-container
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker run -it --name test-container ubuntu /bin/bash
root@61bb5dc8398e:/# cat /data/message.txt
cat: /data/message.txt: No such file or directory
root@61bb5dc8398e:/# ^C
root@61bb5dc8398e:/# exit
exit
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker volume create mydata
mydata
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker volume ls
DRIVER VOLUME NAME
local mydata
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$
```

## Step 1.2 – Named Volumes

Create a named volume, mount it, and verify that data survives container removal.

```
docker volume create mydata
docker volume ls
```





```
docker run -it --name volume-test \
 -v mydata:/data \
 ubuntu /bin/bash
echo "Persistent Data" > /data/test.txt
exit

docker rm volume-test
docker run -it --name volume-test2 \
 -v mydata:/data \
 ubuntu /bin/bash
cat /data/test.txt # "Persistent Data" - data survived
exit
```

```
root@41fd6d38cd00: /
root@626ccbce9652:/# exit
exit
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker start test-container
docker exec test-container cat /data/message.txt
test-container
Hello World
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker run -it --name test-container ubuntu /bin/bash
docker: Error response from daemon: Conflict. The container name "/test-container" is already in use by container "626ccbce9652615899b50eb20f5350665dda67b25d90e1fc311cd1140e9e5fdc". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker rm -f test-container
test-container
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker run -it --name test-container ubuntu /bin/bash
root@61bb5dc8398e:/# cat /data/message.txt
cat: /data/message.txt: No such file or directory
root@61bb5dc8398e:/# ^C
root@61bb5dc8398e:/# exit
exit
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker volume create mydata
mydata
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker volume ls
DRIVER VOLUME NAME
local mydata
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer$ docker run -it --name volume-test \\\n-v mydata:/data \\\nubuntu /bin/bash
root@41fd6d38cd00:/# echo "Persistent Data" > /data/test.txt
exit
```

### Step 1.3 – Bind Mounts (Host Directory)

Share a folder between the host and a container so that both can read and write.

```
mkdir ~/myapp-data
docker run -d --name web3 \
```



```
-v ~/myapp-data:/app/data \
nginx

Write from host
echo "From Host - Varun Experiment 5" > ~/myapp-data/host-file.txt
docker exec web3 cat /app/data/host-file.txt # shows the host file

Write from container
docker exec web3 sh -c "echo 'From Container Side' > /app/data/container-file.txt"
cat ~/myapp-data/container-file.txt # shows the container file
```

```
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ ls ~
myapp-data nginx_lab snap
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ docker run -d \
--name web3 \
-v ~/myapp-data:/app/data \
nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
df0b66c867e4: Pull complete
1ad233904a11: Pull complete
206356c42440: Pull complete
b47f187216b6: Pull complete
35ff83c394d6: Pull complete
eedda9fd8786: Pull complete
17d0911eaf62: Pull complete
c13150b54a53: Download complete
0019bbecc572: Download complete
Digest: sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
Status: Downloaded newer image for nginx:latest
ea97190505f2f1eb53de466ae3087fc114b3d8ed01f647de378c2ba36caa0fab
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ echo "From Host - Varun Experiment 5" > ~/myapp-data/host-file.txt
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ docker exec web3 cat /app/data/host-file.txt
From Host - Varun Experiment 5
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ docker exec web3 sh -c "echo 'From Container Side' > /app/data/container-file.txt"
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ cat ~/myapp-data/container-file.txt
From Container Side
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer$ cd /mnt/c/Users/acer/my-flask-app
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ nano app.py
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ |
```

## Part 2 – Environment Variables

### Step 2.1 – Passing Variables at Runtime

Modify the Flask app ( app.py ) to read configuration from environment variables, then run it with custom values.



```
import os
from flask import Flask

app = Flask(__name__)

@app.route('/config')
def config():
 return {
 'database_host': os.environ.get('DATABASE_HOST', 'localhost'),
 'debug': os.environ.get('DEBUG', 'false'),
 'has_api_key': bool(os.environ.get('API_KEY'))
 }

if __name__ == '__main__':
 port = int(os.environ.get('PORT', 5000))
 app.run(host='0.0.0.0', port=port, debug=os.environ.get('DEBUG') == 'true')
```

Build the image, then run with environment variables.

```
docker build --no-cache -t flask-app .
docker run -d -p 5000:5000 \
 -e DATABASE_HOST="prod-db.example.com" \
 -e DEBUG="true" \
 -e API_KEY="secret123" \
 flask-app
```



(If port 5000 is already in use, remove the old container first.)

```

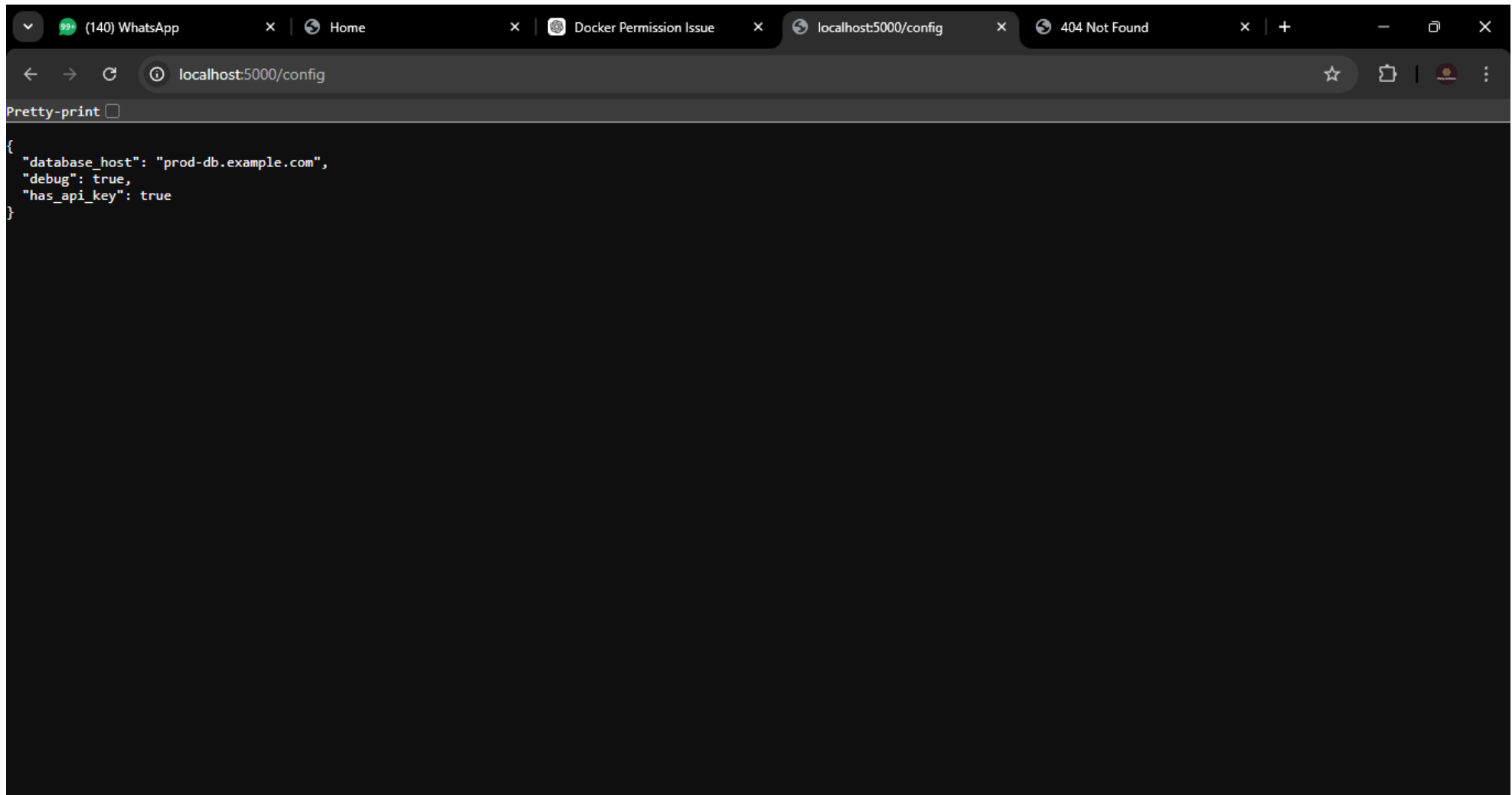
acer@DESKTOP-DOEGC13: /m X + v
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acdale13cf 0.1s
=> [internal] load build context 0.1s
=> => transferring context: 679B 0.1s
=> CACHED [2/5] WORKDIR /app 0.0s
=> CACHED [3/5] COPY requirements.txt . 0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY app.py . 0.0s
=> exporting to image 0.5s
=> => exporting layers 0.2s
=> => exporting manifest sha256:37cf1caea865cfd9a4dde77257b69cfc3be2a873e1a09c0e9dc294799d4d741 0.0s
=> => exporting config sha256:7b07a55cfa75a3d63c6706ddb3268d5e260f9da2b7d3eab76a89dece795f0b15 0.0s
=> => exporting attestation manifest sha256:3c362cfb972b6f8e4998b35ee04d2c8aac07c8e4d99a11863bde9271361cf0a7 0.1s
=> => exporting manifest list sha256:6cca03fac6fae9f0bc5ab3cdcedfd4bd0815717542a66dfadfaf4f0ac5001e5e 0.0s
=> => naming to docker.io/library/flask-app:latest 0.0s
=> => unpacking to docker.io/library/flask-app:latest 0.1s
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker rm -f flask-app 2>/dev/null
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker run -d \
--name flask-app \
-p 5000:5000 \
-e DATABASE_HOST="prod-db.example.com" \
-e DEBUG="true" \
-e API_KEY="secret123" \
flask-app
bc90b4e82bb741b36cdd885bee86865714fdee6cfc1790ed25f303c456239521
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint flask-app (885763f979c2038472cdalaa49164dca2856f26cd8429e31f7c72dccb9187c46): Bind for 0.0.0.0:5000 failed: port is already allocated

Run 'docker run --help' for more information
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ |

```

## Step 2.2 – Test the /config Endpoint

Visit <http://localhost:5000/config> in a browser. The JSON response shows the injected values.



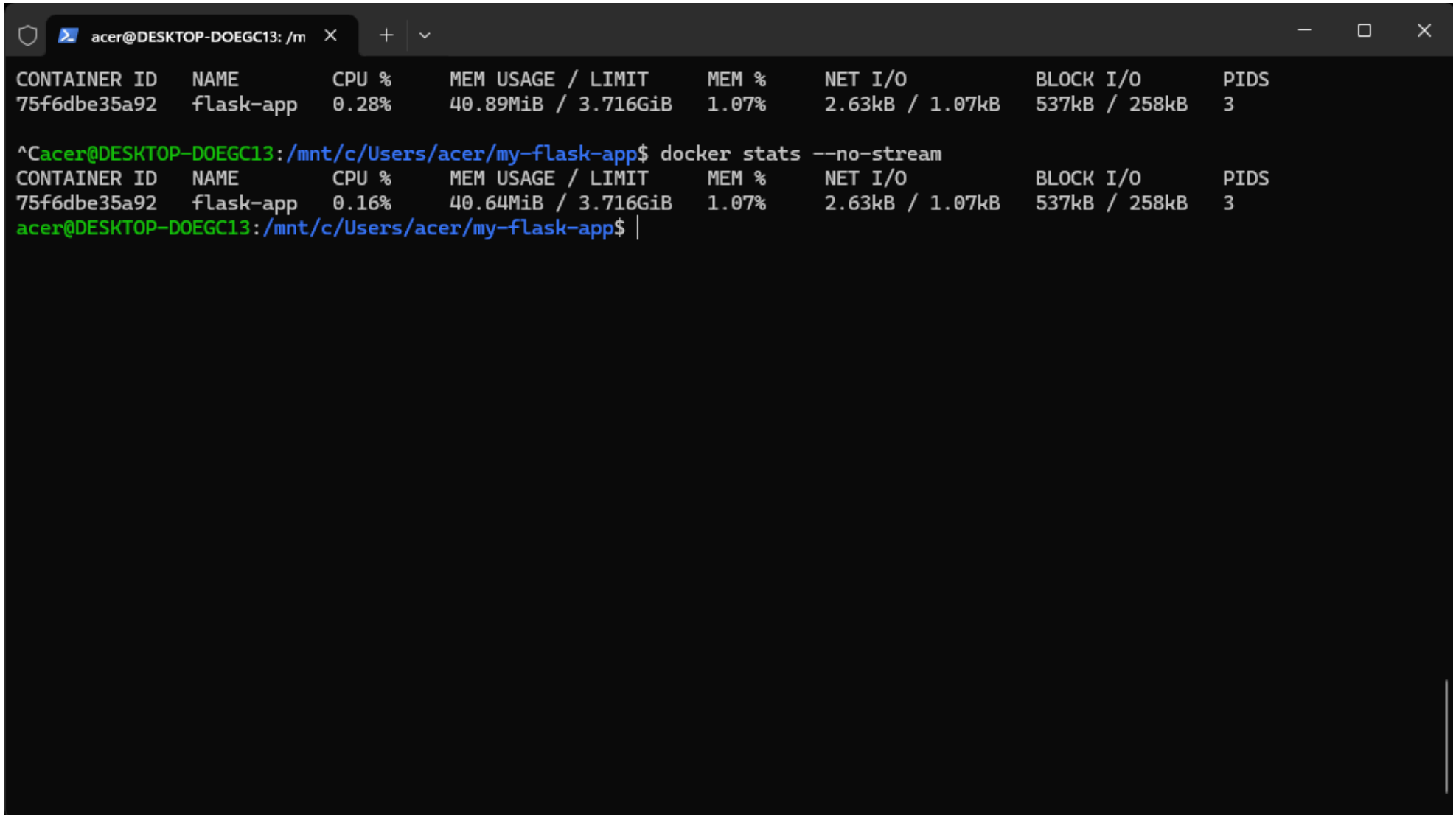
## Part 3 – Docker Monitoring

### Step 3.1 – Real-time Resource Usage with `docker stats`

```
docker stats --no-stream
```



Displays CPU, memory, network I/O for each running container.



```
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker stats --no-stream
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
75f6dbe35a92	flask-app	0.28%	40.89MiB / 3.716GiB	1.07%	2.63kB / 1.07kB	537kB / 258kB	3

```
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$
```

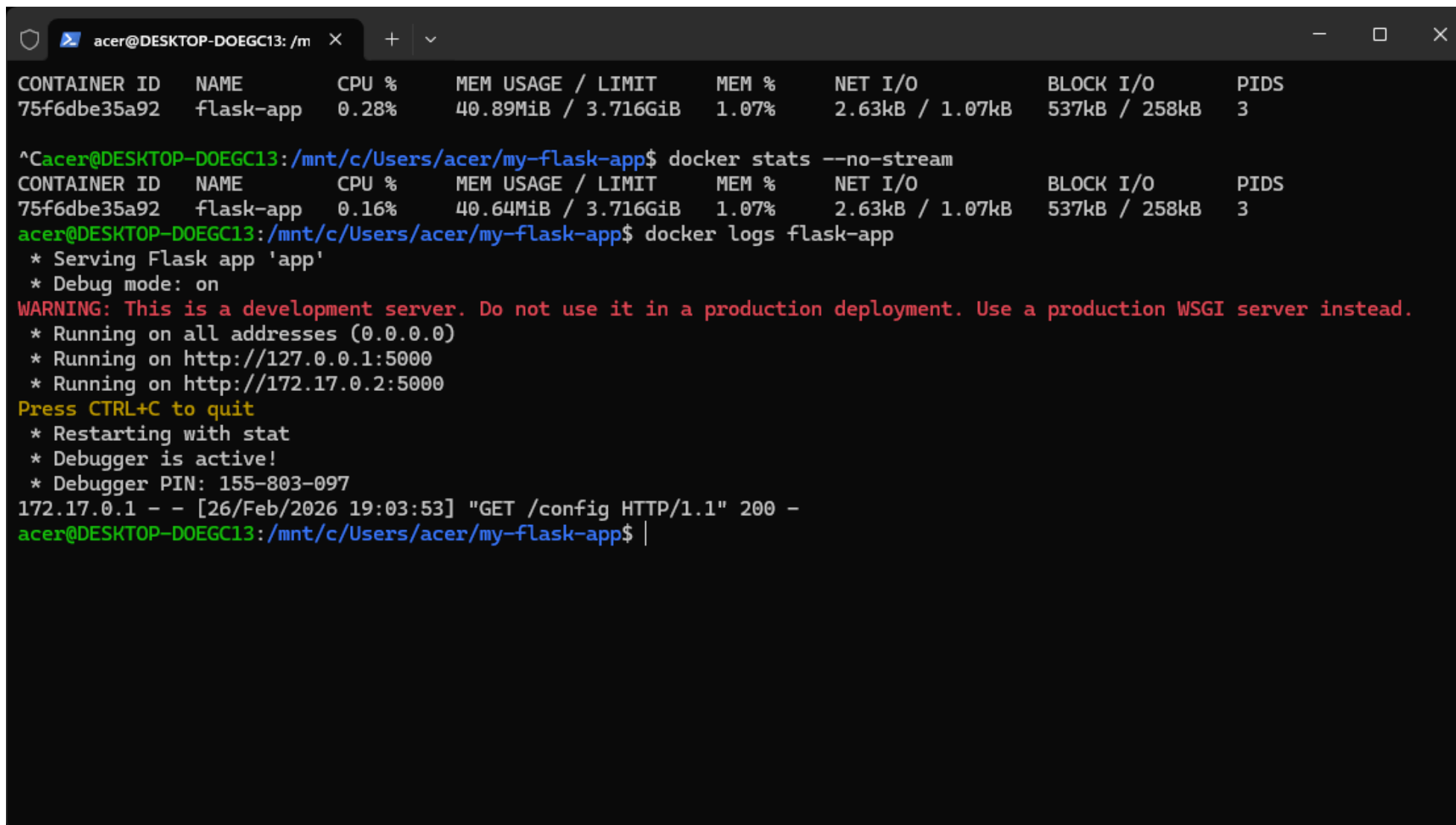
### Step 3.2 – Application Logs with `docker logs`

```
docker logs flask-app
```





Shows stdout/stderr from the Flask app (e.g., incoming requests, debug mode info).



```

acer@DESKTOP-DOEGC13: /m X + v
CONTAINER ID NAME CPU % MEM USAGE / LIMIT MEM % NET I/O BLOCK I/O PIDS
75f6dbe35a92 flask-app 0.28% 40.89MiB / 3.716GiB 1.07% 2.63kB / 1.07kB 537kB / 258kB 3

^Cacer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker stats --no-stream
CONTAINER ID NAME CPU % MEM USAGE / LIMIT MEM % NET I/O BLOCK I/O PIDS
75f6dbe35a92 flask-app 0.16% 40.64MiB / 3.716GiB 1.07% 2.63kB / 1.07kB 537kB / 258kB 3
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker logs flask-app
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 155-803-097
172.17.0.1 - - [26/Feb/2026 19:03:53] "GET /config HTTP/1.1" 200 -
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ |

```

### Step 3.3 – Container Inspection

```

docker inspect flask-app | grep -A 10 "NetworkSettings"
docker inspect --format='{{.State.Status}}' flask-app # "running"

```



```
docker inspect --format='{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' flask-app # 172.17.0.2
```

```
acer@DESKTOP-DOEGC13: /m X + v
 "Aliases": null,
 "DriverOpts": null,
 "GwPriority": 0,
 "NetworkID": "a6113cabcf9b8c1f9be90dd050d06660768586a6f5415a889ffa8e3dbf71465",
 "EndpointID": "700e845dfbae491f0499ec4102f0e552b3c667a1a9faea4d4c05c997daed98c0",
 "Gateway": "172.17.0.1",
 "IPAddress": "172.17.0.2",
 "MacAddress": "0e:d8:b1:8f:c2:52",
 "IPPrefixLen": 16,
 "IPv6Gateway": "",
 "GlobalIPv6Address": "",
 "GlobalIPv6PrefixLen": 0,
 "DNSNames": null
}
},
"ImageManifestDescriptor": {
 "mediaType": "application/vnd.oci.image.manifest.v1+json",
 "digest": "sha256:669ecacb3c15989b3dc99016f092cb493274162afc875d9ce98ca14e0fae842c",
 "size": 1810,
 "platform": {
 "architecture": "amd64",
 "os": "linux"
 }
}
}
]
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker inspect --format='{{.State.Status}}' flask-app
running
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ |
```

## Part 4 – Docker Networks

## Step 4.1 – Create a Custom Bridge Network

```
docker network create my-network
```

```
docker network inspect my-network
```



```
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker inspect --format='{{.State.Status}}' flask-app
running
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$ docker network create my-network
13e48d09628232fa949d59adacfb5b5e1cd3131877857395b16025fa5415a7e6
acer@DESKTOP-DOEGC13: /mnt/c/Users/acer/my-flask-app$
```

```
[{"name": "my-network", "id": "13e48d09628232fa949d59adacfb5b5e1cd3131877857395b16025fa5415a7e6", "scope": "local", "driver": "bridge", "options": {"com.docker.network.bridge.default_bridge": true, "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0", "com.docker.network.bridge.name": "my-network"}, "ipam": {"driver": "default", "options": {}, "config": [{"subnet": "172.17.0.0/16", "gateway": "172.17.0.1"}]}, "containers": {"flask-app": {"id": "13e48d09628232fa949d59adacfb5b5e1cd3131877857395b16025fa5415a7e6", "name": "flask-app", "network_id": "13e48d09628232fa949d59adacfb5b5e1cd3131877857395b16025fa5415a7e6", "endpoint_id": "700e845dfbae491f0499ec4102f0e552b3c667a1a9faea4d4c05c997daed98c0", "gateway": "172.17.0.1", "ip_address": "172.17.0.2", "mac_address": "0e:d8:b1:8f:c2:52", "ip_prefix_len": 16, "ipv6_gateway": "", "global_ipv6_address": "", "global_ipv6_prefix_len": 0, "dns_names": null}}]
}
```

## Step 4.2 – Run Containers on the Custom Network

```
docker run -d --name web1 --network my-network nginx
docker run -d --name web2 --network my-network nginx
docker inspect --format='{{range .NetworkSettings.Networks}}{{.IPAddress}}{{end}}' web2
```



```
acer@DESKTOP-DOEGC13: /m X + v
 "Name": "web2",
 "EndpointID": "efb952b9d4daa1831855e61512f6faa120a7e9971f9224fe1f98df19e049e922",
 "MacAddress": "02:da:20:49:2e:e7",
 "IPv4Address": "172.18.0.3/16",
 "IPv6Address": ""
 },
 "cfd2a398b02e92b544a479d613433c00efada72208802a15a7420de1f959adb2": {
 "Name": "web1",
 "EndpointID": "617c5665b45e5cc06d47ac92dee68a034ccc819b1f1d29216a6a17816e9bb285",
 "MacAddress": "0a:7f:f8:05:ec:c1",
 "IPv4Address": "172.18.0.2/16",
 "IPv6Address": ""
 }
},
"Status": {
 "IPAM": {
 "Subnets": {
 "172.18.0.0/16": {
 "IPsInUse": 5,
 "DynamicIPsAvailable": 65531
 }
 }
 }
}
}
]
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ docker inspect --format='{{range .NetworkSettings.Networks}}{{.IPAd
dress}}{{end}}' web2
172.18.0.3
acer@DESKTOP-DOEGC13:/mnt/c/Users/acer/my-flask-app$ |
```

### Step 4.3 – Test Cross-Container Communication

Because the default `nginx` image does not include `ping`, spin up a temporary Ubuntu container on the same network and install `ping`.

```
docker run -it --network my-network ubuntu /bin/bash
apt update && apt install iputils-ping -y
ping web2 # resolves to 172.18.0.3 and gets replies
```



```
root@7976b8ae7c10: /
Setting up libpam-cap:amd64 (1:2.66-5ubuntu2.2) ...
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/DebianConfig/FrontEnd/Dialog.pm line 79.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.38.2 /usr/local/share/perl/5.38.2 /usr/lib/x86_64-linux-gnu/perl/5.38 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.38 /usr/share/perl/5.38 /usr/local/lib/site_perl) at /usr/share/perl5/DebianConfig/FrontEnd/Readline.pm line 8.)
debconf: falling back to frontend: Teletype
Setting up iputils-ping (3:20240117-1ubuntu0.1) ...
PING web2 (172.18.0.3) 56(84) bytes of data:
64 bytes from web2.my-network (172.18.0.3): icmp_seq=1 ttl=64 time=2.63 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=2 ttl=64 time=0.103 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=3 ttl=64 time=0.136 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=4 ttl=64 time=0.095 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=5 ttl=64 time=0.109 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=6 ttl=64 time=0.124 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=7 ttl=64 time=0.083 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=8 ttl=64 time=0.103 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=9 ttl=64 time=0.138 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=10 ttl=64 time=0.108 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=11 ttl=64 time=0.100 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=12 ttl=64 time=0.134 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=13 ttl=64 time=0.118 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=14 ttl=64 time=0.105 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=15 ttl=64 time=0.104 ms
64 bytes from web2.my-network (172.18.0.3): icmp_seq=16 ttl=64 time=0.135 ms
```

## Quick Reference

Area	Useful Commands
Volumes	<code>docker volume create &lt;name&gt;</code> , <code>docker run -v vol:/path</code> , <code>docker volume ls</code> , <code>docker volume inspect &lt;name&gt;</code>
Bind mounts	<code>-v /host/path:/container/path</code>
Environment	<code>-e VAR=value</code> , <code>--env-file .env</code> , <code>ENV VAR=value</code> (in Dockerfile)
Monitoring	<code>docker stats</code> , <code>docker logs -f &lt;container&gt;</code> , <code>docker top &lt;container&gt;</code> , <code>docker inspect &lt;container&gt;</code>
Networks	<code>docker network create &lt;name&gt;</code> , <code>--network &lt;name&gt;</code> , <code>docker network connect/disconnect</code> , <code>docker network inspect</code>
Port mapping	<code>-p host:container</code> , <code>-p 80</code> (random host port)

## ✓ Key Takeaways

- **Volumes** and **bind mounts** keep data alive beyond the container's lifetime.
- **Environment variables** make containers configurable without rebuilding images.
- **Monitoring** commands help diagnose performance, logs, and current state.
- **Custom networks** allow containers to communicate by name and provide better isolation than the default bridge.



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## ## Docker Lab - Experiment 6: Nginx Container & WordPress with Docker Compose



&lt;hr&gt;

&lt;h4 align="center"&gt; Pre-requisite &lt;/h4&gt;

> **Working Directory Structure**

&gt; All commands in this experiment are executed inside `C:\Users\acer` (the home directory).

&gt; The subfolder `html` holds the custom Nginx page, and `wp-compose` contains the WordPress stack.

&gt;

&gt; ![Directory Structure](./Images/directory-structure.png)

&lt;hr&gt;

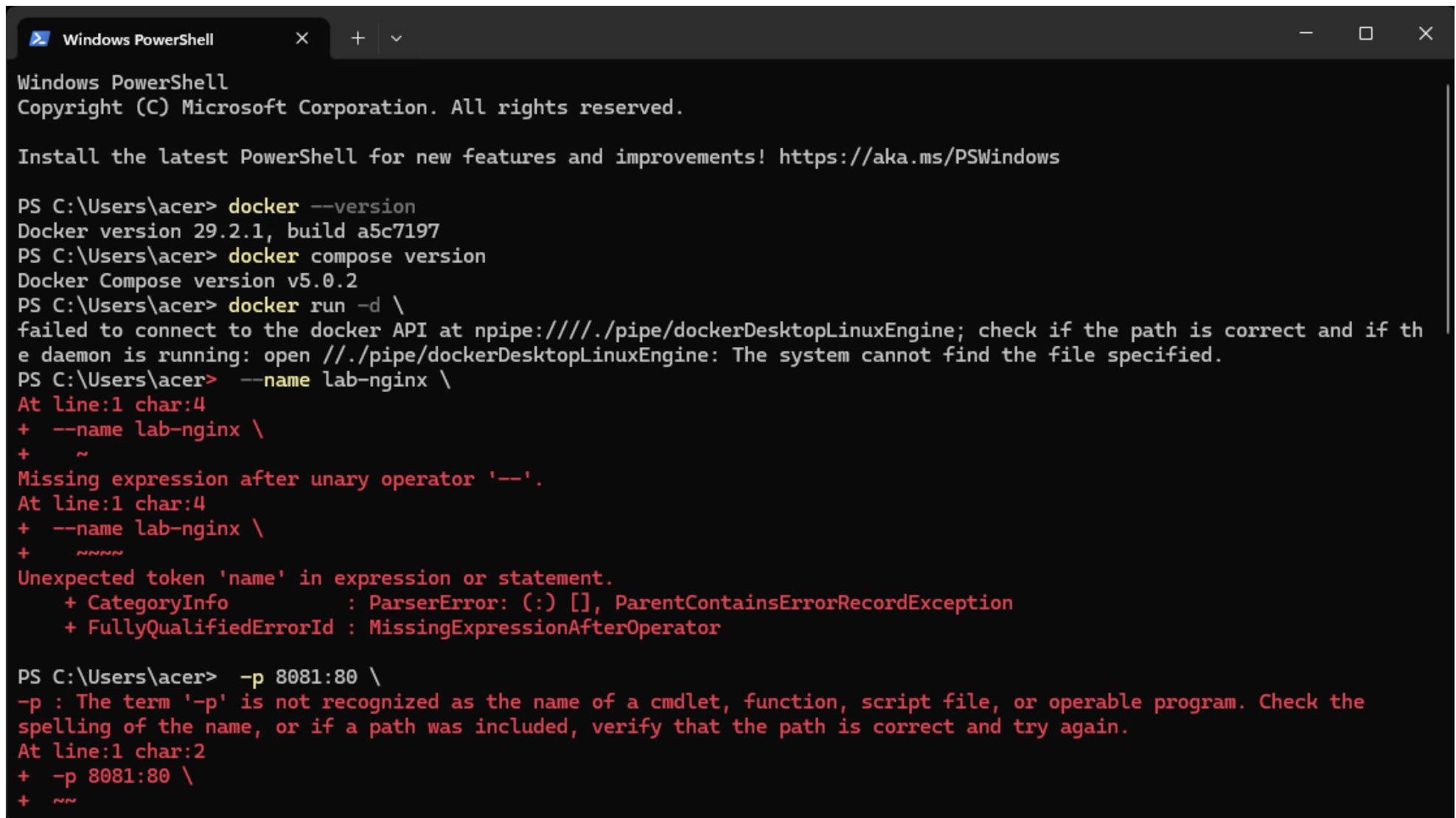
**Step 1 - Verify Docker and Docker Compose installation**

Check that Docker Engine and the Compose plugin are correctly installed and accessible from PowerShell.

```powershell



```
docker --version
docker compose version
```



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\acer> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\acer> docker compose version
Docker Compose version v5.0.2
PS C:\Users\acer> docker run -d \
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\Users\acer> --name lab-nginx \
At line:1 char:4
+ --name lab-nginx \
+ ~
Missing expression after unary operator '--'.
At line:1 char:4
+ --name lab-nginx \
+ ~~~~~
Unexpected token 'name' in expression or statement.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : MissingExpressionAfterOperator

PS C:\Users\acer> -p 8081:80 \
-p : The term '-p' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:2
+ -p 8081:80 \
+ ~~
```

Step2 – First attempt to run a container (Docker daemon not running)

The first `docker run` fails because the Docker Desktop Linux engine is not running. The error message indicates the daemon is unreachable.

```
docker run -d --name lab-nginx -p 8081:80 -v ${PWD}/html:/usr/share/nginx/html nginx:alpine
```



(Error shown in the lower part of the screenshot)

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\acer> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\acer> docker compose version
Docker Compose version v5.0.2
PS C:\Users\acer> docker run -d \
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\Users\acer> --name lab-nginx \
At line:1 char:4
+ --name lab-nginx \
+ ~
Missing expression after unary operator '--'.
At line:1 char:4
+ --name lab-nginx \
+ ~~~~~
Unexpected token 'name' in expression or statement.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : MissingExpressionAfterOperator

PS C:\Users\acer> -p 8081:80 \
-p : The term '-p' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:2
+ -p 8081:80 \
+ ~~
```

Step 3 – Start Docker Desktop and run the Nginx container

After starting Docker Desktop, the same command pulls the `nginx:alpine` image and launches a container named `lab-nginx`.

```
docker run -d --name lab-nginx -p 8081:80 -v ${PWD}/html:/usr/share/nginx/html nginx:alpine
```



```
Windows PowerShell
Run 'docker run --help' for more information
PS C:\Users\acer> --name lab-nginx \
At line:1 char:4
+ --name lab-nginx \
+ ~
Missing expression after unary operator '--'.
At line:1 char:4
+ --name lab-nginx \
+ ~~~~~
Unexpected token 'name' in expression or statement.
+ CategoryInfo          : ParserError: (:) [], ParentContainsErrorRecordException
+ FullyQualifiedErrorId : MissingExpressionAfterOperator

PS C:\Users\acer> docker run -d --name lab-nginx -p 8081:80 -v ${PWD}/html:/usr/share/nginx/html nginx:alpine
Unable to find image 'nginx:alpine' locally
alpine: Pulling from library/nginx
aee4e54b3865: Pull complete
453da7dbc73e: Pull complete
6a0ac1617861: Pull complete
4a8b0b2a5b19: Pull complete
612c0c1df4c5: Pull complete
583599bb7d38: Pull complete
781ff50d2644: Pull complete
82736a35d0e7: Pull complete
e8fc446e336c: Download complete
6192e1e6a438: Download complete
Digest: sha256:5616878291a2eed594aee8db4dade5878cf7edcb475e59193904b198d9b830de
Status: Downloaded newer image for nginx:alpine
d322a8e3591d66f3936e90e8dfe919d579f63a40eb53dd5f103c6a94d0154728
PS C:\Users\acer> |
```

Step 4 – Inspect running containers

List all running containers with `docker ps`. The output shows the `lab-nginx` container, its port mapping, and its name.

docker ps



```

Windows PowerShell
docker: Error response from daemon: Conflict. The container name "/lab-nginx" is already in use by container "d322a8e3591d66f3936e90e8dfe919d579f63a40eb53dd5f103c6a94d0154728". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
PS C:\Users\acer> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
d322a8e3591d   nginx:alpine                        "/docker-entrypoint..." About a minute ago Up About a minute 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
047e52416624   89a0b59e6d72                       "httpd-foreground"      2 minutes ago Up 2 minutes
k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7-3d35-4d41-ac75-ef8ec47d400d_1
8cf8e2297f3e   89a0b59e6d72                       "httpd-foreground"      2 minutes ago Up 2 minutes
k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c-afda-48e8-a630-ef8ec47d400d_1
cd460fe5958b   nginx:latest                       "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
4cc3ad39407b   nginx:latest                       "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
a6d67eed41c0   nginx:latest                       "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
0e5443889059   nginx:latest                       "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
3b8f642df20b   nginx:latest                       "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
ad2fd917697e   nginx:latest                       "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
PS C:\Users\acer> notepad html\index.html
PS C:\Users\acer> notepad html\index.html
PS C:\Users\acer> notepad docker-compose.yml
PS C:\Users\acer> version: '3.8'

```

Step 5 – Edit the default web page

Open the `index.html` file inside the `html` folder with Notepad to customise the Nginx welcome page.

```
notepad html\index.html
```



```
Windows PowerShell
docker: Error response from daemon: Conflict. The container name "/lab-nginx" is already in use by container "d322a8e3591d66f3936e90e8dfe919d579f63a40eb53dd5f103c6a94d0154728". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information
PS C:\Users\acer> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
d322a8e3591d   nginx:alpine                        "/docker-entrypoint..." About a minute ago Up About a minute 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
047e52416624   89a0b59e6d72                        "httpd-foreground"      2 minutes ago Up 2 minutes
8cf8e2297f3e   89a0b59e6d72                        "httpd-foreground"      2 minutes ago Up 2 minutes
cd460fe5958b   nginx:latest                        "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
4cc3ad39407b   nginx:latest                        "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
a6d67eed41c0   nginx:latest                        "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
0e5443889059   nginx:latest                        "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
3b8f642df20b   nginx:latest                        "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
ad2fd917697e   nginx:latest                        "/docker-entrypoint..." 3 minutes ago Up 3 minutes      80/tcp
PS C:\Users\acer> notepad html\index.html
PS C:\Users\acer> notepad html\index.html
PS C:\Users\acer> notepad docker-compose.yml
PS C:\Users\acer> version: '3.8'
```

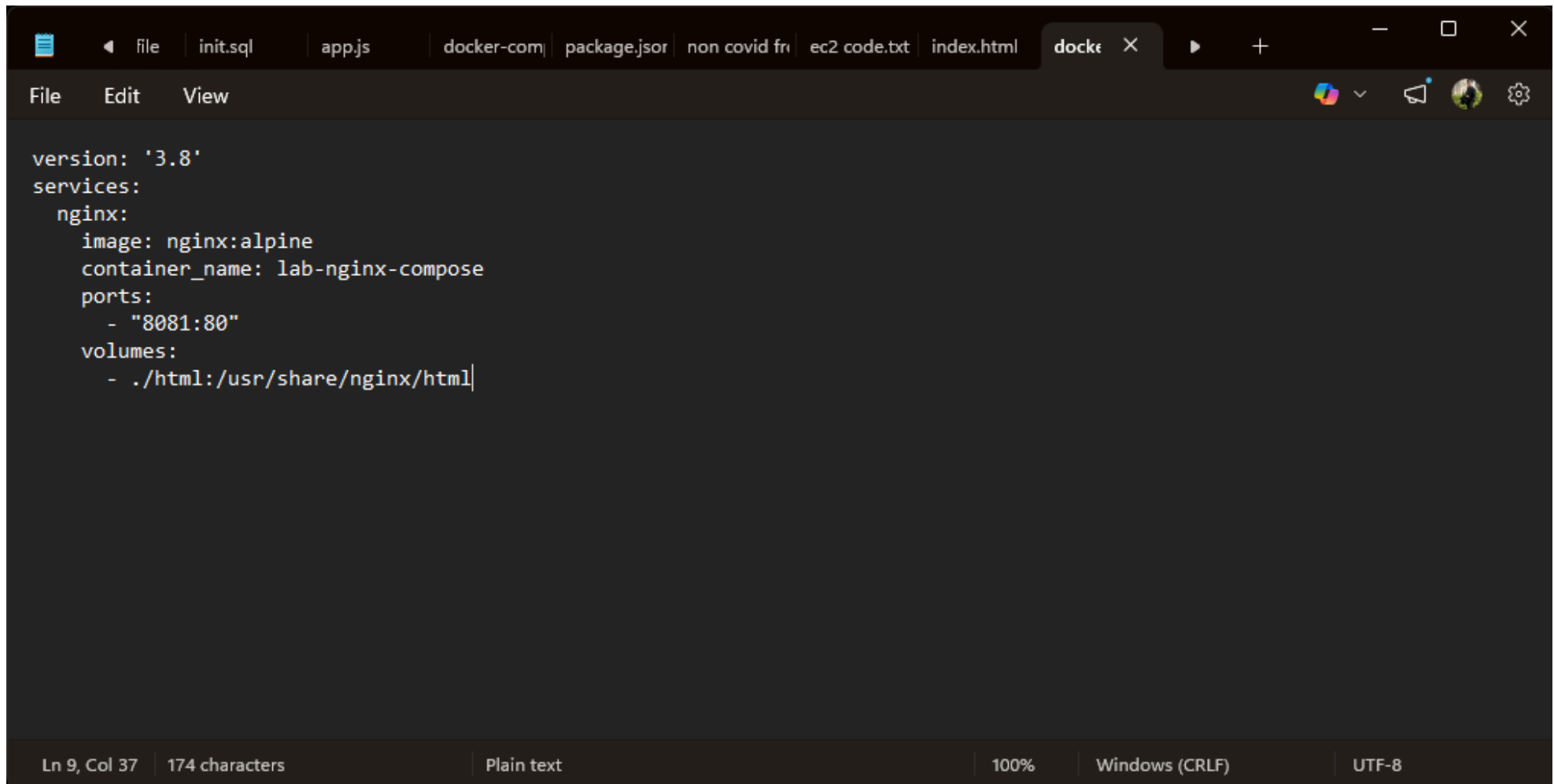
Step 6 – Write a Docker Compose file for Nginx (first attempt)

Trying to type the YAML content directly into the PowerShell terminal causes syntax errors because PowerShell interprets the dashes and colons as invalid commands.

```
version: '3.8'
services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx-compose
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```



PowerShell errors appear in the screenshots: `version: '3.8'` is not recognised, and similar issues for `services:` , `ports:` , and `volumes:` .



The image shows a code editor window with a dark theme. The top bar displays several open files: file, init.sql, app.js, docker-compose.yml, package.json, non covid fr, ec2 code.txt, index.html, and docker-compose.yml (active). The editor content shows a Docker Compose file snippet for nginx. The status bar at the bottom indicates the cursor is at line 9, column 37, with 174 characters, in plain text format, at 100% zoom, using Windows (CRLF) line endings and UTF-8 encoding.

```
version: '3.8'
services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx-compose
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```



```

Windows PowerShell
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer> - "8081:80"
Cannot convert value "8081:80" to type "System.Int32". Error: "Input string was not in a correct format."
At line:1 char:7
+ ~~~~~
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [], RuntimeException
+ FullyQualifiedErrorId : InvalidCastFromStringToInteger

PS C:\Users\acer> volumes:
volumes: : The term 'volumes:' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:5
+ ~~~~~
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (volumes::String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer> notepad docker-compose.yml
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:37:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
✔Network acer_default          Created                                0.1s
✔Container lab-nginx-compose Created                                0.4s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint lab-nginx-compose (198a99b171fa4ccfc8c58193d7c143ee5c886ddae2606ffbb11c0f1cbff7da50e): Bind for 0.0.0.0:8081 failed: port is already allocated
PS C:\Users\acer> |

```

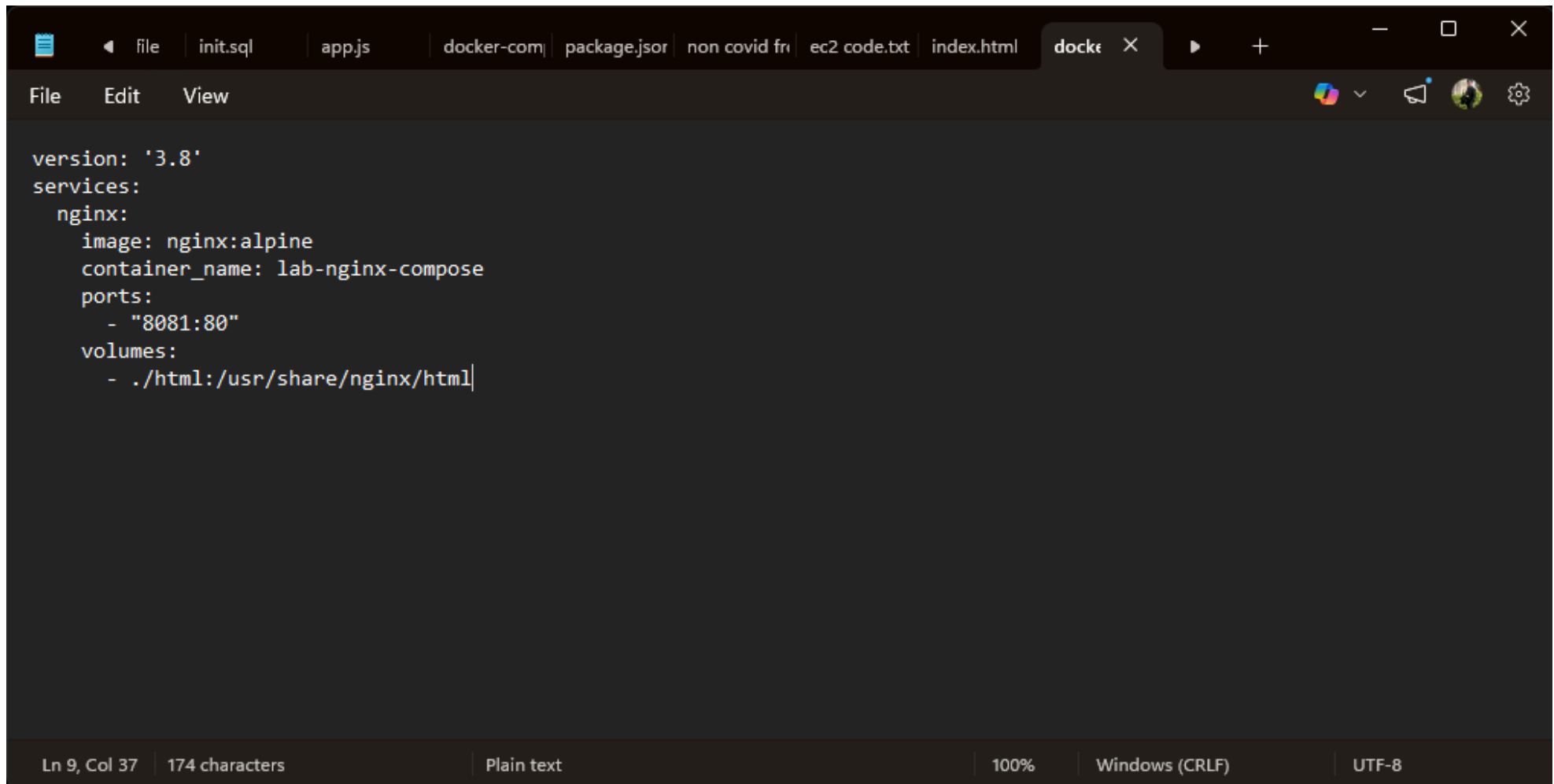
Step 7 – Create the Compose file correctly with a text editor

Use Notepad to write the `docker-compose.yml` file. This avoids shell interpretation problems.

```
notepad docker-compose.yml
```



The correct content (shown in the editor) is saved and then used with `docker compose` .



The screenshot shows a Notepad application window with a dark theme. The title bar at the top displays the file path: "Containerization-and-Devops/lab/Experiment-6/README.md at main · varun124477/Containerization-and-Devops". The menu bar includes "File", "Edit", and "View". The toolbar on the right contains icons for undo, redo, and search. The main text area contains a Docker Compose file for nginx. The status bar at the bottom shows "Ln 9, Col 37", "174 characters", "Plain text", "100%", "Windows (CRLF)", and "UTF-8".

```
version: '3.8'
services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx-compose
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```

(the Notepad screenshot)

```

Windows PowerShell
At line:1 char:7
+ - "8081:80"
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [], RuntimeException
+ FullyQualifiedErrorId : InvalidCastFromStringToInteger

PS C:\Users\acer> volumes:
volumes: : The term 'volumes:' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:5
+ volumes:
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (volumes::String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer> notepad docker-compose.yml
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:37:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
✔Network acer_default          Created                                0.1s
✔Container lab-nginx-compose Created                                0.4s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint lab-nginx-compose (198a99b171fa4ccfc8c58193d7c143ee5c886ddae2606ffb11c0f1cbff7da50e): Bind for 0.0.0.0:8081 failed: port is already allocated
PS C:\Users\acer> docker compose ps
time="2026-04-24T19:38:29+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME      IMAGE      COMMAND      SERVICE      CREATED      STATUS      PORTS
PS C:\Users\acer> |

```

Step 8 – Attempt to start the Nginx service with Compose

When running `docker compose up -d`, the container `lab-nginx-compose` is created but fails because port `8081` is already occupied by the manually started `lab-nginx` container.

docker compose up -d



```

Windows PowerShell
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer> - "8081:80"
Cannot convert value "8081:80" to type "System.Int32". Error: "Input string was not in a correct format."
At line:1 char:7
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (:) [], RuntimeException
+ FullyQualifiedErrorId : InvalidCastFromStringToInteger

PS C:\Users\acer> volumes:
volumes: : The term 'volumes:' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:5
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (volumes::String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer> notepad docker-compose.yml
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:37:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
✓Network acer_default      Created                                0.1s
✓Container lab-nginx-compose Created                                0.4s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint lab-nginx-compose (198a99b171fa4ccfc8c58193d7c143ee5c886ddae2606ffbb11c0f1cbff7da50e): Bind for 0.0.0.0:8081 failed: port is already allocated
PS C:\Users\acer> |

```

Step 9 – Remove the conflicting container and restart the Compose stack

Stop and remove the old `lab-nginx` container, then run Compose again. This time the service starts successfully.

```
docker rm -f lab-nginx
docker compose up -d
```



```
Windows PowerShell
PS C:\Users\acer> notepad docker-compose.yml
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:37:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
  ✓Network acer_default          Created                                0.1s
  ✓Container lab-nginx-compose Created                                0.4s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint lab-nginx-compose (198a99b171fa4ccfc8c58193d7c143ee5c886ddae2606fffb11c0f1cbff7da50e): Bind for 0.0.0.0:8081 failed: port is already allocated
PS C:\Users\acer> docker compose ps
time="2026-04-24T19:38:29+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME          IMAGE          COMMAND          SERVICE  CREATED   STATUS    PORTS
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:39:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
PS C:\Users\acer> docker compose ps
time="2026-04-24T19:40:04+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME          IMAGE          COMMAND          SERVICE  CREATED        STATUS       PORTS
lab-nginx-compose  nginx:alpine  "/docker-entrypoint..."  nginx    2 minutes ago  Up 6 seconds  80/tcp
PS C:\Users\acer> docker rm -f lab-nginx
lab-nginx
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:40:46+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 1/1
  ✓Container lab-nginx-compose Recreated                                0.9s
PS C:\Users\acer>
```

Step 10 – Verify the Compose services

Check that the Compose-managed Nginx container is running.

```
docker compose ps
```



```

Windows PowerShell
PS C:\Users\acer> notepad docker-compose.yml
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:37:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
  ✓Network acer_default          Created                                0.1s
  ✓Container lab-nginx-compose Created                                0.4s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint lab-nginx-compose (198a99b171fa4ccfc8c58193d7c143ee5c886ddae2606fffb11c0f1cbff7da50e): Bind for 0.0.0.0:8081 failed: port is already allocated
PS C:\Users\acer> docker compose ps
time="2026-04-24T19:38:29+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME          IMAGE          COMMAND          SERVICE   CREATED   STATUS    PORTS
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:39:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
PS C:\Users\acer> docker compose ps
time="2026-04-24T19:40:04+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME          IMAGE          COMMAND          SERVICE   CREATED        STATUS       PORTS
lab-nginx-compose  nginx:alpine  "/docker-entrypoint..."  nginx     2 minutes ago  Up 6 seconds  80/tcp
PS C:\Users\acer> docker rm -f lab-nginx
lab-nginx
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:40:46+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 1/1
  ✓Container lab-nginx-compose Recreated                                0.9s
PS C:\Users\acer>

```

(later part)

Step 11 – Prepare the WordPress project

Create a new directory `wp-compose`, navigate into it, and create a fresh `docker-compose.yml` for WordPress.



```
mkdir wp-compose
cd wp-compose
notepad docker-compose.yml
```

```
Windows PowerShell
NAME      IMAGE      COMMAND      SERVICE      CREATED      STATUS      PORTS
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:39:58+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
PS C:\Users\acer> docker compose ps
time="2026-04-24T19:40:04+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME              IMAGE          COMMAND              SERVICE      CREATED        STATUS          PORTS
lab-nginx-compose nginx:alpine    "/docker-entrypoint..." nginx        2 minutes ago  Up 6 seconds    80/tcp
PS C:\Users\acer> docker rm -f lab-nginx
lab-nginx
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:40:46+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 1/1
✔Container lab-nginx-compose Recreated 0.9s
PS C:\Users\acer> mkdir wp-compose

Directory: C:\Users\acer

Mode                LastWriteTime         Length Name
----                -
d-----          24-04-2026    19:42                wp-compose

PS C:\Users\acer> cd wp-compose
PS C:\Users\acer\wp-compose> notepad docker-compose.yml
PS C:\Users\acer\wp-compose> |
```

Step12 – Define the WordPress and MySQL services

The Compose file defines a MySQL 5.7 database and a latest WordPress image. It sets environment variables for credentials and creates named volumes for data persistence.

```
services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

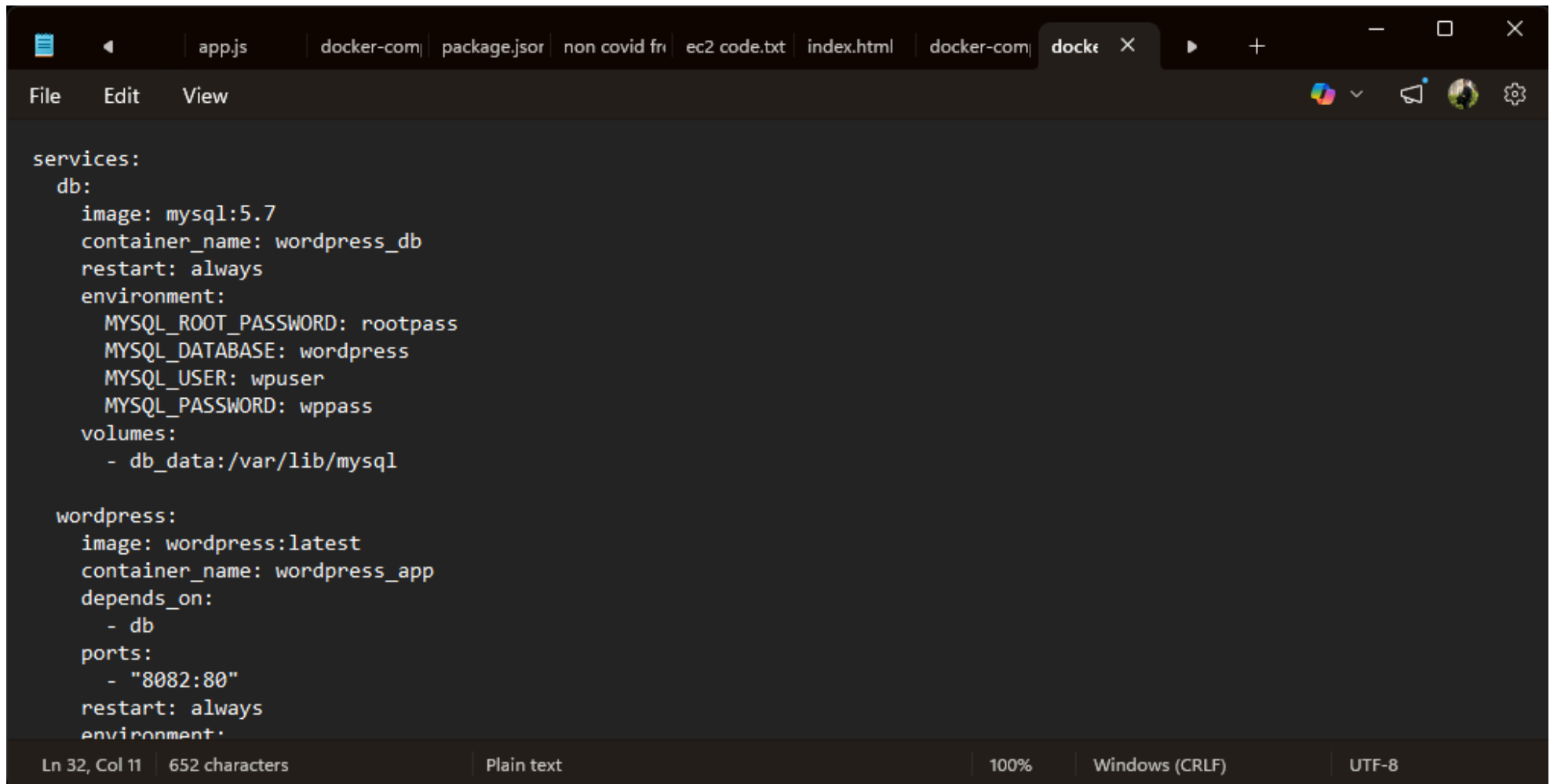
  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8082:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
```



db_data:

wp_data:



The screenshot shows a code editor window with a dark theme. The editor has a menu bar with 'File', 'Edit', and 'View'. Below the menu bar is a toolbar with icons for file operations and a search icon. The main area displays a Docker Compose file with the following content:

```
services:
  db:
    image: mysql:5.7
    container_name: wordpress_db
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress_app
    depends_on:
      - db
    ports:
      - "8082:80"
    restart: always
    environment:
```

The status bar at the bottom of the editor shows 'Ln 32, Col 11', '652 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'.

Step13 – Pull images and launch the WordPress stack

`docker compose up -d` downloads the required images (MySQL 5.7, WordPress latest) and starts both containers.



```
docker compose up -d
```

```

Windows PowerShell
time="2026-04-24T19:40:04+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
lab-nginx-compose   nginx:alpine        "/docker-entrypoint..." nginx      2 minutes ago  Up 6 seconds  80/tcp
PS C:\Users\acer> docker rm -f lab-nginx
lab-nginx
PS C:\Users\acer> docker compose up -d
time="2026-04-24T19:40:46+05:30" level=warning msg="C:\\Users\\acer\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 1/1
✔Container lab-nginx-compose Recreated 0.9s
PS C:\Users\acer> mkdir wp-compose

Directory: C:\Users\acer

Mode                LastWriteTime         Length Name
----                -
d-----         24-04-2026      19:42             wp-compose

PS C:\Users\acer> cd wp-compose
PS C:\Users\acer\wp-compose> notepad docker-compose.yml
PS C:\Users\acer\wp-compose> notepad docker-compose.yml
PS C:\Users\acer\wp-compose> docker compose up -d
[+] up 36/40
- Image mysql:5.7 [#####] 148.3MB / 149.4MB Pulling 66.0s
- Image wordpress:latest [#####] 239.7MB / 245.2MB Pulling 66.0s

```

```

Windows PowerShell
PS C:\Users\acer> mkdir wp-compose

Directory: C:\Users\acer

Mode                LastWriteTime         Length Name
----                -
d-----          24-04-2026    19:42             wp-compose

PS C:\Users\acer> cd wp-compose
PS C:\Users\acer\wp-compose> notepad docker-compose.yml
PS C:\Users\acer\wp-compose> notepad docker-compose.yml
PS C:\Users\acer\wp-compose> docker compose up -d
[+] up 45/45
  ✓Image mysql:5.7                Pulled              71.8s
  ✓Image wordpress:latest         Pulled              77.2s
  ✓Network wp-compose_default     Created             0.1s
  ✓Volume wp-compose_db_data      Created             0.0s
  ✓Volume wp-compose_wp_data      Created             0.0s
  ✓Container wordpress_db         Created             0.7s
  ✓Container wordpress_app        Created             0.5s
PS C:\Users\acer\wp-compose> docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
wordpress_app       wordpress:latest     "docker-entrypoint.s..." wordpress   44 seconds ago Up 42 seconds 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
wordpress_db        mysql:5.7            "docker-entrypoint.s..." db         45 seconds ago Up 42 seconds 3306/tcp, 33060
PS C:\Users\acer\wp-compose> |

```

Step 14 – Confirm the WordPress stack is running

Check the status of the Compose project. Both `wordpress_db` and `wordpress_app` are listed with their respective ports.

```
docker compose ps
```



```

Windows PowerShell
PS C:\Users\acer> mkdir wp-compose

Directory: C:\Users\acer

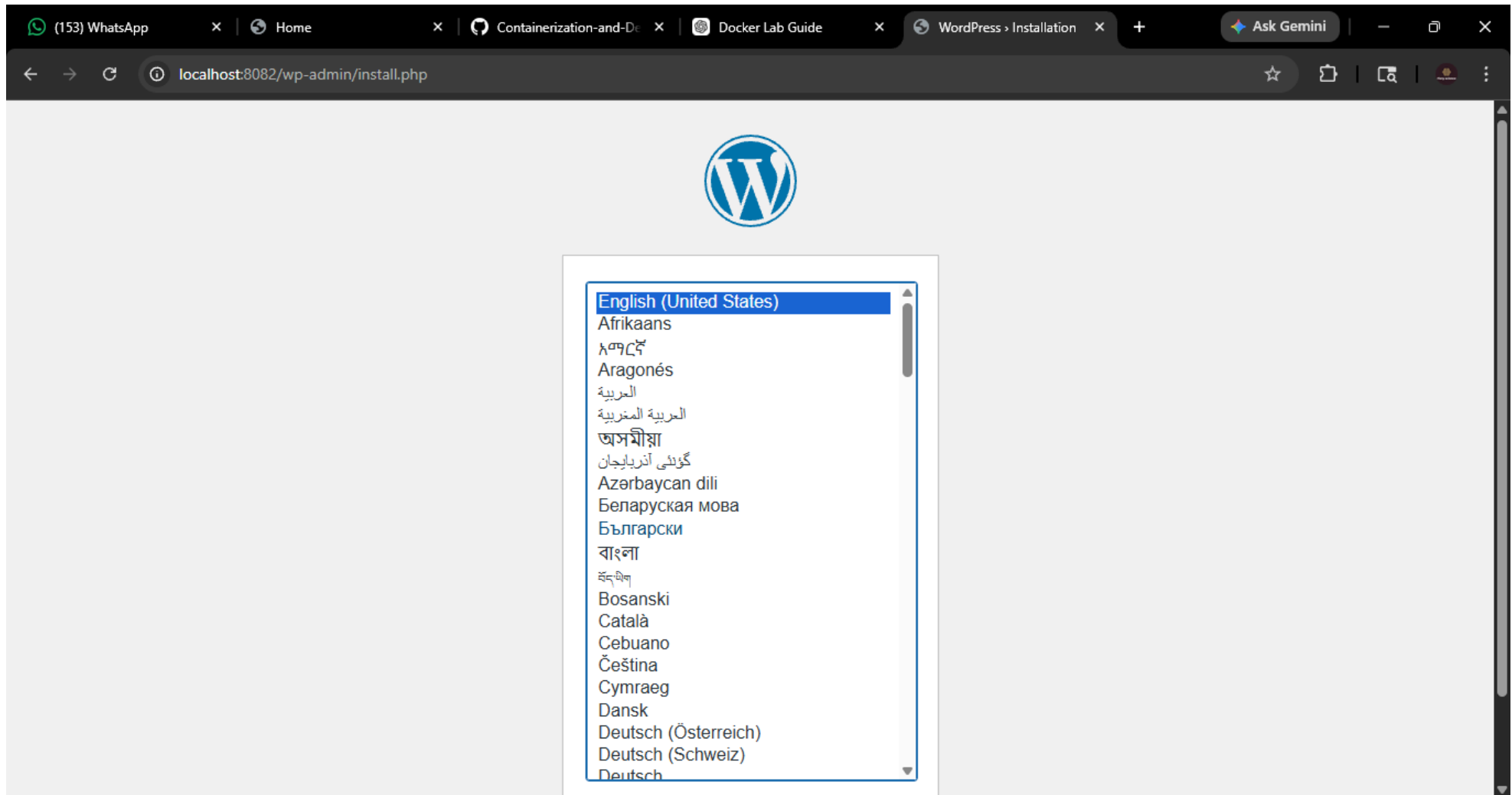
Mode                LastWriteTime         Length Name
----                -
d-----          24-04-2026    19:42             wp-compose

PS C:\Users\acer> cd wp-compose
PS C:\Users\acer\wp-compose> notepad docker-compose.yml
PS C:\Users\acer\wp-compose> notepad docker-compose.yml
PS C:\Users\acer\wp-compose> docker compose up -d
[+] up 45/45
✓Image mysql:5.7           Pulled          71.8s
✓Image wordpress:latest    Pulled          77.2s
✓Network wp-compose_default Created          0.1s
✓Volume wp-compose_db_data Created           0.0s
✓Volume wp-compose_wp_data Created           0.0s
✓Container wordpress_db    Created          0.7s
✓Container wordpress_app    Created          0.5s
PS C:\Users\acer\wp-compose> docker compose ps
NAME                IMAGE                COMMAND                SERVICE    CREATED        STATUS        PORTS
wordpress_app        wordpress:latest      "docker-entrypoint.s..." wordpress    44 seconds ago Up 42 seconds 0.0.0.0:8082->80/tcp
wordpress_db          mysql:5.7             "docker-entrypoint.s..." db          45 seconds ago Up 42 seconds 3306/tcp, 33060/tcp
PS C:\Users\acer\wp-compose> |

```

Step15 – Access the WordPress installation wizard

Open a browser and navigate to <http://localhost:8082> . You are greeted with the WordPress language selection page, indicating the site is ready for configuration.



Additional Notes

- **Port conflicts:** Always ensure a port is not already bound before starting a new service. Use `docker ps` and `docker rm -f <container>` to resolve conflicts.

- **PowerShell vs YAML:** YAML content must be written to a file using a text editor. Pasting it directly into PowerShell will cause parsing errors.
- **Compose version warning:** The `version` attribute in `docker-compose.yml` is obsolete (as shown by the Compose messages). It can be safely removed from the file.
- **Data persistence:** Named volumes (`db_data` , `wp_data`) keep database and website files safe even if containers are removed.





varun124477 / Containerization-and-Devops

Type / to search



<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Containerization-and-Devops / lab / Experiment-7 / README.md



varun124477 Fix Experiment-7: convert submodule to normal folder ✓

e7e661d · 8 minutes ago



History

Preview

Code

Blame

Raw



Docker Lab - Experiment 7: CI/CD using Jenkins, GitHub and Docker Hub



Aim

To design and implement a complete CI/CD pipeline using Jenkins, integrating source code from GitHub, and building & pushing

<hr>

<h4 align="center"> Pre-requisite </h4>

- Docker & Docker Compose installed
- GitHub account (for source code)
- Docker Hub account (for image registry)
- Basic Git and Linux command knowledge

> **Project Directory Structure**

> All source files, the Jenkins Compose file, and the experiment images are kept inside one folder.

>

> ![Local project files](./Images/image.png)

<hr>

Step-by-Step Procedure

****Step 1 - Write the Jenkins Pipeline Definition (`Jenkinsfile`)****

The `Jenkinsfile` contains the entire CI/CD pipeline written in Groovy.

It defines four stages: Clone source, Build Docker image, Login to Docker Hub, and Push.

```
````groovy
pipeline {
 agent any
 environment {
 IMAGE_NAME = "varun99566/myapp"
 }
 stages {
 stage('Clone Source') {
 steps {
 git 'https://github.com/varun124477/my-app.git'
 }
 }
 stage('Build Docker Image') {
 steps {
 sh 'docker build -t $IMAGE_NAME:latest .'
 }
 }
 stage('Login to Docker Hub') {
 steps {
 withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
 sh 'echo $DOCKER_TOKEN | docker login -u varun99566 --password-stdin'
 }
 }
 }
 stage('Push to Docker Hub') {
 steps {
 sh 'docker push $IMAGE_NAME:latest'
 }
 }
 }
}
```

```
}
}
}
```

```
1 pipeline {
2
3 stages {
4
5 stage('Clone Source') {
6 steps {
7 git 'https://github.com/varun124477/my-app.git'
8 }
9 }
10
11 stage('Build Docker Image') {
12 steps {
13 sh 'docker build -t $IMAGE_NAME:latest .'
14 }
15 }
16
17 stage('Login to Docker Hub') {
18 steps {
19 withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
20 sh '''
21 echo $DOCKER_TOKEN | docker login -u varun124477 --password-stdin
22 '''
23 }
24 }
25 }
26
27 stage('Push to Docker Hub') {
28 steps {
29 sh 'docker push $IMAGE_NAME:latest'
30 }
31 }
32 }
33 }
34
35
36
37
38
```

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab\lab\Experiment-7> git push -u origin main --force  
branch 'main' set up to track 'origin/main'.  
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab\lab\Experiment-7>

## Step 2 – Write the Flask Application ( app.py )

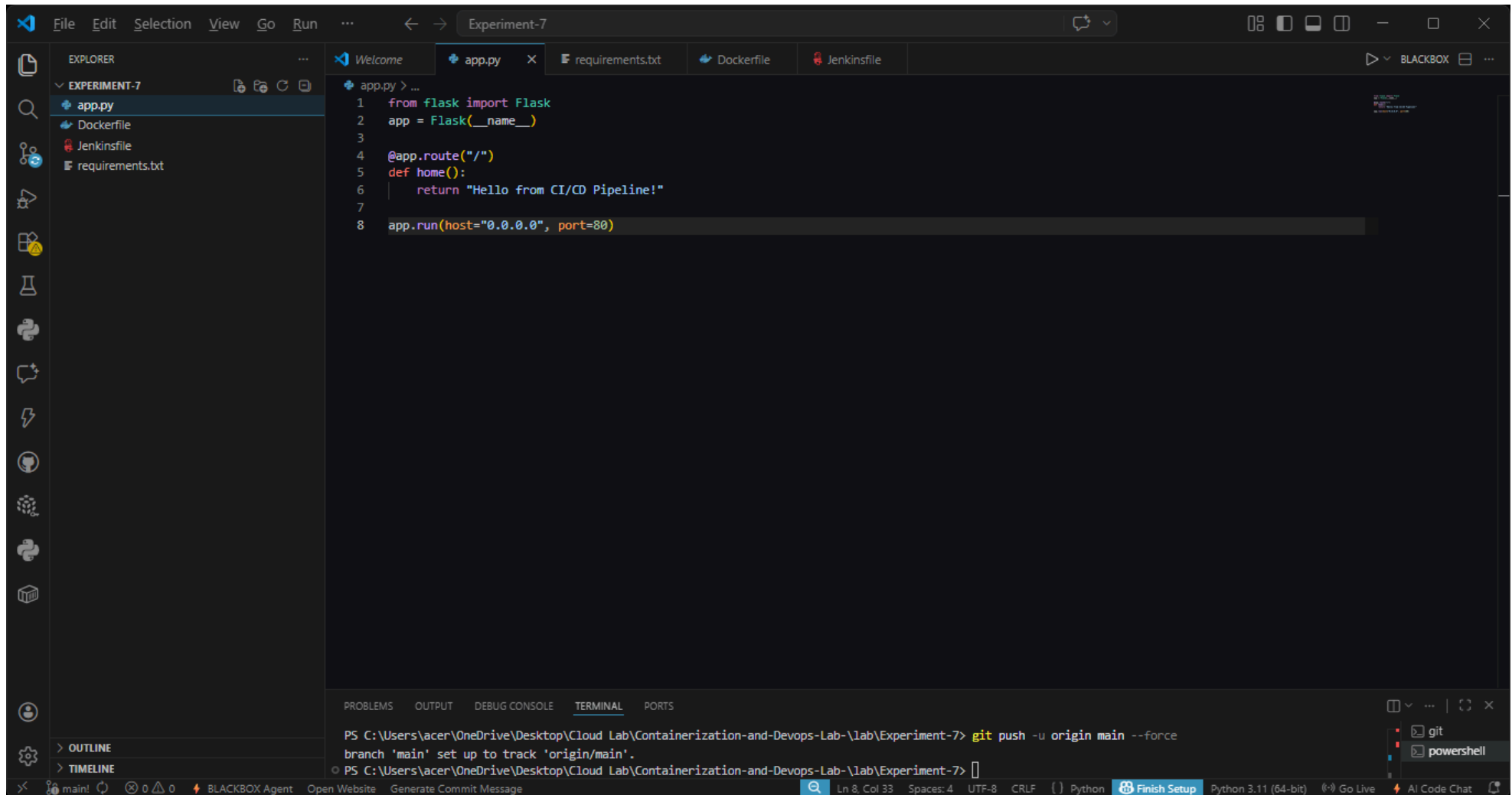
A simple Python web application that returns a greeting message.

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
 return "Hello from CI/CD Pipeline!"

app.run(host="0.0.0.0", port=80)
```





The screenshot shows the Visual Studio Code interface. The Explorer panel on the left shows a project named 'EXPERIMENT-7' with files 'app.py', 'Dockerfile', 'Jenkinsfile', and 'requirements.txt'. The main editor window displays the content of 'app.py':

```
1 from flask import Flask
2 app = Flask(__name__)
3
4 @app.route("/")
5 def home():
6 return "Hello from CI/CD Pipeline!"
7
8 app.run(host="0.0.0.0", port=80)
```

The TERMINAL panel at the bottom shows the following commands and output:

```
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> git push -u origin main --force
branch 'main' set up to track 'origin/main'.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>
```

The status bar at the bottom indicates the current file is 'app.py' at line 8, column 33, with 4 spaces, UTF-8 encoding, CRLF line endings, and Python 3.11 (64-bit) interpreter.

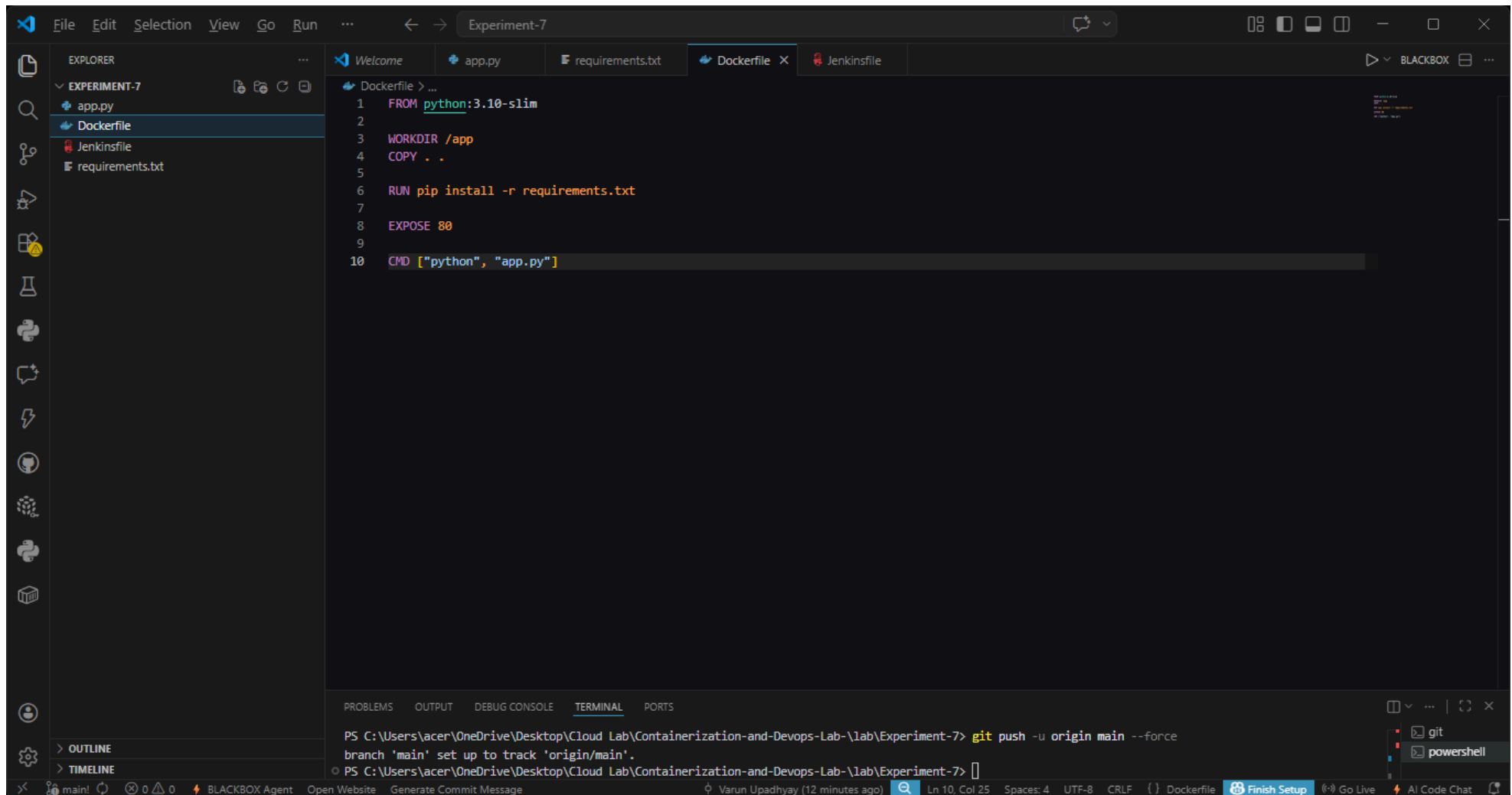
### Step 3 – Create the Dockerfile

The Dockerfile packages the Flask app into a lightweight Python image.

```
FROM python:3.10-slim
WORKDIR /app
COPY . .
```



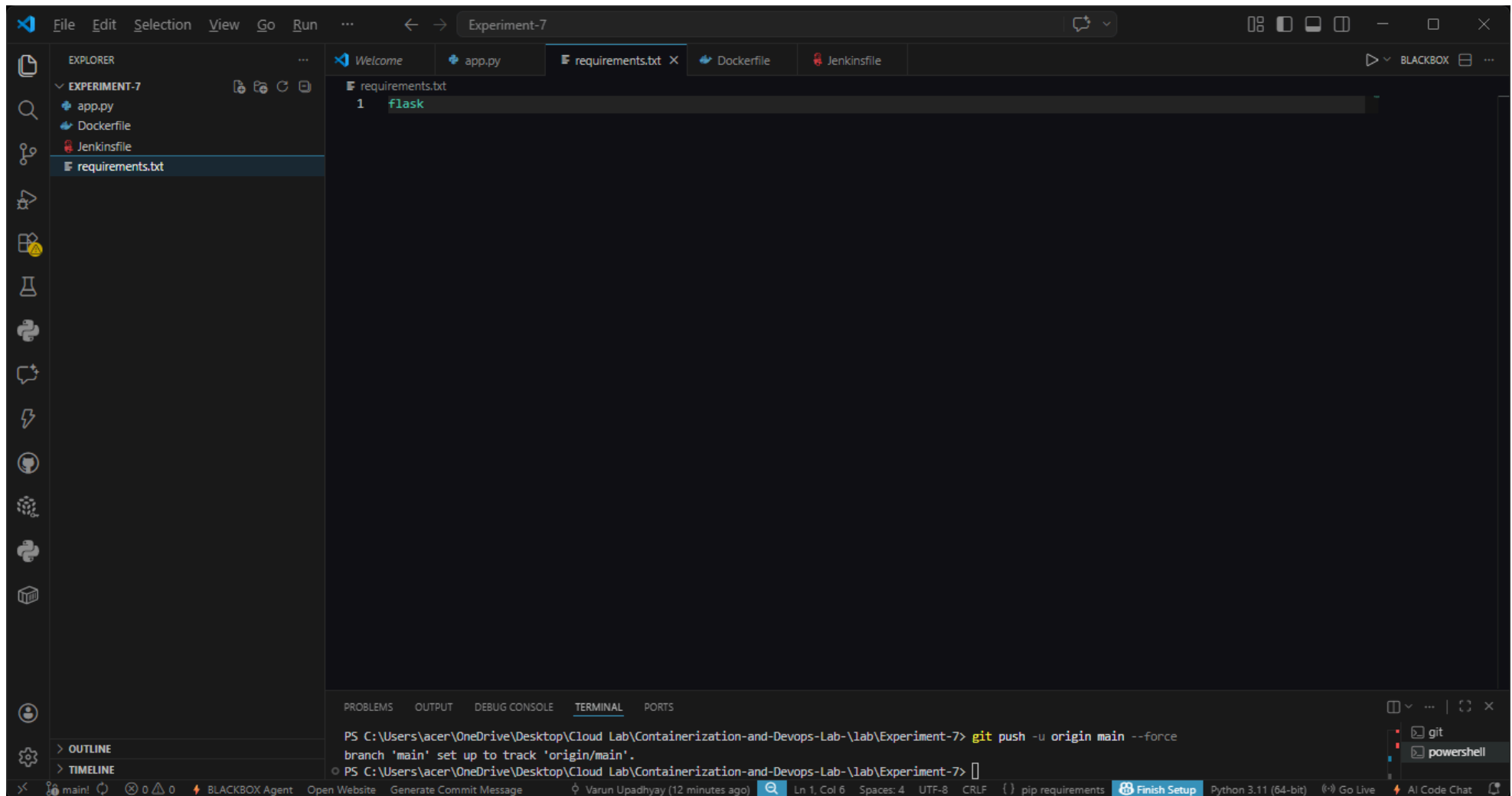
```
RUN pip install -r requirements.txt
EXPOSE 80
CMD ["python", "app.py"]
```



## Step 4 – First Push Attempt to GitHub

After creating all files, the project is pushed to the remote repository.

```
git push -u origin main --force
```



## Step 5 – Successful Push to GitHub

The force push uploads all source code and sets the upstream branch.



```
git push -u origin main --force
```



The screenshot shows the Visual Studio Code (VS Code) interface. The Explorer sidebar on the left shows a project named 'EXPERIMENT-7' with files 'app.py', 'Dockerfile', 'Jenkinsfile', and 'requirements.txt'. The 'requirements.txt' file is open in the editor, showing the text '1 flask'. The Terminal window at the bottom displays the output of two 'git push' commands. The first command was 'git push --set-upstream origin main', which resulted in a hint to see the 'Note about fast-forwards' in 'git push --help'. The second command was 'git push -u origin main --force', which successfully pushed the changes to the 'main' branch of the repository 'https://github.com/varun124477/my-app.git'. The terminal output includes details about enumerating objects, counting objects, delta compression, and writing objects. The status bar at the bottom shows the current file is 'requirements.txt' and the Python interpreter is 'Python 3.11 (64-bit)'.

```
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> git push --set-upstream origin main
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
• PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> git push -u origin main --force
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 929 bytes | 185.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/varun124477/my-app.git
+ 5d73f41...a491d56 main -> main (forced update)
branch 'main' set up to track 'origin/main'.
• PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>
```

## Step 6 – Verify the GitHub Repository

The repository now contains `app.py` , `Dockerfile` , `Jenkinsfile` , and `requirements.txt` .

The screenshot shows a GitHub repository named 'my-app' by user 'varun124477'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows an initial commit by 'varun124477' 13 minutes ago, with files: Dockerfile, Jenkinsfile, app.py, and requirements.txt. The README section is currently empty, with a prompt to 'Add a README'. The right sidebar shows the repository's activity, including 0 stars, 0 watching, and 0 forks. The 'About' section states 'No description, website, or topics provided.' The 'Releases' section shows 'No releases published' with a link to 'Create a new release'. The 'Packages' section shows 'No packages published' with a link to 'Publish your first package'.

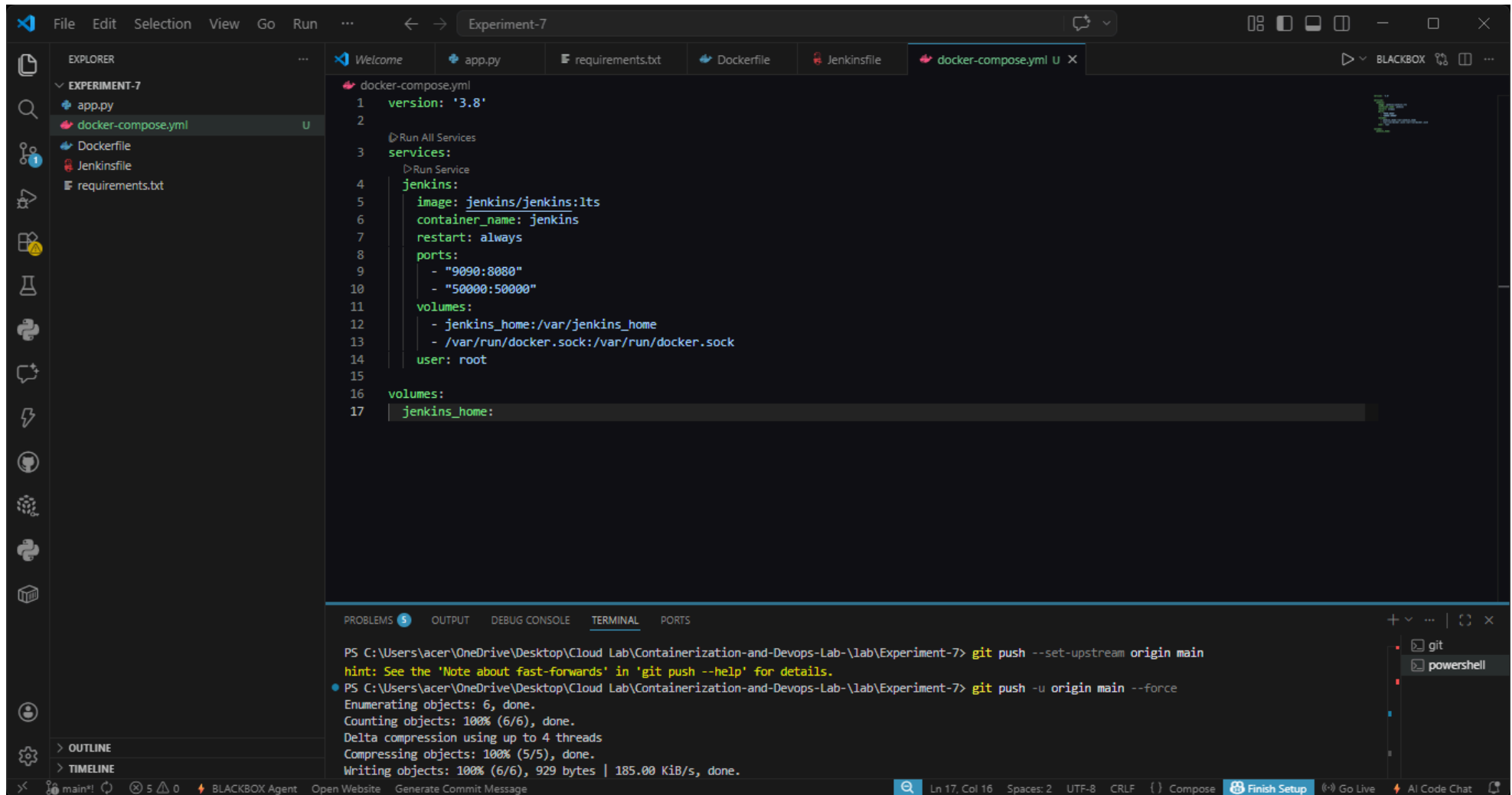
## Step 7 – Write Docker Compose for Jenkins

Create `docker-compose.yml` to run Jenkins with persistent volume and Docker socket access.

```
version: '3.8'
services:
 jenkins:
```



```
image: jenkins/jenkins:lts
container_name: jenkins
restart: always
ports:
 - "9000:8080"
 - "50000:50000"
volumes:
 - jenkins_home:/var/jenkins_home
 - /var/run/docker.sock:/var/run/docker.sock
user: root
volumes:
 jenkins_home:
```



The screenshot shows the Visual Studio Code interface. The Explorer pane on the left shows the file structure for 'EXPERIMENT-7', including 'app.py', 'docker-compose.yml', 'Dockerfile', 'Jenkinsfile', and 'requirements.txt'. The 'docker-compose.yml' file is open in the editor, showing the following content:

```
1 version: '3.8'
2
3 services:
4 jenkins:
5 image: jenkins/jenkins:its
6 container_name: jenkins
7 restart: always
8 ports:
9 - "9090:8080"
10 - "50000:50000"
11 volumes:
12 - jenkins_home:/var/jenkins_home
13 - /var/run/docker.sock:/var/run/docker.sock
14 user: root
15
16 volumes:
17 jenkins_home:
```

The terminal window at the bottom shows the output of a git push command:

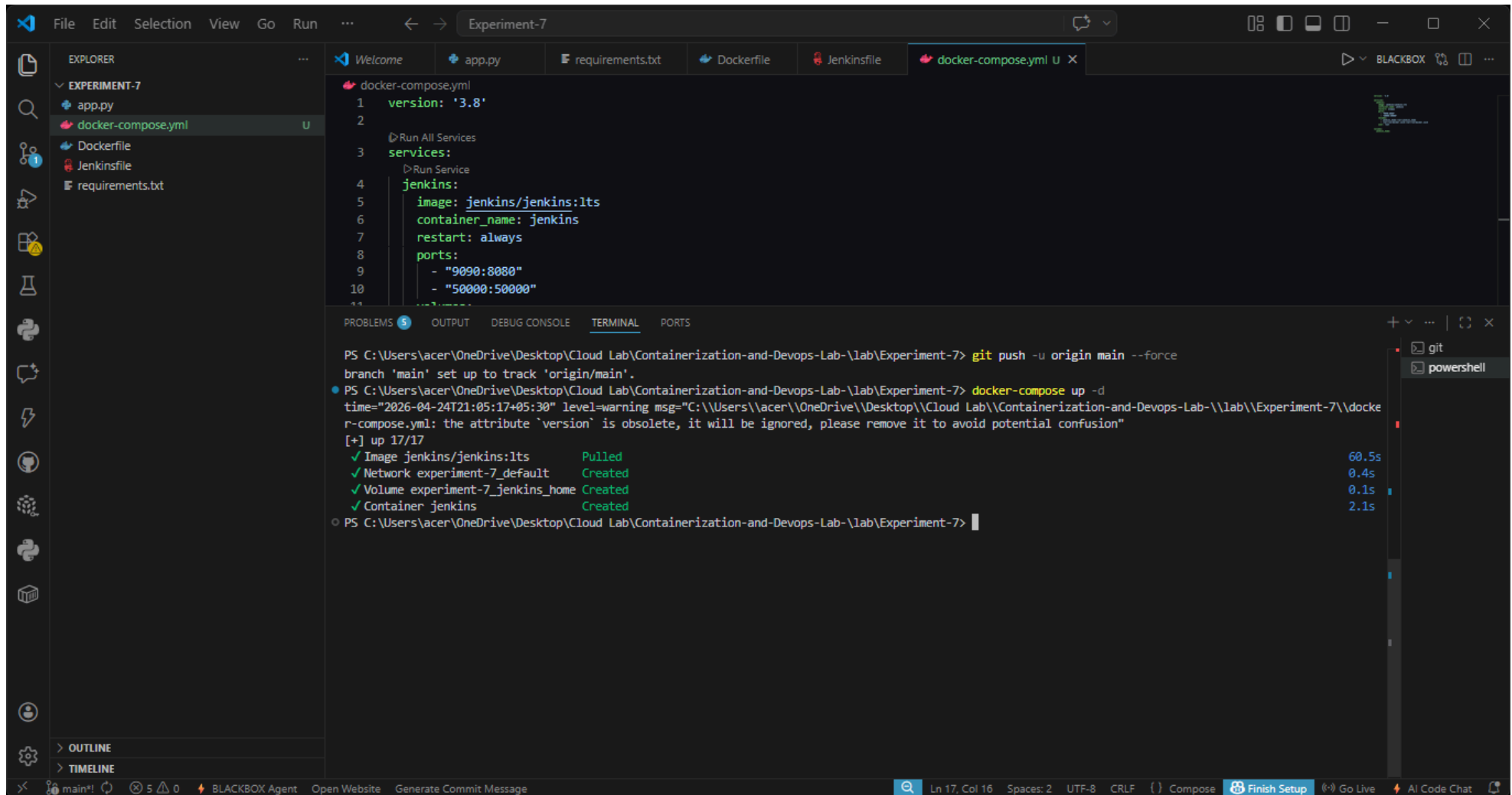
```
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> git push --set-upstream origin main
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
• PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> git push -u origin main --force
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 929 bytes | 185.00 KiB/s, done.
```

## Step 8 – Start Jenkins Container

Launch the Jenkins service using Docker Compose.

```
docker compose up -d
```





The screenshot shows a Visual Studio Code editor with the following components:

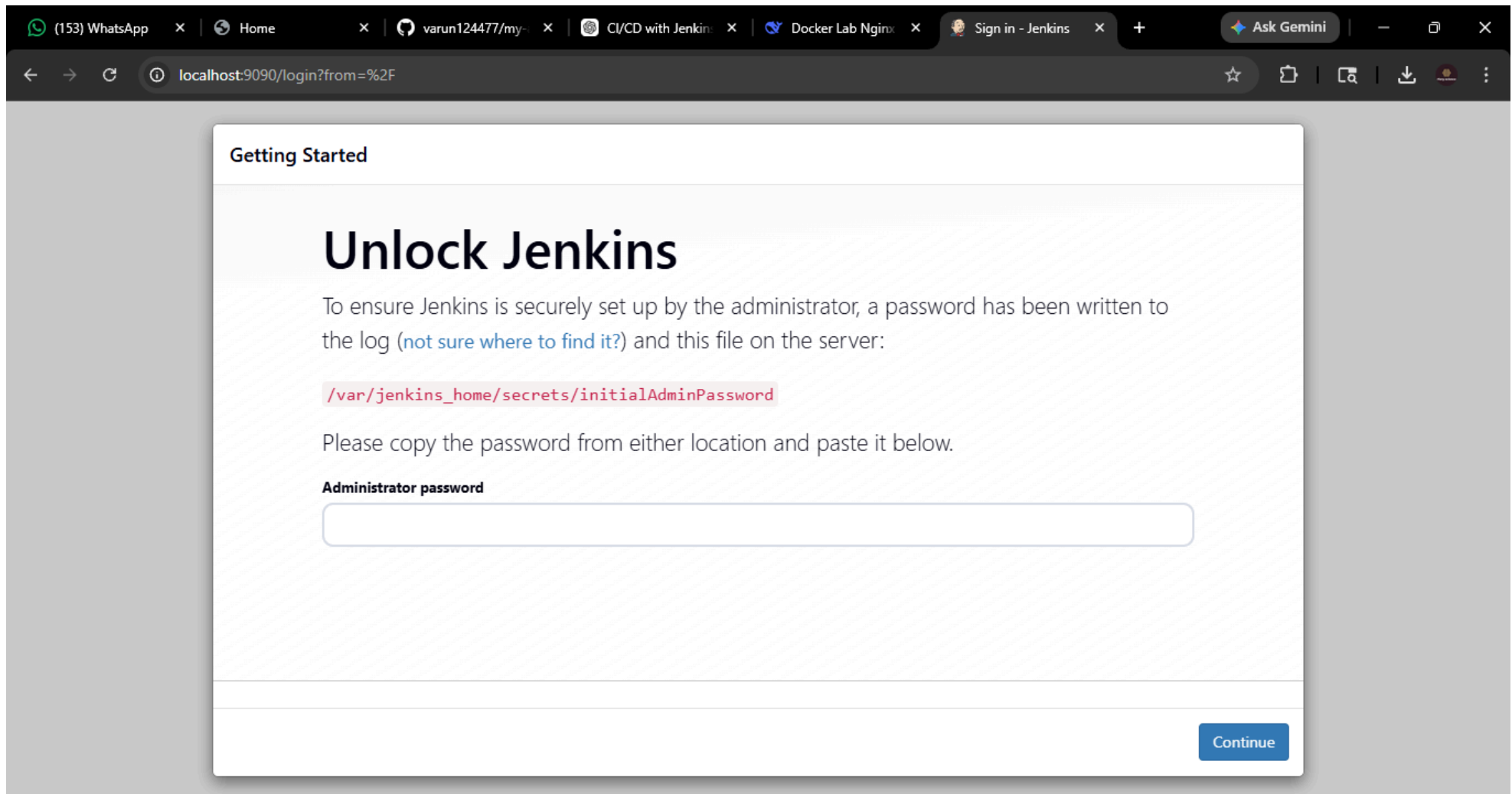
- EXPLORER:** Shows the file structure for 'EXPERIMENT-7' with files: app.py, docker-compose.yml (selected), Dockerfile, Jenkinsfile, and requirements.txt.
- docker-compose.yml:** Contains the following configuration:

```
1 version: '3.8'
2
3 services:
4 jenkins:
5 image: jenkins/jenkins:its
6 container_name: jenkins
7 restart: always
8 ports:
9 - "9090:8080"
10 - "50000:50000"
```
- TERMINAL:** Shows the execution of the following commands and their output:

```
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> git push -u origin main --force
branch 'main' set up to track 'origin/main'.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> docker-compose up -d
time="2026-04-24T21:05:17+05:30" level=warning msg="C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 17/17
✓ Image jenkins/jenkins:its Pulled 60.5s
✓ Network experiment-7_default Created 0.4s
✓ Volume experiment-7_jenkins_home Created 0.1s
✓ Container jenkins Created 2.1s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>
```
- STATUS BAR:** Shows 'main' branch, 5 files, 0 errors, and various tool icons (BLACKBOX Agent, Open Website, Generate Commit Message, etc.).

## Step 9 – Jenkins Unlock Page

Navigate to <http://localhost:9000> (port 9000 as mapped). The unlock page asks for the initial admin password.



## Step 10 – Retrieve Initial Admin Password

Fetch the password from inside the running Jenkins container.

```
docker exec -it jenkins cat /var/jenkins_home/secrets/initialAdminPassword
```



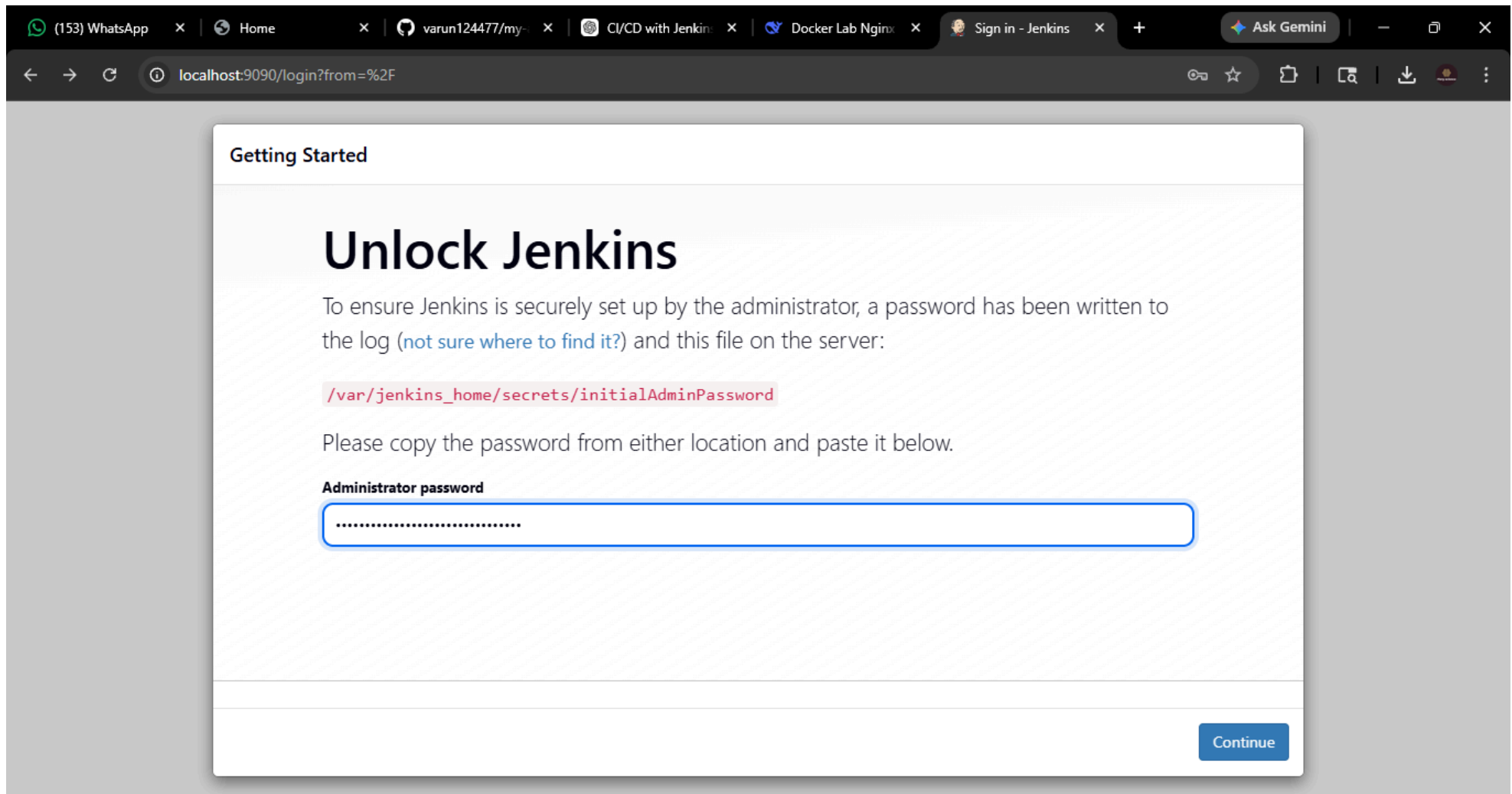
```
EXPERIMENT-7
├── app.py
├── docker-compose.yml
├── Dockerfile
├── Jenkinsfile
└── requirements.txt
```

```
docker-compose.yml
1 version: '3.8'
2
3 services:
4 jenkins:
5 image: jenkins/jenkins:its
6 container_name: jenkins
7 restart: always
8 ports:
9 - "9090:8080"
10 - "50000:50000"
```

```
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> git push -u origin main --force
branch 'main' set up to track 'origin/main'.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> docker-compose up -d
time="2026-04-24T21:05:17+05:30" level=warning msg="C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 17/17
✓ Image jenkins/jenkins:its Pulled 60.5s
✓ Network experiment-7_default Created 0.4s
✓ Volume experiment-7_jenkins_home Created 0.1s
✓ Container jenkins Created 2.1s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> docker exec -it jenkins cat /var/jenkins_home/secrets/initialAdminPassword
dminPassword
720929cf5eee4bb3889f22c2a17691eb
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>
```

## Step 11 – Unlock Jenkins

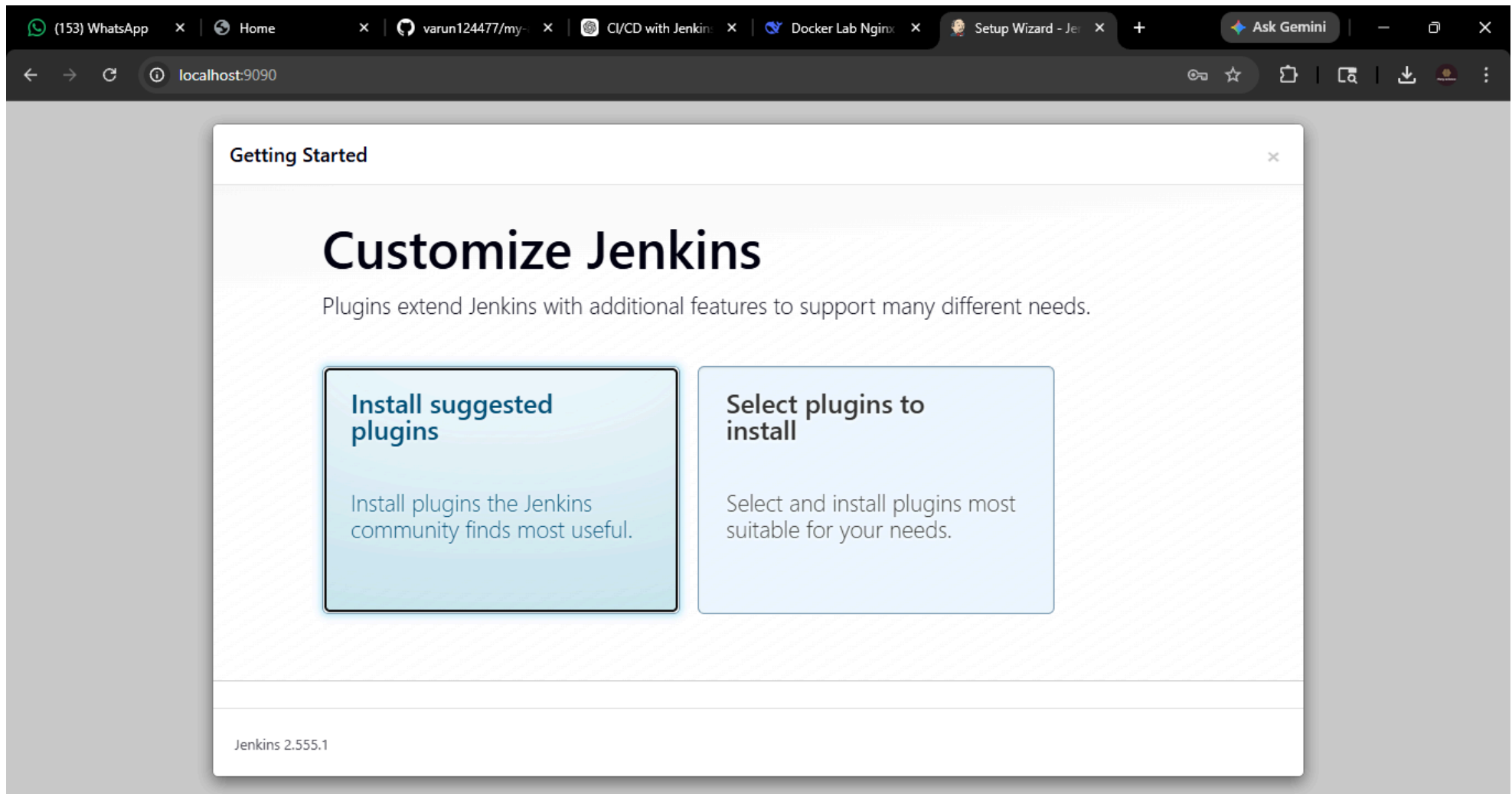
Paste the retrieved password and click **Continue**.



## Step 12 – Install Suggested Plugins

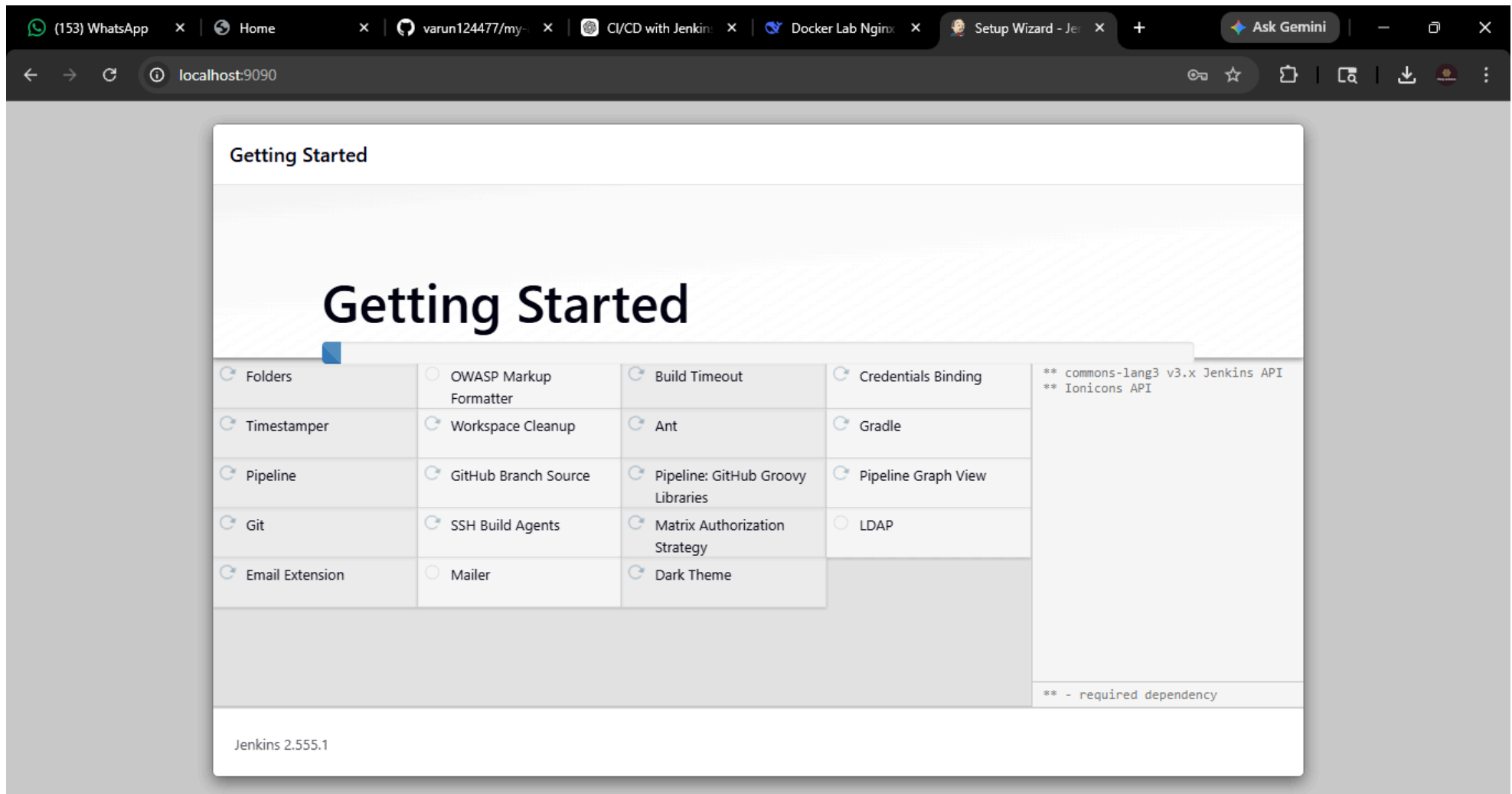
Select **Install suggested plugins** to install Git, Pipeline, and other essential plugins.





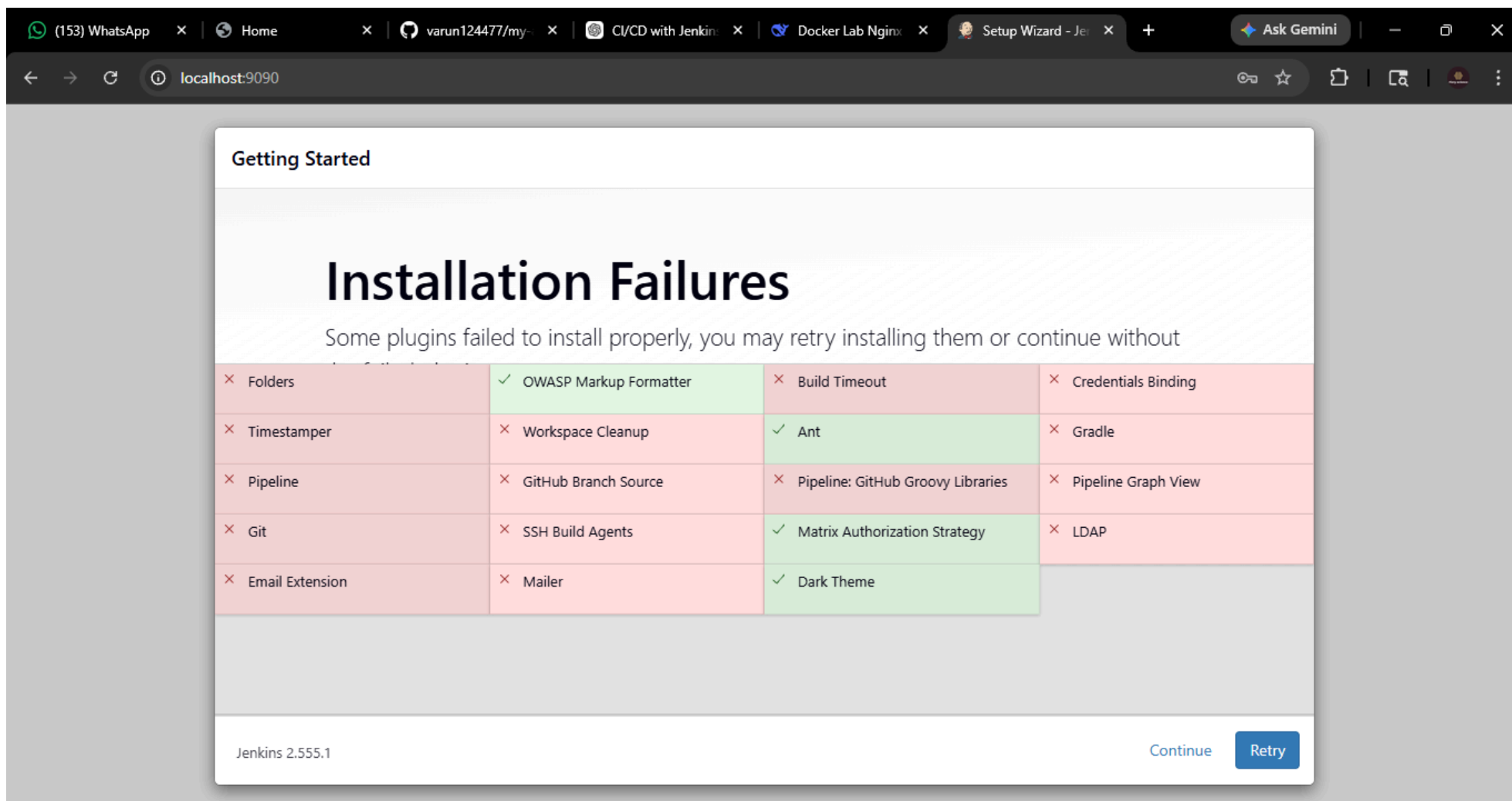
### Step 13 – Plugin Installation Progress

Wait while the plugins are downloaded and installed.



## Step 14 – Handle Plugin Failures

Some plugins may fail to install (non-critical for this lab). Click **Continue** to proceed.



The screenshot shows a web browser window with the Jenkins Setup Wizard interface. The page title is "Getting Started" and the main heading is "Installation Failures". Below the heading, a message states: "Some plugins failed to install properly, you may retry installing them or continue without". A table lists the installation status of various plugins. The table has four columns. The first column lists plugins that failed to install (marked with a red 'X'). The second column lists plugins that were successfully installed (marked with a green checkmark). The third and fourth columns list additional plugins that failed to install (marked with a red 'X'). At the bottom of the table, there are two buttons: "Continue" and "Retry". The Jenkins version "Jenkins 2.555.1" is displayed at the bottom left of the wizard.

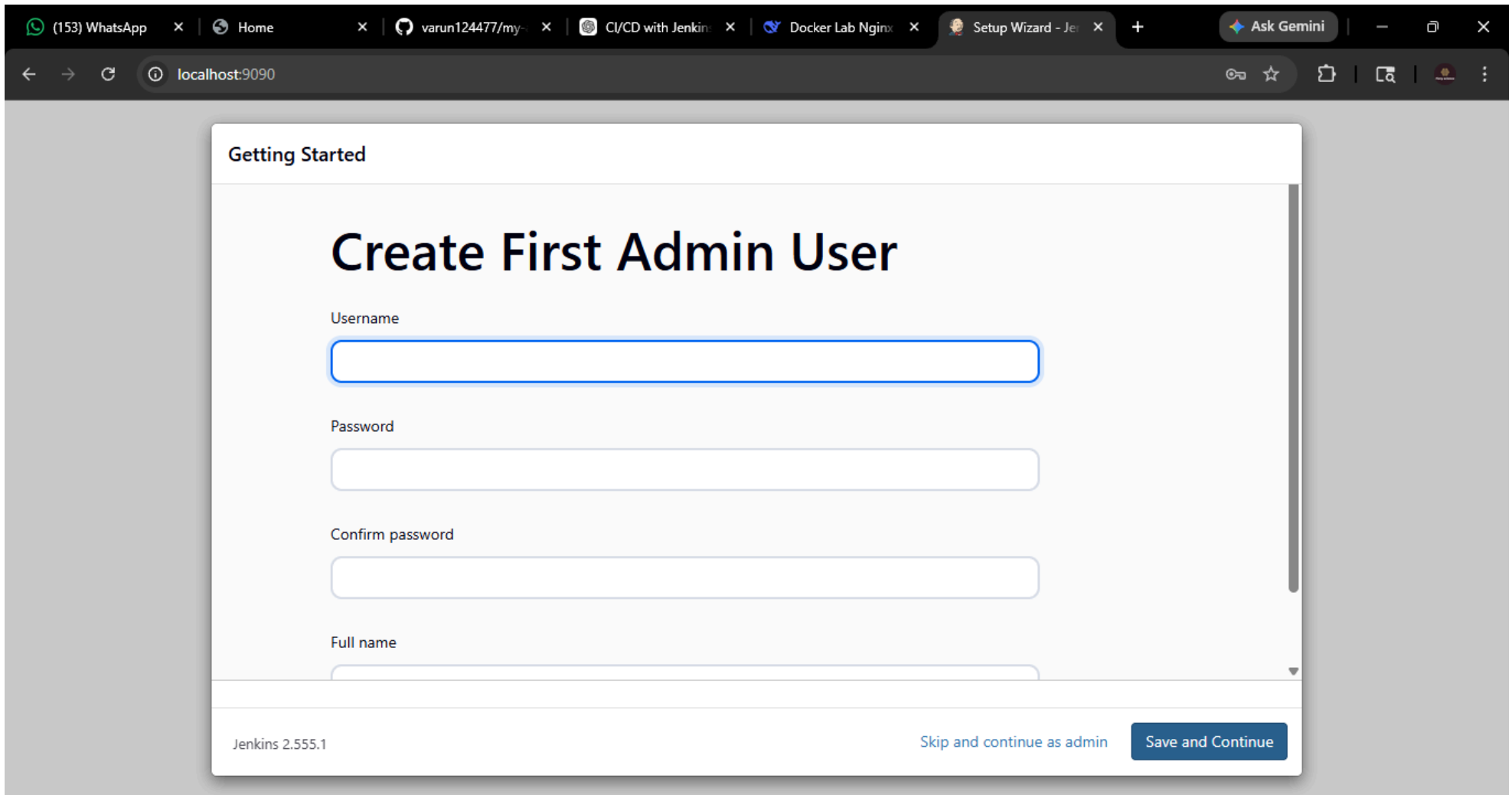
| Failed Plugins    | Successful Plugins       | Failed Plugins                      | Failed Plugins        |
|-------------------|--------------------------|-------------------------------------|-----------------------|
| ✗ Folders         | ✓ OWASP Markup Formatter | ✗ Build Timeout                     | ✗ Credentials Binding |
| ✗ Timestamper     | ✗ Workspace Cleanup      | ✓ Ant                               | ✗ Gradle              |
| ✗ Pipeline        | ✗ GitHub Branch Source   | ✗ Pipeline: GitHub Groovy Libraries | ✗ Pipeline Graph View |
| ✗ Git             | ✗ SSH Build Agents       | ✓ Matrix Authorization Strategy     | ✗ LDAP                |
| ✗ Email Extension | ✗ Mailer                 | ✓ Dark Theme                        |                       |

Jenkins 2.555.1

[Continue](#) [Retry](#)

### Step 15 – Create First Admin User (Blank Form)

Fill in a username, password, and full name for the administrator account.



The screenshot shows a web browser window with the address bar at `localhost:9090`. The browser has several tabs open, including 'Setup Wizard - Jenkins'. The main content area displays the 'Getting Started' wizard for Jenkins 2.555.1. The current step is 'Create First Admin User', which includes four input fields: 'Username', 'Password', 'Confirm password', and 'Full name'. The 'Username' field is currently active, indicated by a blue border. At the bottom of the wizard, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'.

Getting Started

## Create First Admin User

Username

Password

Confirm password

Full name

Jenkins 2.555.1

[Skip and continue as admin](#) [Save and Continue](#)

### Step 16 – Admin User Details Entered

Complete the form and save.

Getting Started

Username

VarunUpadhyay

Password

.....

Confirm password

.....

Full name

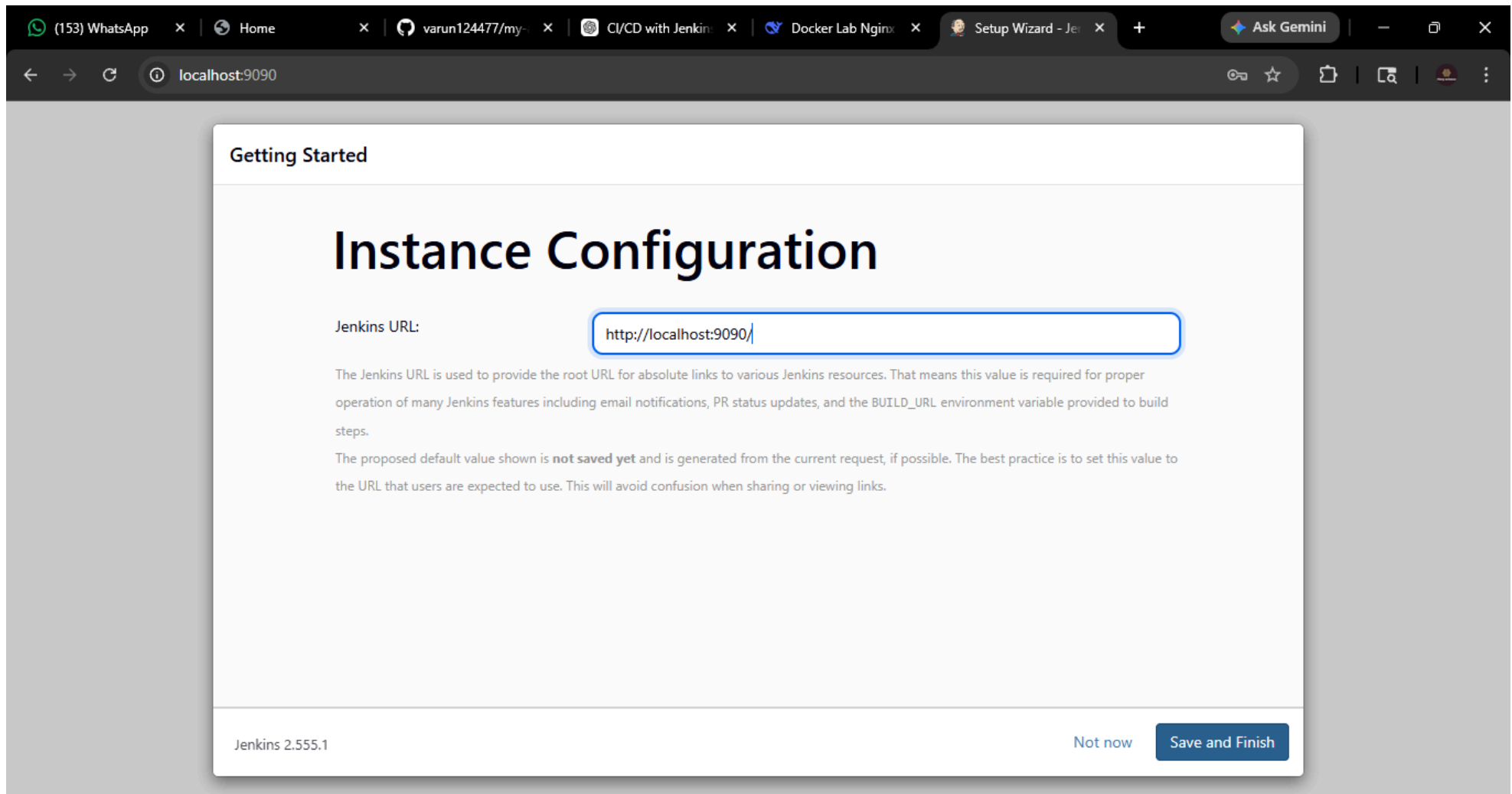
Varun Upadhyay

Jenkins 2.555.1

[Skip and continue as admin](#) [Save and Continue](#)

## Step 17 – Configure Jenkins Instance URL

Accept the default URL ( `http://localhost:9000/` ) and finish the setup.



The screenshot shows a web browser window with the address bar set to `localhost:9090`. The page title is "Getting Started" and the main heading is "Instance Configuration". A text input field for "Jenkins URL:" contains the value `http://localhost:9090/`. Below the input field, there is explanatory text about the Jenkins URL and its default value. At the bottom of the configuration box, the version "Jenkins 2.555.1" is displayed on the left, and two buttons, "Not now" and "Save and Finish", are on the right.

Getting Started

# Instance Configuration

Jenkins URL:

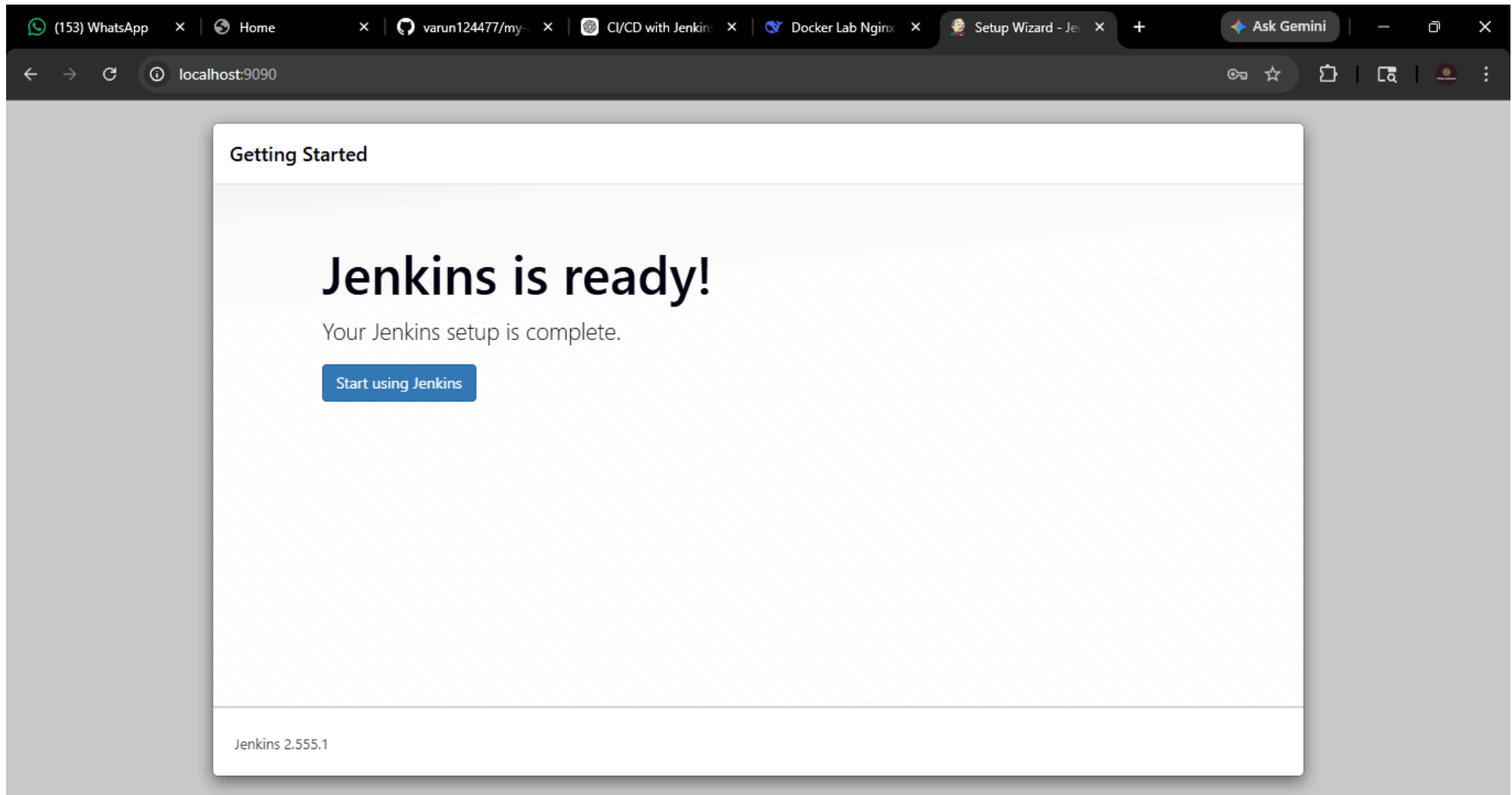
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the `BUILD_URL` environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.555.1 Not now Save and Finish

## Step 18 – Jenkins Setup Complete

You are now ready to use Jenkins.



## Step 19 – Jenkins Dashboard

The main dashboard appears. You can now create jobs.

The screenshot shows the Jenkins web interface in a browser. The browser's address bar displays 'localhost:9090'. The Jenkins logo is in the top left corner. Below it, there are links for 'New Item' and 'Build History'. On the left side, there are two panels: 'Build Queue' showing 'No builds in the queue.' and 'Build Executor Status' showing '0/2'. The main content area has a 'Welcome to Jenkins!' heading, followed by a paragraph explaining the purpose of the page. Below this, there are three sections: 'Start building your software project' with a 'Create a job' button, and 'Set up a distributed build' with three options: 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'.

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status 0/2

## Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

### Start building your software project

Create a job

### Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

## Step 20 – Verify Installed Plugins (Optional)

Check that **Docker Pipeline** and **Git** plugins are present.



The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:9090/pluginManager/`. The page title is "Jenkins / Plugins". On the left sidebar, there are links for "Updates", "Available plugins", "Installed plugins", "Advanced settings", and "Download progress". The "Available plugins" link is highlighted. The main content area has a search bar labeled "Search plugin updates" and a large light blue box with a lock icon and the text "No updates available". Below this box, a message states: "Disabled rows are already upgraded, awaiting restart. Shaded but selectable rows are in progress or failed." At the bottom right, there are links for "REST API" and "Jenkins 2.555.1".

## Step 21 – Create a New Pipeline Job

- Click **New Item**, enter `ci-cd-pipeline`, select **Pipeline**, and click OK.
- Under **Pipeline**, choose **Pipeline script from SCM**.
- Set **SCM** to **Git**.

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:9090/job/ci-cd-pipeline/configure`. The page title is "Jenkins / ci-cd-pipeline / Configure". On the left, there is a sidebar with navigation links: "General", "Triggers", "Pipeline" (which is highlighted), and "Advanced". The main content area is titled "Pipeline" and contains the instruction "Define your Pipeline using Groovy directly or pull it from source control." Below this, the "Definition" section shows a dropdown menu set to "Pipeline script from SCM". Underneath, the "SCM" dropdown is set to "None". The "Script Path" field is set to "Jenkinsfile". The "Lightweight checkout" checkbox is checked. At the bottom of the configuration area, there are "Save" and "Apply" buttons. The "Advanced" section is partially visible below the main configuration area.

## Step 22 – Set GitHub Repository URL

Enter the repository URL ( `https://github.com/varun124477/my-app.git` ) and set **Script Path** to `Jenkinsfile` . Save the job.

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:9090/job/ci-cd-pipeline/configure`. The page title is "Jenkins / ci-cd-pipeline / Configure". On the left, a sidebar menu shows "Configure" with sub-items: "General", "Triggers", "Pipeline" (which is highlighted), and "Advanced". The main content area is titled "Repositories" and contains the following fields:

- Repository URL**: A text input field containing `https://github.com/varun124477/my-app.git`.
- Credentials**: A dropdown menu currently showing `- none -`, with an `+ Add` button to its right.
- Advanced**: A button with a dropdown arrow.
- + Add Repository**: A button to add a new repository.
- Branches to build**: A text input field that is currently empty.

At the bottom of the configuration area, there are two buttons: `Save` (in blue) and `Apply` (in light gray).

## Step 23 – Trigger a Manual Build

Go to the pipeline job and click **Build Now**.

The screenshot shows the Jenkins web interface in a browser. The address bar indicates the URL is `localhost:9090/job/ci-cd-pipeline/`. The page title is "Jenkins / ci-cd-pipeline". On the left sidebar, there are buttons for "Status", "Changes", "Build Now", "Configure", "Delete Pipeline", "Rename", and "Pipeline Syntax". The main content area is titled "ci-cd-pipeline" and includes a "Permalinks" section. Below this, there is a "Builds" section showing a list of builds. The first build is labeled "#1" and occurred at "4:19 PM". The build status is indicated by a blue and white striped bar, followed by a red "X" icon, suggesting a failed build. At the bottom right of the page, there are links for "REST API" and "Jenkins 2.555.1".

## Step 24 – Restart Jenkins Environment (Optional)

If needed, you can take down and re-create the Jenkins container, or access its bash shell.

```
docker compose down
docker compose up -d
```



```
docker exec -it jenkins bash
```

The screenshot shows a VS Code editor with the following components:

- EXPLORER:** Shows the file structure for 'EXPERIMENT-7' with files: app.py, docker-compose.yml (selected), Dockerfile, Jenkinsfile, and requirements.txt.
- EDITOR:** Displays the content of 'docker-compose.yml':
 

```
1 version: '3.8'
2
3 services:
4 jenkins:
5 image: jenkins/jenkins:its
6 container_name: jenkins
7 restart: always
8 user: root
9 ports:
```
- TERMINAL:** Shows the execution of 'docker-compose down' and 'docker-compose up -d'. The output indicates that the 'version' attribute in the docker-compose.yml file is obsolete and will be ignored. It also shows the successful creation of the 'experiment-7\_default' network and the 'jenkins' container.
 

```
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> docker-compose down
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>
Error response from daemon: No such container: jenkins
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> docker-compose up -d
time="2026-04-24T22:01:51+05:30" level=warning msg="C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
 ✓ Network experiment-7_default Created 0.3s
 ✓ Container jenkins Created 0.6s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> docker exec -it jenkins bash
root@58c85f3203ad:/# apt update
apt install -y docker.io
```
- STATUS BAR:** Shows the current file as 'main', 5 lines, 0 characters, and the active extension as 'BLACKBOX Agent'.

## Step 25 – Install Docker CLI Inside Jenkins (Optional)

Installing the Docker client inside the container is **not required** because the host Docker socket is mounted. However, it can be installed for debugging.

```
apt update && apt install -y docker.io
```



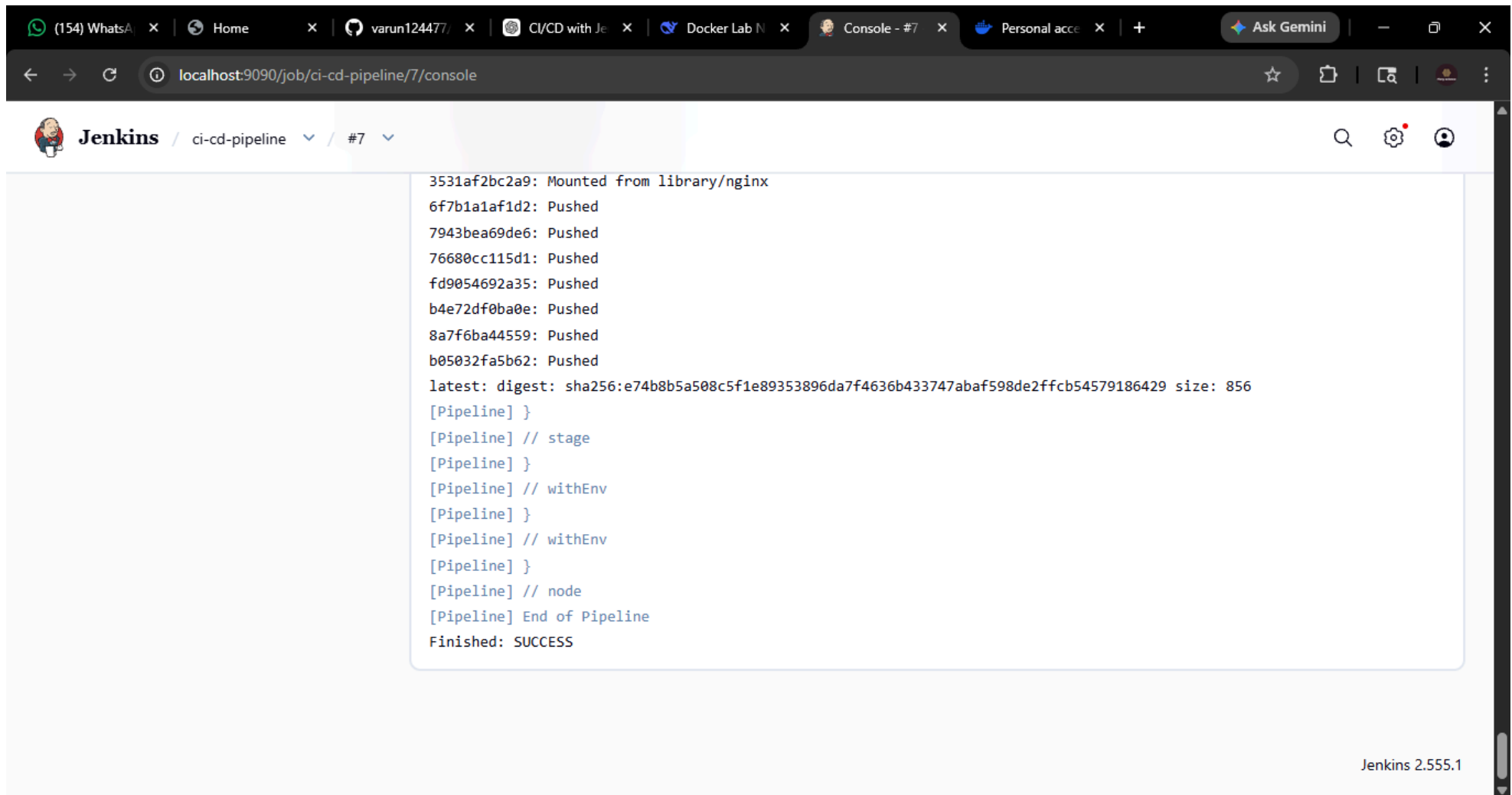
The screenshot shows the VS Code editor interface with the following components:

- EXPLORER:** Shows the file structure for 'EXPERIMENT-7' with files: app.py, docker-compose.yml (selected), Dockerfile, Jenkinsfile, and requirements.txt.
- EDITOR:** Displays the content of 'docker-compose.yml':

```
1 version: '3.8'
2
3 services:
4 jenkins:
5 image: jenkins/jenkins:its
6 container_name: jenkins
7 restart: always
8 user: root
9 ports:
```
- TERMINAL:** Shows the command prompt output for the current directory: `PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>`. It includes a warning about the deprecated 'version' attribute in docker-compose.yml and the successful execution of `docker-compose up -d`, which created the network and container. Subsequent `docker exec -it jenkins bash` commands show the successful installation of `docker.io` on the container.
- PROBLEMS:** Shows a warning about the deprecated 'version' attribute in docker-compose.yml.
- STATUS BAR:** Indicates the current file is 'main\*' and shows various status icons.

## Step 26 – Successful Pipeline Execution

The console output shows all four stages: Clone → Build → Login → Push. The image is pushed to Docker Hub.



The screenshot shows a web browser window with the Jenkins interface. The address bar displays 'localhost:9090/job/ci-cd-pipeline/7/console'. The Jenkins header shows the logo, the job name 'ci-cd-pipeline', and the build number '#7'. The console output is as follows:

```
3531af2bc2a9: Mounted from library/nginx
6f7b1a1af1d2: Pushed
7943bea69de6: Pushed
76680cc115d1: Pushed
fd9054692a35: Pushed
b4e72df0ba0e: Pushed
8a7f6ba44559: Pushed
b05032fa5b62: Pushed
latest: digest: sha256:e74b8b5a508c5f1e89353896da7f4636b433747abaf598de2ffcb54579186429 size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

The Jenkins version 'Jenkins 2.555.1' is visible in the bottom right corner of the console area.

## Step 27 – Build History

The pipeline job page displays the build history, with the latest successful build highlighted.

The screenshot shows the Jenkins web interface for a pipeline named 'ci-cd-pipeline'. The browser address bar indicates the URL is `localhost:9090/job/ci-cd-pipeline/`. The Jenkins logo and the pipeline name are at the top. On the left, there are links for 'Stages', 'Rename', and 'Pipeline Syntax'. The main area displays build history:

- Last unsuccessful build (#6), 3 min 28 sec ago
- Last completed build (#7), 1 min 10 sec ago

A 'Builds' sidebar is open, showing a list of builds for 'Today':

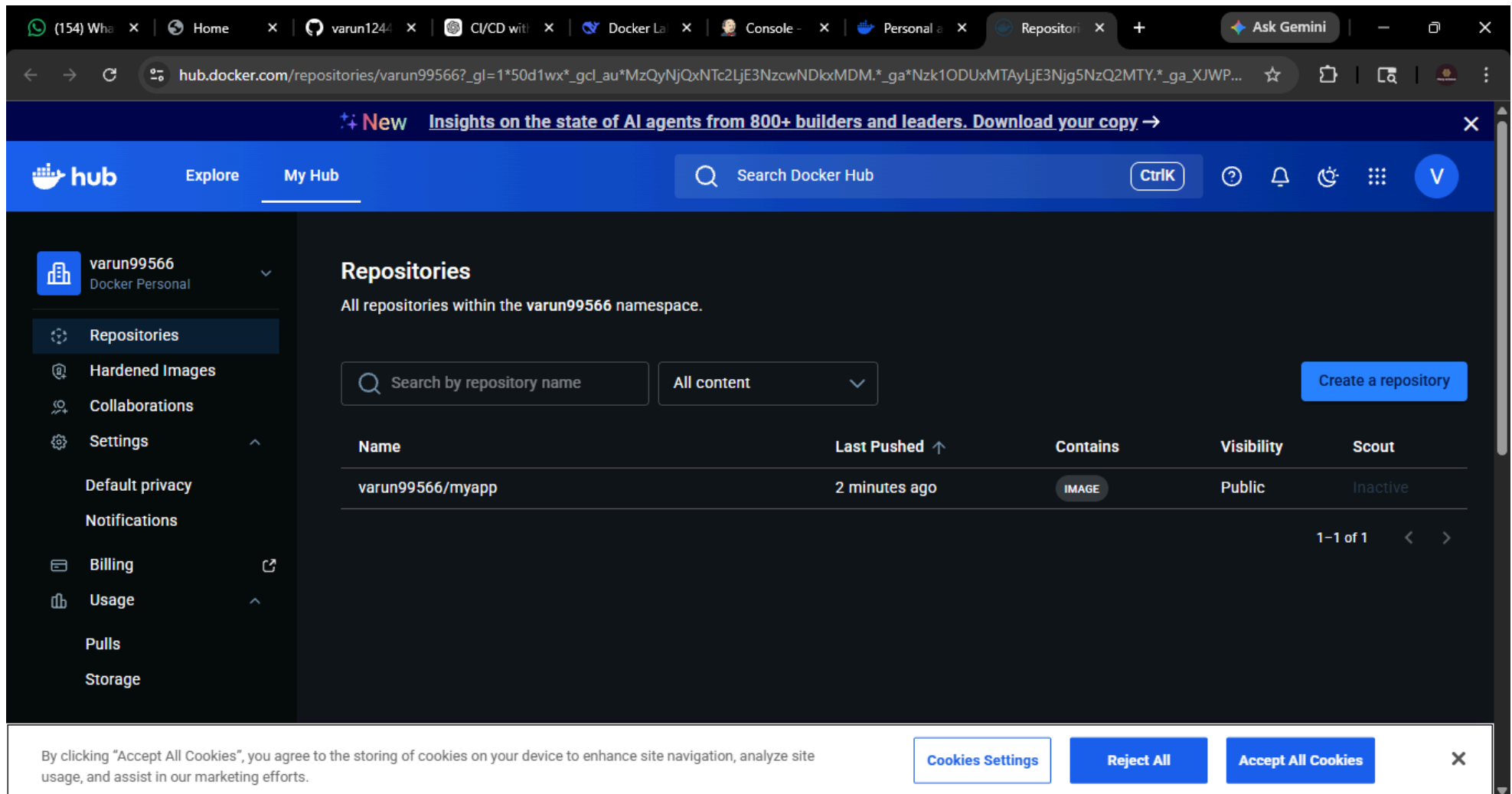
| Build Number | Status                    | Time    |
|--------------|---------------------------|---------|
| #7           | Success (Green checkmark) | 5:06 PM |
| #6           | Failure (Red X)           | 5:04 PM |
| #5           | Failure (Red X)           | 4:55 PM |
| #4           | Failure (Red X)           | 4:39 PM |
| #3           | Failure (Red X)           | 4:27 PM |
| #2           | Failure (Red X)           | 4:20 PM |
| #1           | Failure (Red X)           | 4:19 PM |

At the bottom right, it says 'REST API' and 'Jenkins 2.555.1'.

## Step 28 – Verify Image on Docker Hub

The `varun99566/myapp` repository now contains the freshly pushed image.





hub Explore My Hub Search Docker Hub CtrlK ? 🔔 🔄 ⋮ V

**Repositories**  
All repositories within the **varun99566** namespace.

Search by repository name All content Create a repository

| Name             | Last Pushed ↑ | Contains | Visibility | Scout    |
|------------------|---------------|----------|------------|----------|
| varun99566/myapp | 2 minutes ago | IMAGE    | Public     | Inactive |

1-1 of 1 < >

By clicking "Accept All Cookies", you agree to the storing of cookies on your device to enhance site navigation, analyze site usage, and assist in our marketing efforts.

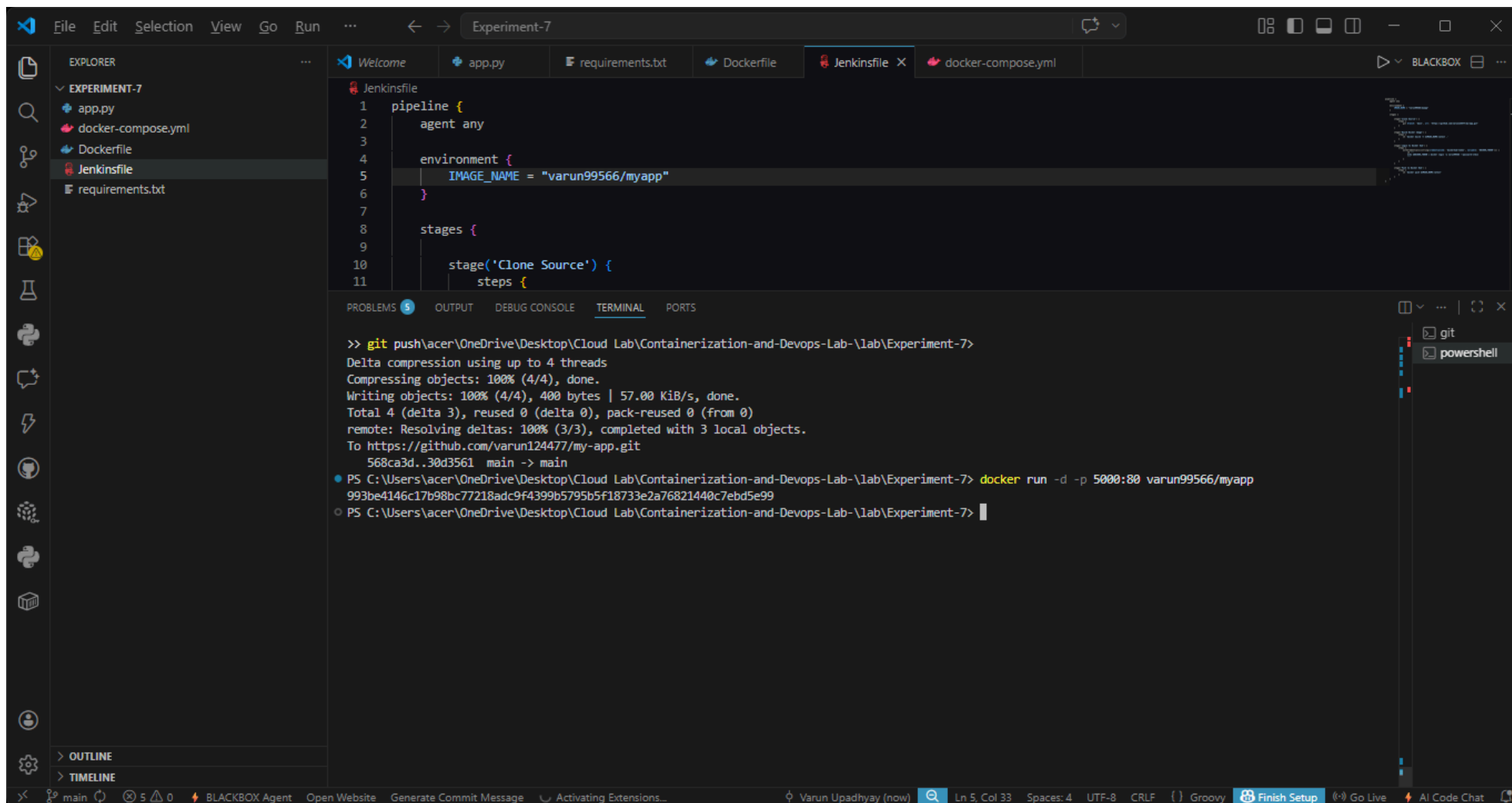
Cookies Settings Reject All Accept All Cookies X

## Step 29 – Run the Docker Image from Docker Hub

Pull and run the image locally. The container exposes port 80, mapped to port 5000 on the host.

```
docker run -d -p 5000:80 varun99566/myapp
```





The screenshot shows the Visual Studio Code interface with the following components:

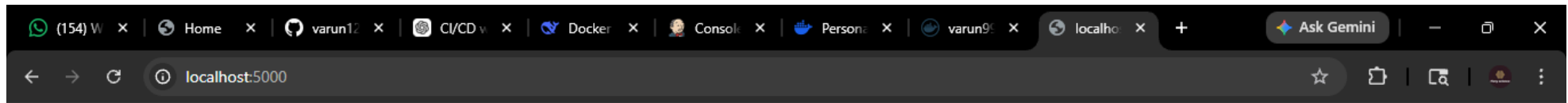
- EXPLORER:** Lists files for 'EXPERIMENT-7': app.py, docker-compose.yml, Dockerfile, Jenkinsfile, and requirements.txt.
- Jenkinsfile:** Contains a pipeline configuration:

```
1 pipeline {
2 agent any
3
4 environment {
5 IMAGE_NAME = "varun99566/myapp"
6 }
7
8 stages {
9 stage('Clone Source') {
10 steps {
```
- TERMINAL:** Shows the output of a git push and a docker run command.

```
>> git push acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 400 bytes | 57.00 KiB/s, done.
Total 4 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/varun124477/my-app.git
 568ca3d..30d3561 main -> main
● PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7> docker run -d -p 5000:80 varun99566/myapp
993be4146c17b98bc77218adc9f4399b5795b5f18733e2a76821440c7ebd5e99
○ PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-7>
```
- STATUS BAR:** Shows 'main' branch, 5 files, 0 errors, and various extension status icons.

## Step 30 – Test the Application

Open a browser and navigate to <http://localhost:5000> . You should see the message “Hello from CI/CD Pipeline!”.



Hello from CI/CD Pipeline!

## Key Concepts

- **Jenkinsfile** – Defines the pipeline as code; stored alongside the application.
- **Webhook (optional)** – Set up in GitHub to trigger Jenkins automatically on every push.
- **Credentials Store** – Docker Hub token is stored securely in Jenkins; `withCredentials` injects it temporarily.

- **Host Docker Socket** – Jenkins container shares the host's Docker daemon via `-v /var/run/docker.sock`, allowing it to build and push images without running Docker inside.

## Notes

- The `version` warning from Docker Compose can be ignored (it is obsolete and can be removed).
- In the provided setup, Jenkins is accessed on **port 9000** instead of the default 8080.
- The pipeline uses `git push --force` only during initial setup; normally, regular pushes are sufficient.





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varun124477 Add Experiments 11 &amp; 12, update Experiments 6 &amp; 7, and update homepage ✓

abb8bef · 22 minutes ago

History

Preview

Code

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## ## Lab Experiment 9: Ansible - Configuration Management with Docker Containers



### ### Aim

To demonstrate infrastructure automation using Ansible. A control node (WSL Ubuntu) manages four Docker-based Ubuntu server

<hr>

<h4 align="center"> Pre-requisite </h4>

- Ubuntu environment (WSL) on Windows
- Docker installed and running (Docker Desktop)
- Basic Linux and SSH knowledge
- Python 3 and pip (available inside WSL)

> **Experiment Directory Structure**

> All files (Dockerfile, inventory, playbook, SSH keys) are kept in the same working directory.

>

> ![Directory Structure](./Images/image.png)

<hr>

### ### Step-by-Step Procedure

---

**\*\*Step 1 - Install Ansible via `apt`\*\***

Update package lists and install Ansible from the official Ubuntu repositories.

```bash

sudo apt update -y

sudo apt install ansible -y

```
acer@DESKTOP-DOEGC13: ~  
sudo apt update -y  
sudo apt install ansible -y  
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
System information as of Wed Mar 25 14:10:13 UTC 2026  
  
System load:  1.22          Processes:            56  
Usage of /:   0.3% of 1006.85GB Users logged in:      0  
Memory usage: 9%          IPv4 address for eth0: 172.20.118.106  
Swap usage:   0%  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
  https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
This message is shown once a day. To disable it please create the  
/home/acer/.hushlogin file.  
acer@DESKTOP-DOEGC13:~$ sudo apt update -y  
[sudo] password for acer:  
Sorry, try again.  
[sudo] password for acer:  
Sorry, try again.  
[sudo] password for acer:  
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
```

```
acer@DESKTOP-DOEGC13: ~  
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.6 kB]  
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]  
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]  
Get:40 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1484 B]  
Get:41 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]  
Get:42 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [748 B]  
Get:43 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]  
Fetched 16.1 MB in 17s (919 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
153 packages can be upgraded. Run 'apt list --upgradable' to see them.  
acer@DESKTOP-DOEGC13:~$ sudo apt install ansible -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud  
  python3-lockfile python3-netaddr python3-ntlm-auth python3-packaging python3-passlib python3-requests-ntlm  
  python3-resolvelib python3-selinux python3-simplejson python3-winrm python3-xlrd  
Suggested packages:  
  cowsay sshpass python3-trio python3-aioquic python3-h2 python3-httpx python3-httpcore python-lockfile-doc ipython3  
  python-netaddr-docs  
The following NEW packages will be installed:  
  ansible ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos  
  python3-libcloud python3-lockfile python3-netaddr python3-ntlm-auth python3-packaging python3-passlib  
  python3-requests-ntlm python3-resolvelib python3-selinux python3-simplejson python3-winrm python3-xlrd  
0 upgraded, 19 newly installed, 0 to remove and 153 not upgraded.  
Need to get 22.0 MB of archives.  
After this operation, 330 MB of additional disk space will be used.
```

Step2 – Verify Ansible Installation

Check the installed version and test the local connection with the `ping` module.



```
ansible --version
ansible localhost -m ping
```

Expected output: "ping": "pong"

```
acer@DESKTOP-DOEGC13: ~  
Setting up python3-packaging (24.0-1) ...  
Setting up python3-jmespath (1.0.1-1) ...  
Setting up ieee-data (20220827.1) ...  
Setting up python3-dnspython (2.6.1-1ubuntu1) ...  
Setting up python3-selinux (3.5-2ubuntu2.1) ...  
Setting up python3-argcomplete (3.1.4-1ubuntu0.1) ...  
Setting up python3-requests-ntlm (1.1.0-3) ...  
Setting up python3-libcloud (3.4.1-5) ...  
Setting up python3-netaddr (0.8.0-2ubuntu1) ...  
Setting up python3-winrm (0.4.3-2) ...  
Setting up ansible-core (2.16.3-0ubuntu2) ...  
Setting up ansible (9.2.0+dfsg-0ubuntu5) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
acer@DESKTOP-DOEGC13:~$ ansible --version  
ansible [core 2.16.3]  
  config file = None  
  configured module search path = ['/home/acer/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']  
  ansible python module location = /usr/lib/python3/dist-packages/ansible  
  ansible collection location = /home/acer/.ansible/collections:/usr/share/ansible/collections  
  executable location = /usr/bin/ansible  
  python version = 3.12.3 (main, Jun 18 2025, 17:59:45) [GCC 13.3.0] (/usr/bin/python3)  
  jinja version = 3.1.2  
  libyaml = True  
acer@DESKTOP-DOEGC13:~$ ansible localhost -m ping  
[WARNING]: No inventory was parsed, only implicit localhost is available  
localhost | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
acer@DESKTOP-DOEGC13:~$ |
```

Step 3 – Generate an SSH Key Pair

Create an RSA 4096-bit key pair. This key will be used to authenticate to the managed containers without a password.

```
rm -rf ~/.ssh  
mkdir ~/.ssh  
chmod 700 ~/.ssh  
ssh-keygen -t rsa -b 4096
```



(Accept default file location and leave passphrase empty for this lab.)

```
acer@DESKTOP-DOEGC13: ~  
ansible collection location = /home/acer/.ansible/collections:/usr/share/ansible/collections  
executable location = /usr/bin/ansible  
python version = 3.12.3 (main, Jun 18 2025, 17:59:45) [GCC 13.3.0] (/usr/bin/python3)  
jinja version = 3.1.2  
libyaml = True  
acer@DESKTOP-DOEGC13:~$ ansible localhost -m ping  
[WARNING]: No inventory was parsed, only implicit localhost is available  
localhost | SUCCESS => {  
    "changed": false,  
    "ping": "pong"  
}  
acer@DESKTOP-DOEGC13:~$ ssh-keygen -t rsa -b 4096  
Generating public/private rsa key pair.  
Enter file in which to save the key (/home/acer/.ssh/id_rsa): cp ~/.ssh/id_rsa.pub .  
cp ~/.sshEnter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Passphrases do not match. Try again.  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Saving key "cp ~/.ssh/id_rsa.pub ." failed: No such file or directory  
acer@DESKTOP-DOEGC13:~$ cp ~/.ssh/id_rsa.pub .  
cp ~/.ssh/id_rsa .  
cp: cannot stat '/home/acer/.ssh/id_rsa.pub': No such file or directory  
cp: cannot stat '/home/acer/.ssh/id_rsa': No such file or directory  
acer@DESKTOP-DOEGC13:~$ rm -rf ~/.ssh  
mkdir ~/.ssh  
chmod 700 ~/.ssh  
acer@DESKTOP-DOEGC13:~$ ssh-keygen -t rsa -b 4096  
Generating public/private rsa key pair.  
Enter file in which to save the key (/home/acer/.ssh/id_rsa): |
```

```
acer@DESKTOP-DOEGC13: ~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/acer/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/acer/.ssh/id_rsa
Your public key has been saved in /home/acer/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:nL7L4273DI4PyMcAIZd4b/ZW+GATwPfmOTCewq+UsXs acer@DESKTOP-DOEGC13
The key's randomart image is:
+---[RSA 4096]-----+
|  ..oo.. |
| .oo.. o |
| ... . + |
| .+ B + |
| +S+ @ . |
| ooO= = |
| O+= .. |
| o =E=.o |
| +**+o.o |
+-----[SHA256]-----+
acer@DESKTOP-DOEGC13:~$ ls ~/.ssh
id_rsa id_rsa.pub
acer@DESKTOP-DOEGC13:~$ cp ~/.ssh/id_rsa .
cp ~/.ssh/id_rsa.pub .
acer@DESKTOP-DOEGC13:~$ ls
id_rsa id_rsa.pub myapp-data nginx_lab snap
acer@DESKTOP-DOEGC13:~$ docker --version
Docker version 29.2.1, build a5c7197
acer@DESKTOP-DOEGC13:~$ \
```

Step 4 – Copy Keys to Working Directory and Verify Docker

Copy the newly generated keys into the current directory. They will be baked into the Docker image. Also confirm that Docker is accessible.



```
cp ~/.ssh/id_rsa .
cp ~/.ssh/id_rsa.pub .
docker --version
```

(If Docker is not installed in WSL, install it with `sudo apt install docker.io -y`.)

```
acer@DESKTOP-DOEGC13: ~  
Your identification has been saved in /home/acer/.ssh/id_rsa  
Your public key has been saved in /home/acer/.ssh/id_rsa.pub  
The key fingerprint is:  
SHA256:nL7L4273DI4PyMcAIZd4b/ZW+GATwPfmOTCewq+UsXs acer@DESKTOP-DOEGC13  
The key's randomart image is:  
+---[RSA 4096]-----+  
| ..oo.. |  
|.oo.. o |  
|... . + |  
|. + B + |  
| +S+ @ . |  
| oo0= = |  
| O+= .. |  
| o =E=.o |  
| +**+o.o |  
+-----[SHA256]-----+  
acer@DESKTOP-DOEGC13:~$ ls ~/.ssh  
id_rsa id_rsa.pub  
acer@DESKTOP-DOEGC13:~$ cp ~/.ssh/id_rsa .  
cp ~/.ssh/id_rsa.pub .  
acer@DESKTOP-DOEGC13:~$ ls  
id_rsa id_rsa.pub myapp-data nginx_lab snap  
acer@DESKTOP-DOEGC13:~$ docker --version  
Docker version 29.2.1, build a5c7197  
acer@DESKTOP-DOEGC13:~$ sudo apt update  
sudo apt install docker.io -y  
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
```

Step 5 – Create a Dockerfile for an SSH-enabled Ubuntu Image

Write a Dockerfile that:

- Starts from the official Ubuntu image
- Installs Python 3, OpenSSH server
- Configures SSH to allow root login with key authentication only
- Copies the private key and public key into the container

```
FROM ubuntu
```

```
RUN apt update -y
```

```
RUN apt install -y python3 python3-pip openssh-server
```

```
RUN mkdir -p /var/run/sshd
```

```
RUN mkdir -p /run/sshd && \  
  echo 'root:password' | chpasswd && \  
  sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \  
  sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \  
  sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config
```

```
RUN mkdir -p /root/.ssh && chmod 700 /root/.ssh
```

```
COPY id_rsa /root/.ssh/id_rsa
```

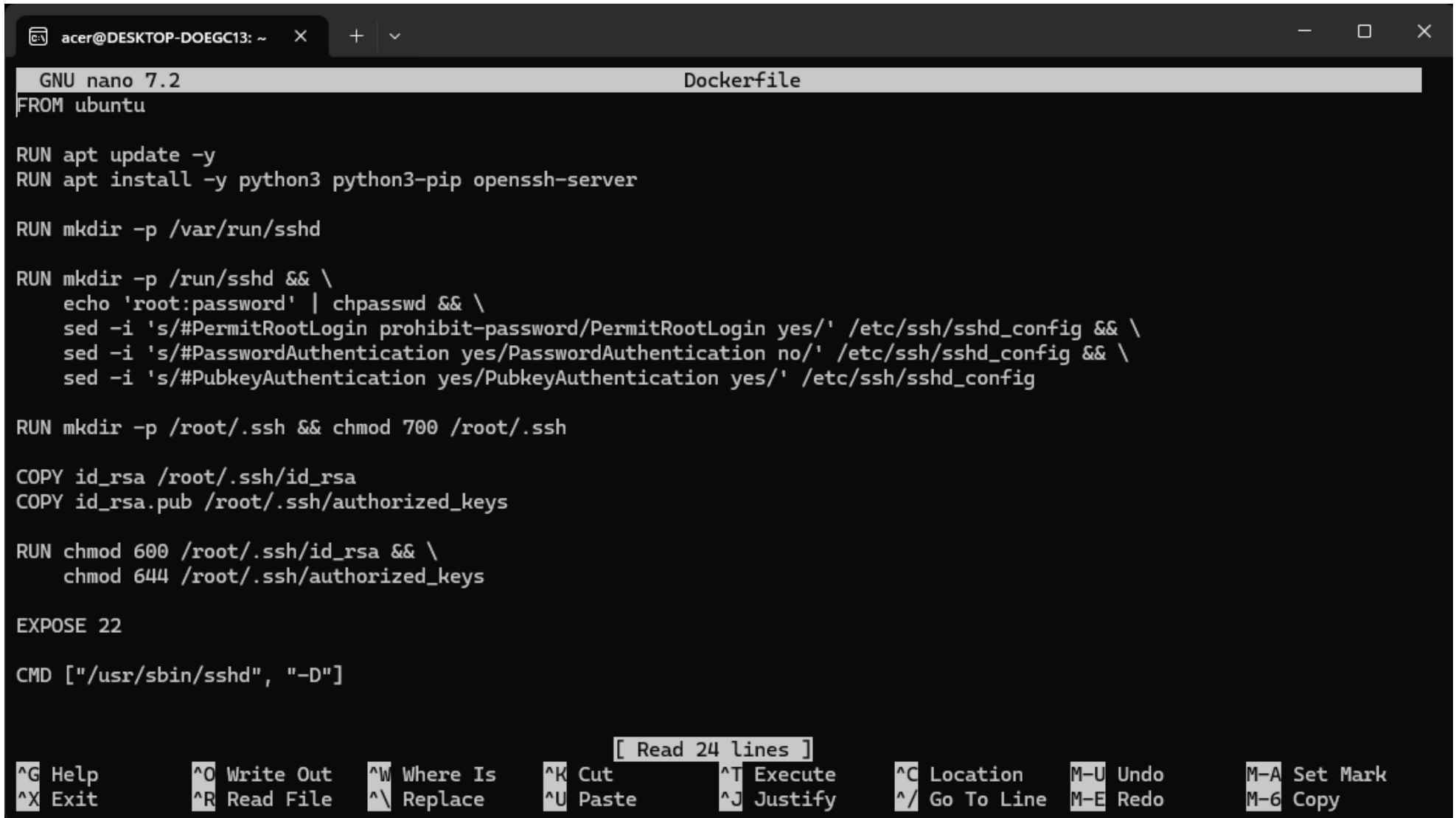
```
COPY id_rsa.pub /root/.ssh/authorized_keys
```

```
RUN chmod 600 /root/.ssh/id_rsa && \  
  chmod 644 /root/.ssh/authorized_keys
```

```
EXPOSE 22
```



CMD ["/usr/sbin/sshd", "-D"]



```
GNU nano 7.2 Dockerfile
FROM ubuntu

RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server

RUN mkdir -p /var/run/ssh

RUN mkdir -p /run/ssh && \
    echo 'root:password' | chpasswd && \
    sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \
    sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \
    sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config

RUN mkdir -p /root/.ssh && chmod 700 /root/.ssh

COPY id_rsa /root/.ssh/id_rsa
COPY id_rsa.pub /root/.ssh/authorized_keys

RUN chmod 600 /root/.ssh/id_rsa && \
    chmod 644 /root/.ssh/authorized_keys

EXPOSE 22

CMD ["/usr/sbin/sshd", "-D"]
```

[Read 24 lines]

| | | | | | | | |
|----------------|---------------------|--------------------|-----------------|-------------------|----------------------|-----------------|---------------------|
| ^G Help | ^O Write Out | ^W Where Is | ^K Cut | ^T Execute | ^C Location | M-U Undo | M-A Set Mark |
| ^X Exit | ^R Read File | ^_ Replace | ^U Paste | ^J Justify | ^/ Go To Line | M-E Redo | M-6 Copy |

Step 6 – Build the Docker Image

Tag the image as `ubuntu-server` .

```
docker build -t ubuntu-server .
```



```

acer@DESKTOP-DOEGC13: ~
+
-
[+] Building 213.1s (13/14)                                docker:default
=> [internal] load build definition from Dockerfile        0.3s
=> => transferring dockerfile: 746B                        0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest 4.8s
=> [internal] load .dockerignore                          0.2s
=> => transferring context: 2B                             0.0s
=> [1/9] FROM docker.io/library/ubuntu:latest@sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc 8.9s
=> => resolve docker.io/library/ubuntu:latest@sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc 0.2s
=> => sha256:b40150c1c2717d324cdb17278c8efdfa4dfcd2ffe083e976f0bcedf31115f081 29.73MB / 29.73MB 4.0s
=> => extracting sha256:b40150c1c2717d324cdb17278c8efdfa4dfcd2ffe083e976f0bcedf31115f081 4.4s
=> [internal] load build context                          0.3s
=> => transferring context: 4.21kB                        0.0s
=> [2/9] RUN apt update -y                               14.4s
=> [3/9] RUN apt install -y python3 python3-pip openssh-server 149.6s
=> [4/9] RUN mkdir -p /var/run/sshd                      1.0s
=> [5/9] RUN mkdir -p /run/sshd && echo 'root:password' | chpasswd && sed -i 's/#PermitRootLogin prohibi 0.7s
=> [6/9] RUN mkdir -p /root/.ssh && chmod 700 /root/.ssh 1.9s
=> [7/9] COPY id_rsa /root/.ssh/id_rsa                   0.2s
=> [8/9] COPY id_rsa.pub /root/.ssh/authorized_keys      0.2s
=> [9/9] RUN chmod 600 /root/.ssh/id_rsa && chmod 644 /root/.ssh/authorized_keys 0.5s
=> exporting to image                                    30.0s
=> => exporting layers                                    30.0s

```


Step 7 – Run Four Server Containers

Launch four containers (server1 to server4) with port mappings from host ports 2201-2204 to container port 22.

```
for i in {1..4}; do
    docker run -d -p 220${i}:22 --name server${i} ubuntu-server
done
```



```

acer@DESKTOP-DOEGC13: ~ X + v
=> [1/9] FROM docker.io/library/ubuntu:latest@sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc 8.9s
=> => resolve docker.io/library/ubuntu:latest@sha256:c4a8d5503dfb2a3eb8ab5f807da5bc69a85730fb49b5cfca2330194ebcc 0.2s
=> => sha256:b40150c1c2717d324cdb17278c8efdafa4dfcd2ffe083e976f0bcdcf31115f081 29.73MB / 29.73MB 4.0s
=> => extracting sha256:b40150c1c2717d324cdb17278c8efdafa4dfcd2ffe083e976f0bcdcf31115f081 4.4s
=> [internal] load build context 0.3s
=> => transferring context: 4.21kB 0.0s
=> [2/9] RUN apt update -y 14.4s
=> [3/9] RUN apt install -y python3 python3-pip openssh-server 149.6s
=> [4/9] RUN mkdir -p /var/run/sshd 1.0s
=> [5/9] RUN mkdir -p /run/sshd && echo 'root:password' | chpasswd && sed -i 's/#PermitRootLogin prohibi 0.7s
=> [6/9] RUN mkdir -p /root/.ssh && chmod 700 /root/.ssh 1.9s
=> [7/9] COPY id_rsa /root/.ssh/id_rsa 0.2s
=> [8/9] COPY id_rsa.pub /root/.ssh/authorized_keys 0.2s
=> [9/9] RUN chmod 600 /root/.ssh/id_rsa && chmod 644 /root/.ssh/authorized_keys 0.5s
=> exporting to image 56.6s
=> => exporting layers 30.7s
=> => exporting manifest sha256:bec0d8fa591fda43edd89450ca49a40f5136834d19cba33f399290d203d87765 0.1s
=> => exporting config sha256:ced6554ca27f5401af5d3808da86050ac709bc710cbbb0c58c427984e0b7e2fd 0.3s
=> => exporting attestation manifest sha256:9941b499e420d591ea056acff24b34548ec68326af64f55891493b7a57539f48 0.2s
=> => exporting manifest list sha256:41865f2382d508229649263580b0a7258bcb29a872eda502d6f1535629f6ae70 0.1s
=> => naming to docker.io/library/ubuntu-server:latest 0.0s
=> => unpacking to docker.io/library/ubuntu-server:latest 24.9s
acer@DESKTOP-DOEGC13:~$ for i in {1..4}; do
  docker run -d -p 220${i}:22 --name server${i} ubuntu-server
done
25c61e7336ef167631142a3b38e2ec570820409203c3e3398efbcd46e94c01c7
7ced115f20e12fb4d41f198a77520e415aba0d66491a08a329447d10555d94c1
d5d56df5ca975af592358470d570f732fb10040c81dcd63653e1d74238be22aa
ac155046d2eb6f66a74def378c613fb617ddf8edc6fc2152fe428690838a223f
acer@DESKTOP-DOEGC13:~$ |

```

Step 8 – Verify Running Containers

List all active Docker containers.

```
docker ps
```



```

acer@DESKTOP-DOEGC13: ~ X + v
=> [7/9] COPY id_rsa /root/.ssh/id_rsa 0.2s
=> [8/9] COPY id_rsa.pub /root/.ssh/authorized_keys 0.2s
=> [9/9] RUN chmod 600 /root/.ssh/id_rsa && chmod 644 /root/.ssh/authorized_keys 0.5s
=> exporting to image 56.6s
=> => exporting layers 30.7s
=> => exporting manifest sha256:bec0d8fa591fda43edd89450ca49a40f5136834d19cba33f399290d203d87765 0.1s
=> => exporting config sha256:ced6554ca27f5401af5d3808da86050ac709bc710cbbb0c58c427984e0b7e2fd 0.3s
=> => exporting attestation manifest sha256:9941b499e420d591ea056acff24b34548ec68326af64f55891493b7a57539f48 0.2s
=> => exporting manifest list sha256:41865f2382d508229649263580b0a7258bcb29a872eda502d6f1535629f6ae70 0.1s
=> => naming to docker.io/library/ubuntu-server:latest 0.0s
=> => unpacking to docker.io/library/ubuntu-server:latest 24.9s
acer@DESKTOP-DOEGC13:~$ for i in {1..4}; do
  docker run -d -p 220${i}:22 --name server${i} ubuntu-server
done
25c61e7336ef167631142a3b38e2ec570820409203c3e3398efbcd46e94c01c7
7ced115f20e12fb4d41f198a77520e415aba0d66491a08a329447d10555d94c1
d5d56df5ca975af592358470d570f732fb10040c81dcd63653e1d74238be22aa
ac155046d2eb6f66a74def378c613fb617dd8edc6fc2152fe428690838a223f
acer@DESKTOP-DOEGC13:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
ac155046d2eb   ubuntu-server  "/usr/sbin/sshd -D"     10 seconds ago Up 9 seconds  0.0.0.0:2204->22/tcp, [::]:2204->22/tcp
server4
d5d56df5ca97   ubuntu-server  "/usr/sbin/sshd -D"     11 seconds ago Up 10 seconds  0.0.0.0:2203->22/tcp, [::]:2203->22/tcp
server3
7ced115f20e1   ubuntu-server  "/usr/sbin/sshd -D"     12 seconds ago Up 11 seconds  0.0.0.0:2202->22/tcp, [::]:2202->22/tcp
server2
25c61e7336ef   ubuntu-server  "/usr/sbin/sshd -D"     14 seconds ago Up 13 seconds  0.0.0.0:2201->22/tcp, [::]:2201->22/tcp
server1
acer@DESKTOP-DOEGC13:~$ |

```

Step 9 – Test Password-Based SSH Login

Connect to `server1` on port 2201 using the root password `password`. This confirms the SSH server is running.



```
ssh root@localhost -p 2201
# When prompted, enter password: password
```

```
acer@DESKTOP-DOEGC13: ~  X + v
25c61e7336ef  ubuntu-server  "/usr/sbin/sshd -D"  14 seconds ago  Up 13 seconds  0.0.0.0:2201->22/tcp, [::]:2201->22/tcp  server1
acer@DESKTOP-DOEGC13:~$ ssh -i ~/.ssh/id_rsa root@localhost -p 2201
The authenticity of host '[localhost]:2201 ([127.0.0.1]:2201)' can't be established.
ED25519 key fingerprint is SHA256:Tsjsda6U6CbJImVuztostoZcfWytvgRAKaGpYJvydv8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[localhost]:2201' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@25c61e7336ef:~# exit
logout
Connection to localhost closed.
acer@DESKTOP-DOEGC13:~$ |
```

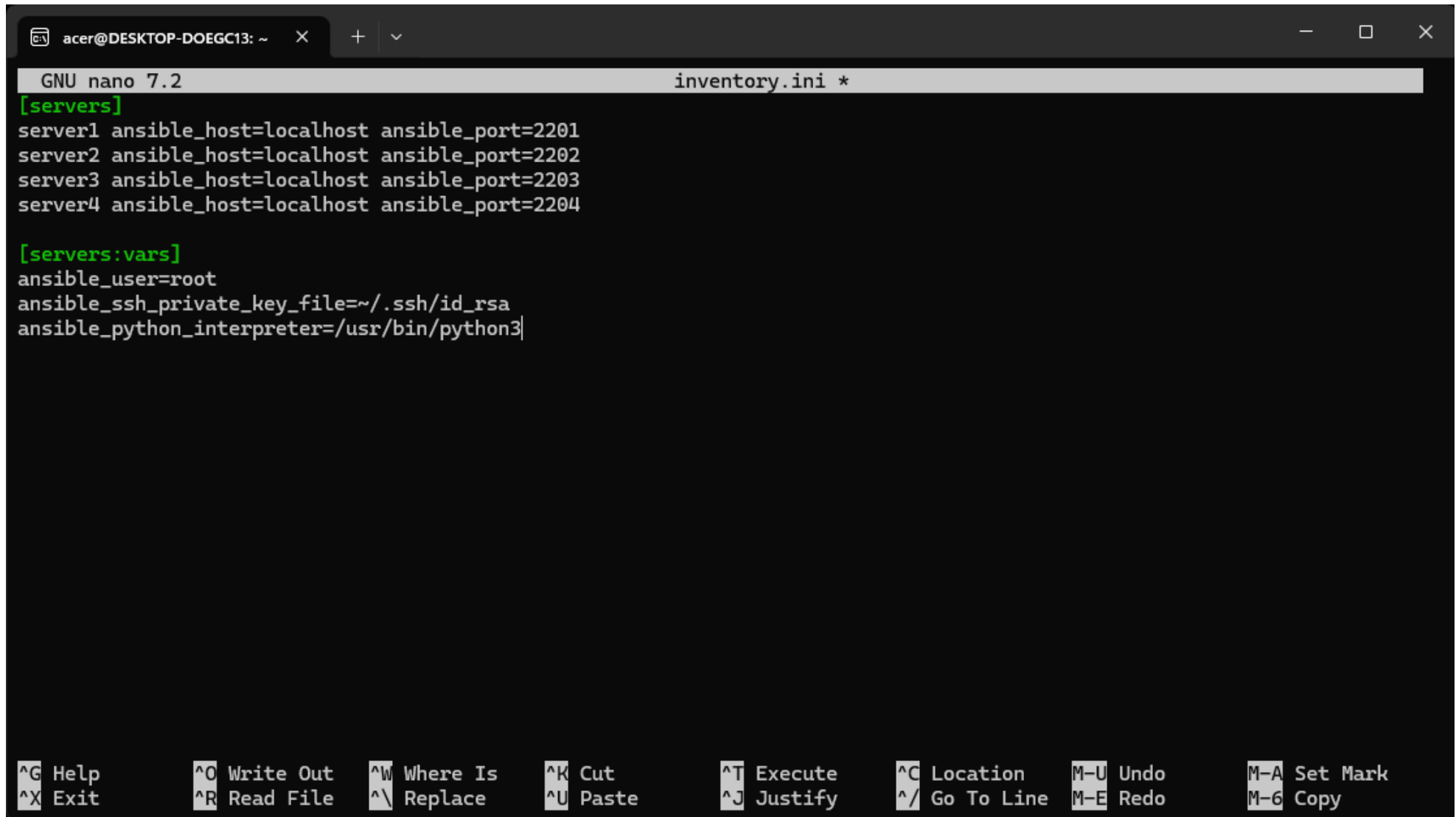
Step 10 – Create the Ansible Inventory File

Define the four servers using their localhost-mapped ports and the private key for authentication.

```
[servers]
server1 ansible_host=localhost ansible_port=2201
server2 ansible_host=localhost ansible_port=2202
server3 ansible_host=localhost ansible_port=2203
server4 ansible_host=localhost ansible_port=2204
```

```
[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
```





```
GNU nano 7.2 inventory.ini *
[servers]
server1 ansible_host=localhost ansible_port=2201
server2 ansible_host=localhost ansible_port=2202
server3 ansible_host=localhost ansible_port=2203
server4 ansible_host=localhost ansible_port=2204

[servers:vars]
ansible_user=root
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo      M-6 Copy
```

Step 11 – Test Key-Based SSH Authentication

Use the private key to log in without a password. If host key verification prompts appear, type `yes` to add the host.

```
ssh -i ~/.ssh/id_rsa root@localhost -p 2201
```



```
acer@DESKTOP-DOEGC13: ~  
2/tcp server1  
acer@DESKTOP-DOEGC13:~$ ssh -i ~/.ssh/id_rsa root@localhost -p 2201  
The authenticity of host '[localhost]:2201 ([127.0.0.1]:2201)' can't be established.  
ED25519 key fingerprint is SHA256:Tsjdoa6U6CbJImVuztostoZcfWytvgRAKaGpYJvydv8.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '[localhost]:2201' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/pro  
  
This system has been minimized by removing packages and content that are  
not required on a system that users do not log into.  
  
To restore this content, you can run the 'unminimize' command.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
root@25c61e7336ef:~# exit  
logout  
Connection to localhost closed.  
acer@DESKTOP-DOEGC13:~$ nano inventory.ini  
acer@DESKTOP-DOEGC13:~$ |
```

Step 12 – Run Ansible Ping Test

Use the `ping` module to check connectivity to all managed nodes.

```
ansible all -i inventory.ini -m ping
```



(Answer `yes` when prompted to accept SSH host keys.)


```
acer@DESKTOP-DOEGC13: ~  
server2 | UNREACHABLE! => {  
  "changed": false,  
  "msg": "Failed to connect to the host via ssh: Host key verification failed.",  
  "unreachable": true  
}  
acer@DESKTOP-DOEGC13:~$ ansible all -i inventory.ini -m ping  
The authenticity of host '[localhost]:2202 ([127.0.0.1]:2202)' can't be established.  
ED25519 key fingerprint is SHA256:Tsjsdo6U6CbJImVuztostoZcfWytvgRAKaGpYJvydv8.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [hashed name]  
  ~/.ssh/known_hosts:4: [hashed name]  
  ~/.ssh/known_hosts:5: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? server1 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
server4 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
server3 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
yes  
server2 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
acer@DESKTOP-DOEGC13:~$ |
```

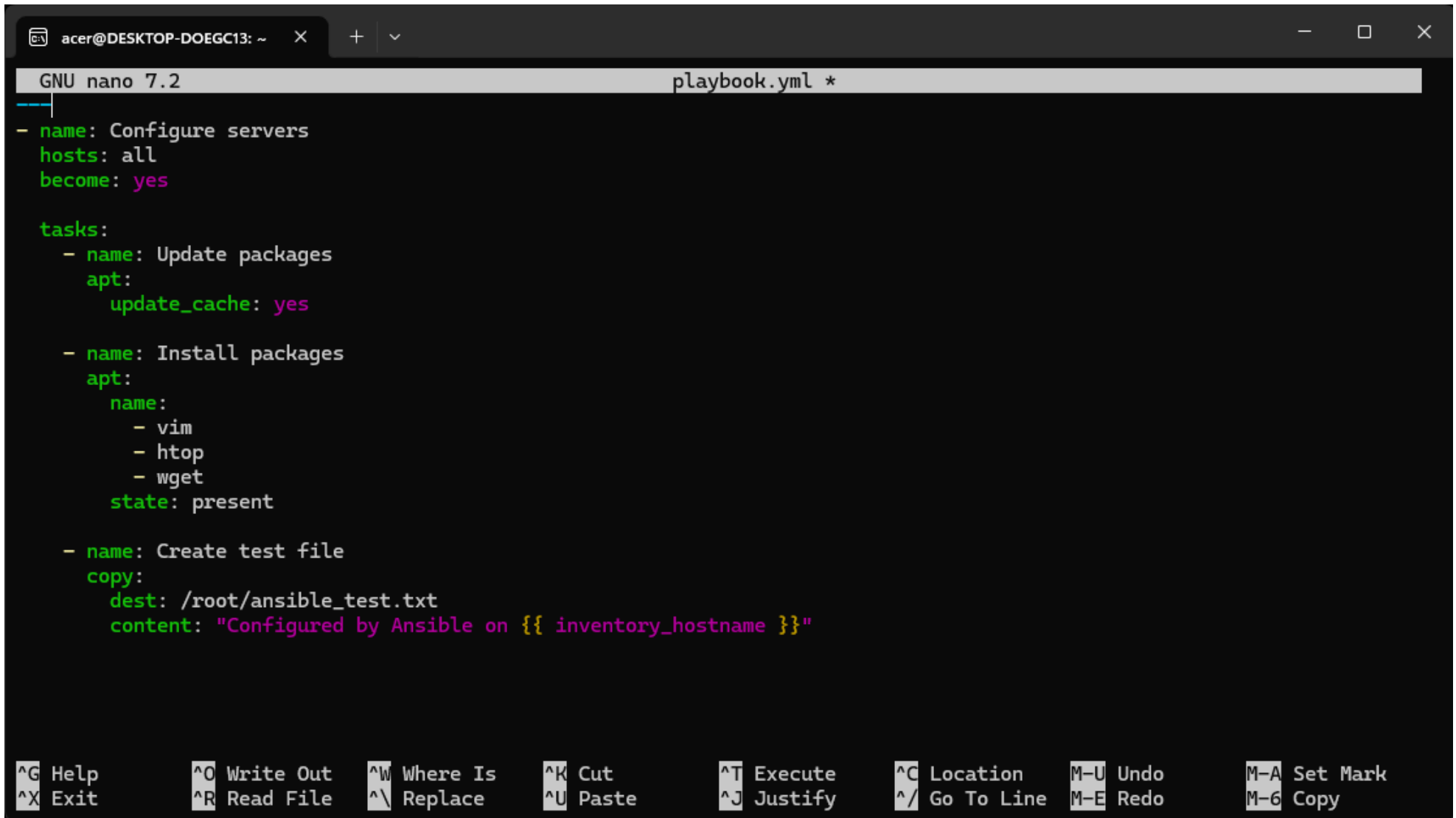
```
acer@DESKTOP-DOEGC13: ~  
"changed": false,  
"msg": "Failed to connect to the host via ssh: Host key verification failed.",  
"unreachable": true  
}  
acer@DESKTOP-DOEGC13:~$ ansible all -i inventory.ini -m ping  
The authenticity of host '[localhost]:2202 ([127.0.0.1]:2202)' can't be established.  
ED25519 key fingerprint is SHA256:Tsjsda6U6CbJImVuztostoZcfWytvgRAKaGpYJvydv8.  
This host key is known by the following other names/addresses:  
  ~/.ssh/known_hosts:1: [hashed name]  
  ~/.ssh/known_hosts:4: [hashed name]  
  ~/.ssh/known_hosts:5: [hashed name]  
Are you sure you want to continue connecting (yes/no/[fingerprint])? server1 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
server4 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
server3 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
yes  
server2 | SUCCESS => {  
  "changed": false,  
  "ping": "pong"  
}  
acer@DESKTOP-DOEGC13:~$ nano playbook.yml  
acer@DESKTOP-DOEGC13:~$ |
```

Step 13 – Write an Ansible Playbook

Create `playbook.yml` to update apt cache, install `vim`, `htop`, `wget`, and place a marker file on each server.



```
---  
- name: Configure servers  
  hosts: all  
  become: yes  
  tasks:  
    - name: Update apt package index  
      apt:  
        update_cache: yes  
  
    - name: Install required packages  
      apt:  
        name:  
          - vim  
          - htop  
          - wget  
        state: present  
  
- name: Create test file  
  copy:  
    dest: /root/ansible_test.txt  
    content: "Configured by Ansible on {{ inventory_hostname }}"
```



The screenshot shows a terminal window with a dark background. At the top, the window title bar indicates the user is 'acer' on a machine named 'DESKTOP-DOEGC13'. Below the title bar, the terminal shows the 'GNU nano 7.2' editor interface. The file being edited is 'playbook.yml'. The content of the file is an Ansible playbook with the following structure:

```
- name: Configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update packages
      apt:
        update_cache: yes

    - name: Install packages
      apt:
        name:
          - vim
          - htop
          - wget
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on {{ inventory_hostname }}"
```

At the bottom of the terminal window, there is a row of keyboard shortcuts for the nano editor, including Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, and Copy.

Step 14 – Execute the Playbook

Run the playbook against the inventory.

```
ansible-playbook -i inventory.ini playbook.yml
```



Step 15 – Verify the Configuration

Check the content of the test file on every server using the Ansible `command` module.

```
ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
```



```

acer@DESKTOP-DOEGC13: ~
ok: [server3]

TASK [Install packages] *****
changed: [server2]
changed: [server4]
changed: [server1]
changed: [server3]

TASK [Create test file] *****
changed: [server2]
changed: [server4]
changed: [server3]
changed: [server1]

PLAY RECAP *****
server1      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server4      : ok=4    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

acer@DESKTOP-DOEGC13:~$ ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
server1 | CHANGED | rc=0 >>
Configured by Ansible on server1
server4 | CHANGED | rc=0 >>
Configured by Ansible on server4
server2 | CHANGED | rc=0 >>
Configured by Ansible on server2
server3 | CHANGED | rc=0 >>
Configured by Ansible on server3
acer@DESKTOP-DOEGC13:~$ |

```

Step 16 – Clean Up (Optional)

Stop and remove the containers when finished.

```
for i in {1..4}; do docker rm -f server${i}; done
```



Key Concepts

- **Agentless:** No software installed on managed nodes – Ansible uses SSH.
- **Idempotency:** Running the playbook multiple times yields the same desired state.
- **Inventory:** A simple file listing target hosts and their connection details.
- **Playbook:** YAML-based definition of desired configuration.
- **Modules:** Pre-built units of work (e.g., `apt` , `copy` , `command`) that Ansible executes.

Observations

- SSH key-based authentication enables seamless, password-less connections.
- Ansible's declarative syntax keeps automation readable and repeatable.
- Using Docker containers as targets provides a lightweight, disposable lab environment.
- Host key verification can be bypassed for testing, but in production it should be properly managed.

References

- [Ansible Official Documentation](https://docs.ansible.com/ansible/latest/)
- [Ansible Tutorial – spacelift.io](https://spacelift.io/docs/ansible-tutorial)





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varun124477 Add Experiments 11 & 12, update Experiments 6 & 7, and update homepage ✓

abb8bef · 22 minutes ago

History

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Raw



Lab Experiment 10: SonarQube - Static Code Analysis



Aim

To set up SonarQube Server and Sonar Scanner using Docker Compose, analyse a sample Java application containing intentional

<hr>

<h4 align="center"> Pre-requisite </h4>

- Docker and Docker Compose installed
- Java JDK and Maven (already present in the lab environment)
- Docker Hub account (to pull images)
- Basic understanding of Java and Maven

> **Experiment Directory Structure**

> Contains the Docker Compose file, the sample Java project (`sample-java-app`), and all screenshots.

>

> ![Folder structure](./Images/image.png)

<hr>

Step-by-Step Procedure

****Step 1 – Verify Environment (Maven & Docker)****

Check that Maven is installed and Docker containers are running (previous experiments may still be active).

```
```powershell
mvn -version
docker ps
```

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> mvn -version
Apache Maven 3.9.15 (98b2cdbfdb5f1ac8781f537ea9accaed7922349)
Maven home: C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10\apache-maven-3.9.15
Java version: 20.0.2, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-20
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
993be4146c17 varun99566/myapp "python app.py" 2 hours ago Up 2 hours 0.0.0.0:5000->80/tcp, [::]:5000->80/tcp
cranky_dirac
58c85f3203ad jenkins/jenkins:lts "/usr/bin/tini -- /u..." 2 hours ago Up 2 hours 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp
jenkins
70dee297b4cf wordpress:latest "docker-entrypoint.s..." 5 hours ago Up 5 hours 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
wordpress_app
f529d17bd67b mysql:5.7 "docker-entrypoint.s..." 5 hours ago Up 5 hours 3306/tcp, 33060/tcp
wordpress_db
d13122c341f3 nginx:alpine "/docker-entrypoint...." 5 hours ago Up 5 hours 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
lab-nginx-compose
047e52416624 89a0b59e6d72 "httpd-foreground" 5 hours ago Up 5 hours k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7-3d35-4d41-ac75-ef8ec47d400d_1
8cf8e2297f3e 89a0b59e6d72 "httpd-foreground" 5 hours ago Up 5 hours k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c-afda-

```

## Step 2 – Create the Docker Compose File

Create `docker-compose.yml` to define two services:

- sonar-db (PostgreSQL 13)

- sonarqube (SonarQube LTS Community edition)

notepad docker-compose.yml



(The file is then filled with the YAML code below.)

```

Windows PowerShell
50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp jenkins
70dee297b4cf wordpress:latest "docker-entrypoint.s..." 5 hours ago Up 5 hours 0.0.0.0:8082->80/tcp, [::]:8082
->80/tcp wordpress_app
f529d17bd67b mysql:5.7 "docker-entrypoint.s..." 5 hours ago Up 5 hours 3306/tcp, 33060/tcp
d13122c341f3 nginx:alpine "/docker-entrypoint...." 5 hours ago Up 5 hours 0.0.0.0:8081->80/tcp, [::]:8081
->80/tcp lab-nginx-compose
047e52416624 89a0b59e6d72 "httpd-foreground" 5 hours ago Up 5 hours
4d41-ac75-ef8ec47d400d_1 k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7-3d35-
8cf8e2297f3e 89a0b59e6d72 "httpd-foreground" 5 hours ago Up 5 hours
48e8-a630-fefa6e976b02_1 k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c-afda-
cd460fe5958b nginx:latest "/docker-entrypoint...." 5 hours ago Up 5 hours 80/tcp
4cc3ad39407b nginx:latest "/docker-entrypoint...." 5 hours ago Up 5 hours 80/tcp
a6d67eed41c0 nginx:latest "/docker-entrypoint...." 5 hours ago Up 5 hours 80/tcp
0e5443889059 nginx:latest "/docker-entrypoint...." 5 hours ago Up 5 hours 80/tcp
3b8f642df20b nginx:latest "/docker-entrypoint...." 5 hours ago Up 5 hours 80/tcp
ad2fd917697e nginx:latest "/docker-entrypoint...." 5 hours ago Up 5 hours 80/tcp
webapp.4.ob9ggn8zszapqjalb7tv1s28
webapp.6.yx647vcopqm0cslptl73hhlr5
webapp.5.562iwch0c47b1vq8tpkco247t
webapp.2.3fnysvi4ray5oohr9hdz2k8zq
webapp.1.lw4nsrw844uf2lv12mub1zpxe
webapp.3.thx48e35b6cqmh7m0hfy70p
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker-compose up -d
no configuration file provided: not found
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> notepad docker-compose.y
ml
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> |

```



```
version: '3.8'
```

```
services:
```

```
 sonar-db:
```

```
 image: postgres:13
```

```
 container_name: sonar-db
```

```
 environment:
```

```
 POSTGRES_USER: sonar
```

```
 POSTGRES_PASSWORD: sonar
```

```
 POSTGRES_DB: sonarqube
```

```
 POSTGRES_HOST_AUTH_METHOD: trust
```

```
 volumes:
```

```
 - sonar-db-data:/var/lib/postgresql/data
```

```
 networks:
```

```
 - sonarqube-lab
```

```
 sonarqube:
```

```
 image: sonarqube:lts-community
```

```
 container_name: sonarqube
```

```
 ports:
```

```
 - "9000:9000"
```

```
 environment:
```

```
 SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
```

```
 SONAR_JDBC_USERNAME: sonar
```

```
 SONAR_JDBC_PASSWORD: sonar
```

```
 volumes:
```

```
 - sonar-data:/opt/sonarqube/data
```

```
 - sonar-extensions:/opt/sonarqube/extensions
```

```
 depends_on:
```

```
 - sonar-db
```

```
 networks:
```

```
 - sonarqube-lab
```

```
volumes:
```

```
 sonar-db-data:
```

```
sonar-data:
sonar-extensions:

networks:
 sonarqube-lab:
 driver: bridge
```

### Step 3 – First Start Attempt (Docker Hub Authentication Required)

Running `docker-compose up -d` fails because the images cannot be pulled without proper Docker Hub login.

```
docker-compose up -d
```



```

Windows PowerShell
3b8f642df20b nginx:latest "/docker-entrypoint..." 5 hours ago Up 5 hours 80/tcp
ad2fd917697e nginx:latest "/docker-entrypoint..." 5 hours ago Up 5 hours 80/tcp
webapp.2.3fnysvi4ray5oohr9hdz2k8zq
webapp.1.lw4nsrw844uf2lv12mublzpze
webapp.3.thx48e35b6cqmh7m0hfyt70p

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker-compose up -d
no configuration file provided: not found
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> notepad docker-compose.y
ml
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> dir

Directory: C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10

Mode LastWriteTime Length Name
---- -
da---l 25-04-2026 00:17 apache-maven-3.9.15
-a----- 25-04-2026 00:32 778 docker-compose.yml

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker-compose up -d
time="2026-04-25T00:32:35+05:30" level=warning msg="C:\\Users\\acer\\OneDrive\\Desktop\\Cloud Lab\\Containerization-and-
Devops-Lab-\\lab\\Experiment-10\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please rem
ove it to avoid potential confusion"
[+] up 2/2
! Image sonarqube:lts-community Interrupted 2.9s
X Image postgres:13 Error authentication required - incorrect username or password 2.9s
Error response from daemon: authentication required - incorrect username or password
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> |

```

## Step 4 – Log in to Docker Hub

Use the web-based login flow to authenticate. A device code is generated; pressing ENTER opens the browser to complete the login.

```
docker login
```



```
Windows PowerShell
da---l 25-04-2026 00:17 apache-maven-3.9.15
-a----- 25-04-2026 00:32 778 docker-compose.yml

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker-compose up -d
time="2026-04-25T00:32:35+05:30" level=warning msg="C:\\Users\\acer\\OneDrive\\Desktop\\Cloud Lab\\Containerization-and-Devops-Lab-\\lab\\Experiment-10\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 2/2
! Image sonarqube:lts-community Interrupted 2.9s
X Image postgres:13 Error authentication required - incorrect username or password 2.9s
Error response from daemon: authentication required - incorrect username or password
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker login
Authenticating with existing credentials... [Username: varun99566]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Stored credentials invalid or expired

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: XRDP-WWPR
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
|
```

## Step 5 – Successful Docker Hub Login

Once the browser confirms the login, the terminal shows Login Succeeded .

docker login



```
Windows PowerShell
! Image sonarqube:lts-community Interrupted 2.9s
X Image postgres:13 Error authentication required - incorrect username or password 2.9s
Error response from daemon: authentication required - incorrect username or password
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker login
Authenticating with existing credentials... [Username: varun99566]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Stored credentials invalid or expired

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: XRDP-WWPR
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

Login Succeeded
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker login
Authenticating with existing credentials... [Username: varun99566]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> |
```



## Step 6 – Start SonarQube and PostgreSQL Successfully

Now `docker compose up -d` pulls both images and creates the containers.

```
docker-compose up -d
```



```
Windows PowerShell
Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: XRDP-WWPR
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

Login Succeeded
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker login
Authenticating with existing credentials... [Username: varun99566]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> docker-compose up -d
time="2026-04-25T00:39:29+05:30" level=warning msg="C:\\Users\\acer\\OneDrive\\Desktop\\Cloud Lab\\Containerization-and-Devops-Lab-\\lab\\Experiment-10\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 34/34
✓Image postgres:13 Pulled 82.2s
✓Image sonarqube:lts-community Pulled 84.0s
✓Network experiment-10_default Created 0.3s
✓Volume experiment-10_sonar-extensions Created 0.0s
✓Volume experiment-10_sonar-db-data Created 0.0s
✓Volume experiment-10_sonar-data Created 0.0s
✓Container sonar-db Created 1.2s
✓Container sonarqube Created 0.4s
```

## Step 7 – Verify SonarQube is Operational

Check the logs of the `sonarqube` container until you see `SonarQube is operational`.

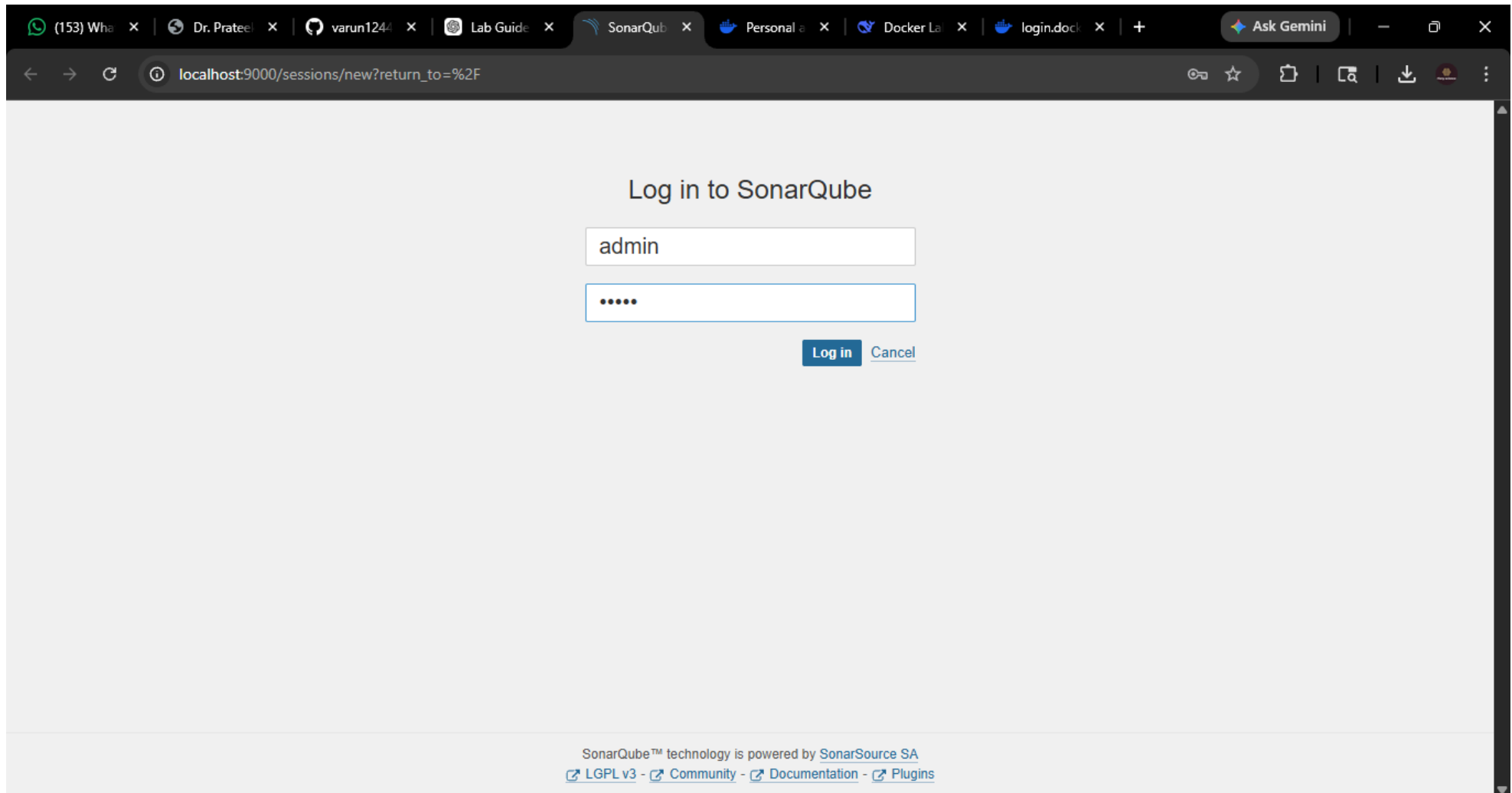
```
docker compose logs -f sonarqube
```



```
Windows PowerShell
g.sonar.ce.app.CeServer /opt/sonarqube/temp/sq-process17402324338050567945properties
sonarqube | 2026.04.24 19:13:09 INFO web[][o.s.s.q.ProjectsInWarningDaemon] Counting number of projects in warning wil
l be disabled as there are no more projects in warning.
sonarqube | 2026.04.24 19:13:10 WARN app[][startup] #####
#####
sonarqube | 2026.04.24 19:13:10 WARN app[][startup] Default Administrator credentials are still being used. Make sure
to change the password or deactivate the account.
sonarqube | 2026.04.24 19:13:10 WARN app[][startup] #####
#####
sonarqube | 2026.04.24 19:13:10 INFO web[][o.s.s.p.Platform] Web Server is operational
sonarqube | WARNING: A terminally deprecated method in java.lang.System has been called
sonarqube | WARNING: System::setSecurityManager has been called by org.sonar.process.PluginSecurityManager (file:/opt/s
onarqube/lib/sonar-application-9.9.8.100196.jar)
sonarqube | WARNING: Please consider reporting this to the maintainers of org.sonar.process.PluginSecurityManager
sonarqube | WARNING: System::setSecurityManager will be removed in a future release
sonarqube | 2026.04.24 19:13:11 INFO ce[][o.s.p.ProcessEntryPoint] Starting Compute Engine
sonarqube | 2026.04.24 19:13:11 INFO ce[][o.s.ce.app.CeServer] Compute Engine starting up...
sonarqube | 2026.04.24 19:13:13 INFO ce[][o.sonar.db.Database] Create JDBC data source for jdbc:postgresql://sonar-db:
5432/sonarqube
sonarqube | 2026.04.24 19:13:13 INFO ce[][c.z.h.HikariDataSource] HikariPool-1 - Starting...
sonarqube | 2026.04.24 19:13:13 INFO ce[][c.z.h.p.HikariPool] HikariPool-1 - Added connection org.postgresql.jdbc.PgCo
nnection@240687f7
sonarqube | 2026.04.24 19:13:13 INFO ce[][c.z.h.HikariDataSource] HikariPool-1 - Start completed.
sonarqube | 2026.04.24 19:13:19 INFO ce[][o.s.s.p.ServerFileSystemImpl] SonarQube home: /opt/sonarqube
sonarqube | 2026.04.24 19:13:20 INFO ce[][o.s.c.c.CePluginRepository] Load plugins
sonarqube | 2026.04.24 19:13:31 INFO ce[][o.s.c.c.ComputeEngineContainerImpl] Running Community edition
sonarqube | 2026.04.24 19:13:31 INFO ce[][o.s.ce.app.CeServer] Compute Engine is started
sonarqube | 2026.04.24 19:13:31 INFO app[][o.s.a.SchedulerImpl] Process[ce] is up
sonarqube | 2026.04.24 19:13:31 INFO app[][o.s.a.SchedulerImpl] SonarQube is operational
```

## Step 8 – Access the Web Dashboard

Open `http://localhost:9000` in a browser. The default credentials are `admin / admin`.



## Step 9 – Create a New Project Manually

After login, choose **Create a project manually**. Give it a project key (e.g., `sample-java-app`) and a display name.

(153) Wha x | Dr. Prateel x | varun1244 x | Lab Guide x | How do y x | Personal a x | Docker La x | login.dock x | + | Ask Gemini | - | | | | |

localhost:9000/projects/create

You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately. [Learn More](#)

sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration ? Search for projects... A

How do you want to create your project?

Do you want to benefit from all of SonarQube's features (like repository import and Pull Request decoration)? Create your project from your favorite DevOps platform. First, you need to set up a DevOps platform configuration.

**From Azure DevOps**  
Set up global configuration

**From Bitbucket Server**  
Set up global configuration

**From Bitbucket Cloud**  
Set up global configuration

**From GitHub**  
Set up global configuration

**From GitLab**  
Set up global configuration

Are you just testing or have an advanced use-case? Create a project manually.

**Manually**

**Get the most out of SonarQube!**  
Take advantage of the whole ecosystem by using SonarLint, a free IDE plugin that helps you find and fix issues earlier in your workflow. Connect SonarLint to SonarQube to sync rule sets and issue states. [Learn More](#)  
Dismiss

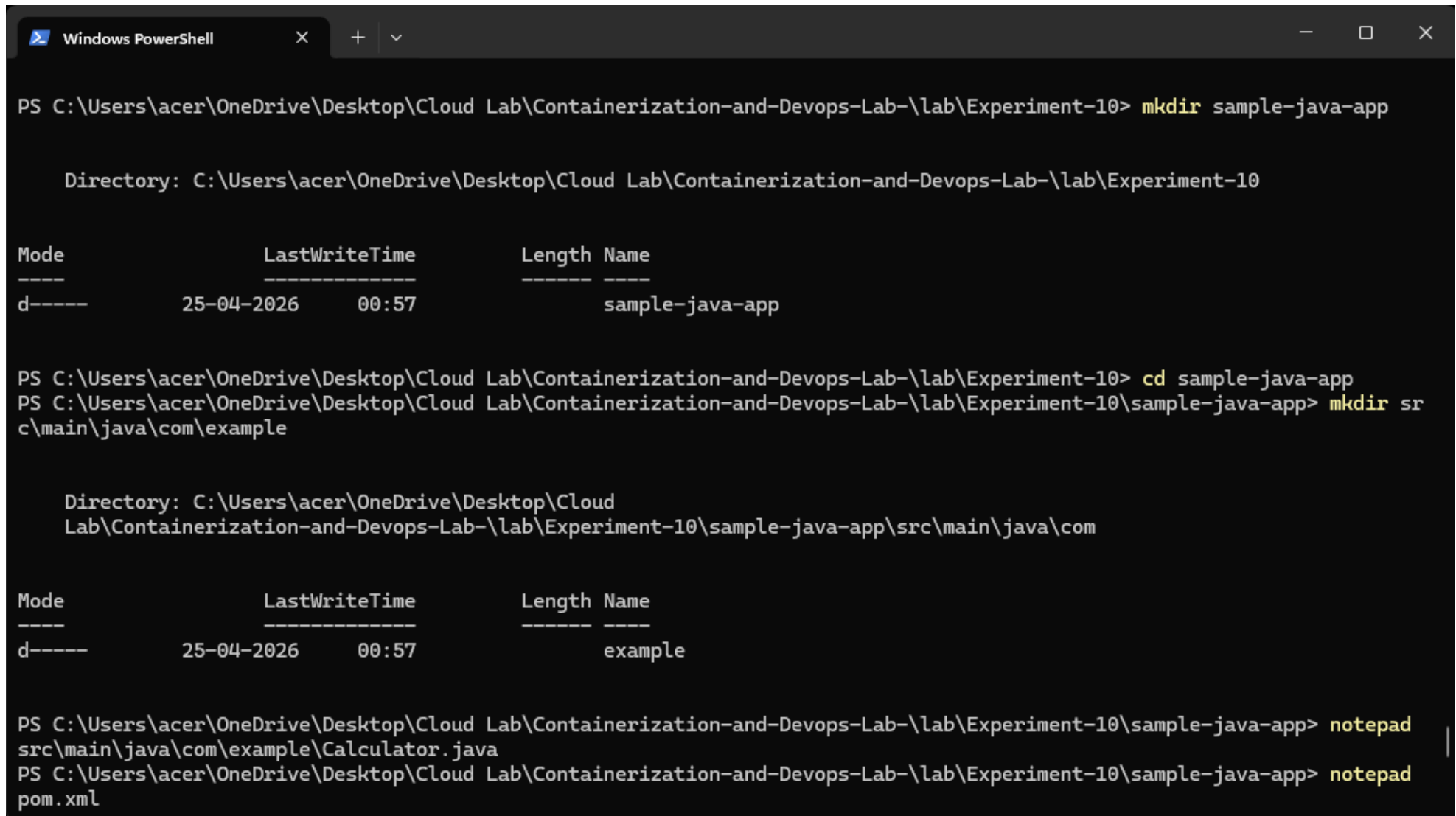
## Step 10 – Generate an Authentication Token

Tokens are used by the scanner instead of passwords.

1. Click your user icon → **My Account** → **Security**
2. Under **Generate Tokens**, enter `scanner-token` and click **Generate**

### 3. Copy the token immediately – it is shown only once.

(The token looks like `sqa_...` and will be used in the next steps.)



```
Windows PowerShell
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> mkdir sample-java-app

Directory: C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10

Mode LastWriteTime Length Name
---- -
d----- 25-04-2026 00:57 sample-java-app

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10> cd sample-java-app
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10\sample-java-app> mkdir src\main\java\com\example

Directory: C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10\sample-java-app\src\main\java\com

Mode LastWriteTime Length Name
---- -
d----- 25-04-2026 00:57 example

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10\sample-java-app> notepad src\main\java\com\example\Calculator.java
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-10\sample-java-app> notepad pom.xml
```

## Step 11 – Write the Sample Java Application with Intentional Flaws

Navigate to the experiment directory and create the Maven project structure and source file.

```
mkdir -p sample-java-app\src\main\java\com\example
cd sample-java-app
notepad src\main\java\com\example\Calculator.java
```



**Calculator.java** (bugs, vulnerabilities, code smells):

```
package com.example;

public class Calculator {

 // BUG: division by zero not handled
 public int divide(int a, int b) {
 return a / b;
 }

 // CODE SMELL: unused variable
 public int add(int a, int b) {
 int result = a + b;
 int unused = 100; // never used
 return result;
 }

 // VULNERABILITY: SQL Injection
 public String getUser(String userId) {
 String query = "SELECT * FROM users WHERE id = '" + userId + "'";
 return query;
 }

 // DUPLICATED CODE: two identical methods
 public int multiply(int a, int b) {
```



```
 int result = 0;
 for (int i = 0; i < b; i++) {
 result = result + a;
 }
 return result;
 }

 public int multiplyAlt(int a, int b) {
 int result = 0;
 for (int i = 0; i < b; i++) {
 result = result + a; // duplicate
 }
 return result;
 }

 // BUG: NullPointerException risk
 public String getName(String name) {
 return name.toUpperCase();
 }
}
```

```

1 package com.example;
2
3 public class Calculator {
4
5 public int divide(int a, int b) {
6 return a / b;
7 }
8
9 public int add(int a, int b) {
10 int result = a + b;
11 int unused = 100;
12 return result;
13 }
14
15 public String getUser(String userId) {
16 String query = "SELECT * FROM users WHERE id = " + userId;
17 return query;
18 }
19
20 public int multiply(int a, int b) {
21 int result = 0;
22 for (int i = 0; i < b; i++) {
23 result = result + a;
24 }
25 return result;
26 }
27
28 public int multiplyAlt(int a, int b) {
29 int result = 0;
30 for (int i = 0; i < b; i++) {
31 result = result + a;
32 }
33 return result;
34 }
35
36 public String getName(String name) {
37 return name.toUpperCase();
38 }
39

```

Also create `pom.xml` with Maven and SonarQube plugin configuration (make sure to replace `YOUR_TOKEN_HERE` with the actual token later):

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>

```



```
<groupId>com.example</groupId>
<artifactId>sample-app</artifactId>
<version>1.0-SNAPSHOT</version>

<properties>
 <maven.compiler.source>11</maven.compiler.source>
 <maven.compiler.target>11</maven.compiler.target>
 <sonar.projectKey>sample-java-app</sonar.projectKey>
 <sonar.host.url>http://localhost:9000</sonar.host.url>
 <sonar.login>YOUR_TOKEN_HERE</sonar.login>
</properties>

<dependencies>
 <dependency>
 <groupId>junit</groupId>
 <artifactId>junit</artifactId>
 <version>4.13.2</version>
 <scope>test</scope>
 </dependency>
</dependencies>

<build>
 <plugins>
 <plugin>
 <groupId>org.sonarsource.scanner.maven</groupId>
 <artifactId>sonar-maven-plugin</artifactId>
 <version>3.9.1.2184</version>
 </plugin>
 </plugins>
</build>
</project>
```

## Step 12 – Run the Scanner (First Attempt – Authentication Failure)

Without the correct token, the Maven plugin cannot authenticate to the SonarQube server.

```
mvn sonar:sonar -Dsonar.login=wrong-token
```



```
Windows PowerShell
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.14.2/junit-bom-5.14.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.14.2/junit-bom-5.14.2.pom (5.7 kB at 91 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.jar (84 kB at 942 kB/s)
[INFO] Default locale: "en_IN", source code encoding: "UTF-8" (analysis is platform dependent)
[INFO] Load global settings
[INFO]
[INFO] -----
[INFO] Skipping sample-app
[INFO] This project has been banned from the build due to previous failures.
[INFO] -----
[INFO] BUILD FAILURE
[INFO] -----
[INFO] Total time: 10.653 s
[INFO] Finished at: 2026-04-25T00:58:43+05:30
[INFO] -----
[ERROR] Failed to execute goal org.sonarsource.scanner.maven:sonar-maven-plugin:3.9.1.2184:sonar (default-cli) on project sample-app: Not authorized. Analyzing this project requires authentication. Please provide a user token in sonar.login or other credentials in sonar.login and sonar.password. -> [Help 1]
[ERROR]
[ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch.
[ERROR] Re-run Maven using the -X switch to enable full debug logging.
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoExecutionException
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab\lab\Experiment-10\sample-java-app>
```

### Step 13 – Provide the Correct Token and Scan Successfully

Replace `YOUR_TOKEN_HERE` in `pom.xml` with the token generated in Step 10, or pass it via command line.

```
mvn sonar:sonar -Dsonar.login=<your-copied-token>
```



Now the scan completes and the results are sent to the SonarQube server.

### Step 14 – View the Analysis Dashboard

Open `http://localhost:9000/dashboard?id=sample-java-app`.

The dashboard shows:

- **Bugs** (e.g., division by zero)
- **Vulnerabilities** (SQL injection)
- **Code Smells** (unused variable, duplicated blocks)
- **Quality Gate status** (FAILED because of the critical issues)

You can click on each number to see the exact line and a description.

The screenshot shows the SonarQube web interface in a browser. The address bar shows 'localhost:9000/projects/create'. A red banner at the top states: 'You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately. Learn More'. The navigation bar includes 'sonarqube', 'Projects', 'Issues', 'Rules', 'Quality Profiles', 'Quality Gates', and 'Administration'. A search bar on the right says 'Search for projects...'. The main content area is titled 'How do you want to create your project?'. It asks if the user wants to benefit from all features (like repository import and Pull Request decoration) and suggests creating a project from a favorite DevOps platform. Below this, there are five cards for different DevOps platforms: 'From Azure DevOps', 'From Bitbucket Server', 'From Bitbucket Cloud', 'From GitHub', and 'From GitLab'. Each card has a logo and the text 'Set up global configuration'. Below these cards, there is a section for 'Manually' creating a project, with a large blue double arrow icon. At the bottom right, there is a dark banner for 'Get the most out of SonarQube!' which promotes SonarLint, a free IDE plugin. It includes a 'Learn More' button and a 'Dismiss' button.

## Key Concepts

- **SonarQube Server** – The central analysis engine and web UI; stores all results.
- **Sonar Scanner** – A command-line tool (or Maven/Gradle plugin) that reads source code and sends reports to the server.
- **Token** – An authentication string generated in the SonarQube UI; used by the scanner instead of a password.

- **Quality Gate** – A set of pass/fail rules; prevents code that fails to meet standards from being deployed.
- **Static Analysis** – Examining code without executing it to find bugs, vulnerabilities, and maintainability issues.

## Observations

- Docker Hub login was required because the images are pulled from a private or rate-limited registry (or previous credentials expired).
- The first scan failed due to missing token – the scanner needs explicit authentication.
- The sample code contained realistic issues: unhandled exception, SQL injection, code duplication, and a null pointer risk.
- SonarQube identifies these issues and presents them in a structured, actionable format.

## Integration with CI/CD

The Jenkins pipeline (from Experiment 7) can be extended with a **SonarQube Analysis** stage:

- After checkout, `mvn clean verify sonar:sonar` runs the scanner.
- The **Quality Gate** stage waits for the SonarQube server to return a pass/fail decision.
- If the gate fails, the pipeline aborts, preventing the broken code from being built or deployed.

**Note:** Always store the SonarQube token in Jenkins credentials (e.g., a “Secret text” credential) and inject it via `withSonarQubeEnv` or environment variables. Never hardcode tokens in source files.

## Result

Successfully deployed a SonarQube server using Docker Compose, scanned a faulty Java application, observed identified issues, and understood how Quality Gates act as a safety net in the software delivery process.







varun124477 / Containerization-and-Devops

Type / to search



&lt;&gt; Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Containerization-and-Devops / lab / Experiment-11 / README.md



varun124477 Add Experiments 11 &amp; 12, update Experiments 6 &amp; 7, and update homepage ✓

abb8bef · 22 minutes ago



History

Preview

Code

Blame

Raw



Below is the complete **Experiment-11** documentation. Place these two files in your `lab/Experiment-11/` folder.

## lab/Experiment-11/README.md

### # Experiment 11: Orchestration using Docker Compose & Docker Swarm



#### ## Aim

To transition from Docker Compose to Docker Swarm, deploying the WordPress + MySQL stack as a Swarm stack, scaling services

#### ## Pre-requisites

- Docker with Swarm mode enabled (comes with Docker Desktop)
- The ``docker-compose.yml`` from Experiment 6 (WordPress + MySQL)
- All previous experiment containers stopped (``docker compose down -v``)

---

#### ## Step-by-Step Procedure

**\*\*Step 1 – Check Current State (No Swarm)\*\***

Stop any existing Compose services and verify nothing is running.

```
```bash
```

```
docker compose down -v
```

```
docker ps
```

Expected: empty list or only unrelated containers.


```
Windows PowerShell
Run 'docker --help' for more information
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker -version
unknown shorthand flag: 'e' in -ersion

Usage:  docker [OPTIONS] COMMAND [ARG...]

Run 'docker --help' for more information
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker compose down -v
time="2026-05-01T12:44:17+05:30" level=warning msg="C:\\Users\\acer\\OneDrive\\Desktop\\Cloud Lab\\Containerization-and-Devops-Lab-\\lab\\Experiment-11\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker swarm init
Error response from daemon: This node is already part of a swarm. Use "docker swarm leave" to leave this swarm and join another one.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker swarm leave --force
Node left the swarm.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker swarm init
Swarm initialized: current node (2xd43xdlcwoqcpnd6w76xouu) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-38m7l9g8sims6yfkbgglz60a5zruj10lio9o2jle819rqemzm-6xlsfe5vmrzyusmb0lpt09fon 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11>
```

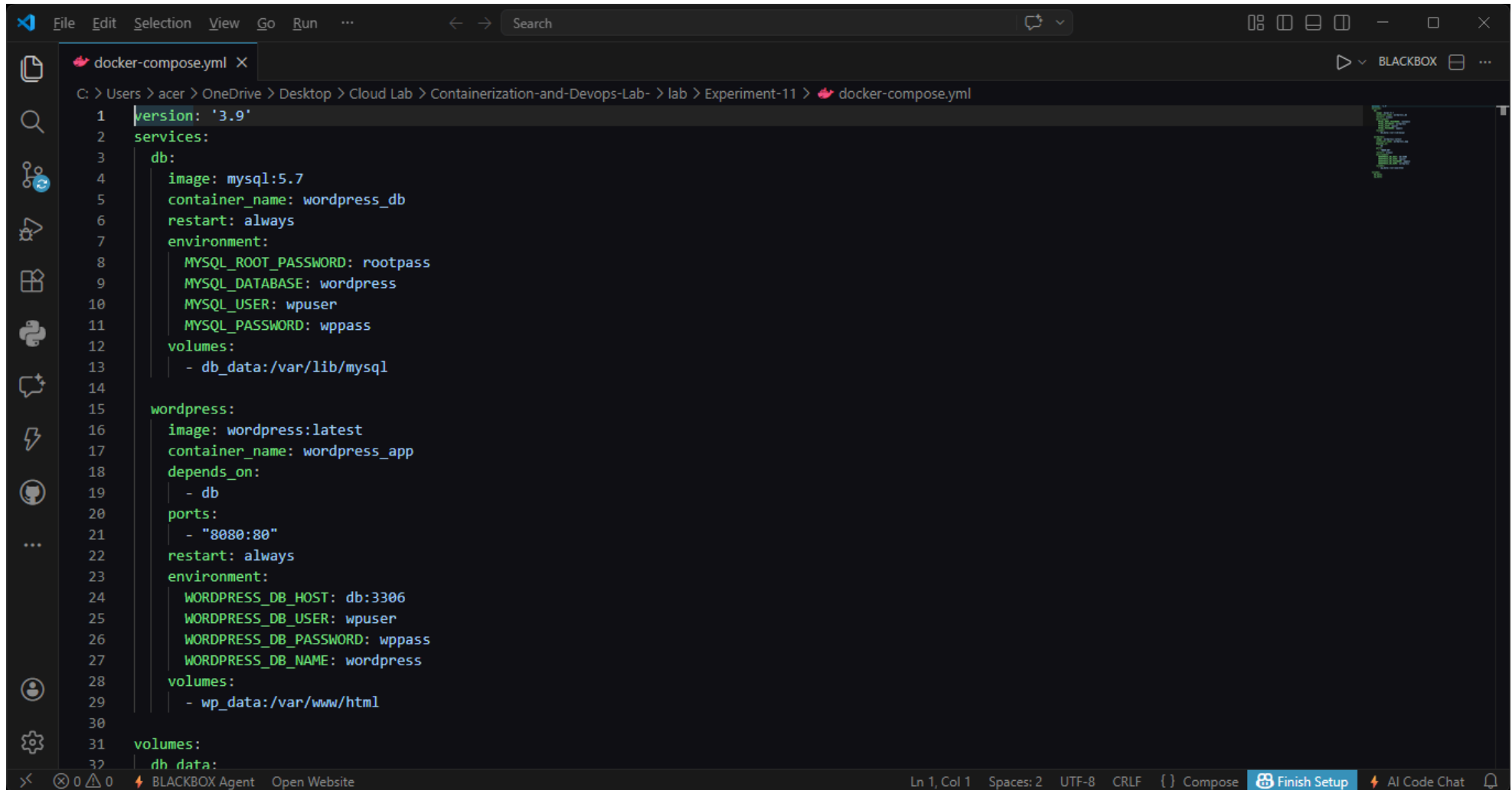
Step 2 – Initialize Docker Swarm

Turn the current Docker engine into a single-node Swarm manager.

```
docker swarm init
```



Output: "Swarm initialized: current node ... is now a manager."



```
1 version: '3.9'
2 services:
3   db:
4     image: mysql:5.7
5     container_name: wordpress_db
6     restart: always
7     environment:
8       MYSQL_ROOT_PASSWORD: rootpass
9       MYSQL_DATABASE: wordpress
10      MYSQL_USER: wpuser
11      MYSQL_PASSWORD: wppass
12     volumes:
13       - db_data:/var/lib/mysql
14
15   wordpress:
16     image: wordpress:latest
17     container_name: wordpress_app
18     depends_on:
19       - db
20     ports:
21       - "8080:80"
22     restart: always
23     environment:
24       WORDPRESS_DB_HOST: db:3306
25       WORDPRESS_DB_USER: wpuser
26       WORDPRESS_DB_PASSWORD: wppass
27       WORDPRESS_DB_NAME: wordpress
28     volumes:
29       - wp_data:/var/www/html
30
31 volumes:
32   db_data:
33   wp_data:
```

Step 3 – Verify Swarm is Active

List the Swarm nodes.

`docker node ls`



You should see one node with status **Ready** and availability **Active**.

```
Windows PowerShell
Error response from daemon: This node is already part of a swarm. Use "docker swarm leave" to leave this swarm and join another one.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker swarm leave --force
Node left the swarm.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker swarm init
Swarm initialized: current node (ygv7eo2u0x8brawzt113gpzn9) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-54916pbsnpaxmbs0yonkpdq3soneikx5hackdl9ptc0phnu2db-279mdmibnburn2m1qu12ey0ub 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker node ls
ID                                HOSTNAME          STATUS    AVAILABILITY    MANAGER STATUS    ENGINE VERSION
ygv7eo2u0x8brawzt113gpzn9 *      docker-desktop    Ready     Active           Leader             29.2.1
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Ignoring deprecated options:

container_name: Setting the container name is not supported.

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_wordpress
```

Step 4 – Deploy the WordPress Stack to Swarm

Use the same Compose file to deploy a **stack** (group of services managed by Swarm).

```
docker stack deploy -c docker-compose.yml wpstack
```



Expected output: Creating network wpstack_default, Creating service wpstack_db, Creating service wpstack_wordpress.

```
Windows PowerShell
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker swarm init
Swarm initialized: current node (ygv7eo2u0x8brawzt113gpzn9) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-54916pbsnpaxmbs0yonkpdq3soneikx5hackdl9ptc0phnu2db-279mdmibnburn2m1qu12ey0ub 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker node ls
ID                               HOSTNAME          STATUS    AVAILABILITY    MANAGER STATUS  ENGINE VERSION
ygv7eo2u0x8brawzt113gpzn9 *    docker-desktop    Ready     Active           Leader           29.2.1
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Ignoring deprecated options:

container_name: Setting the container name is not supported.

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_wordpress
Creating service wpstack_db
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker service ls
ID                NAME                MODE                REPLICAS    IMAGE                PORTS
ttz1lxq9fd0g      wpstack_db           replicated          1/1          mysql:5.7
wudfpci7alzz      wpstack_wordpress    replicated          1/1          wordpress:latest     *:8080->80/tcp
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> |
```

Step 5 – List the Services

Services are Swarm's higher-level objects that manage containers.

```
docker service ls
```



You will see `wpstack_db (1/1)` and `wpstack_wordpress (1/1)`.

```

Windows PowerShell
Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_wordpress
Creating service wpstack_db
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker service ls
ID                NAME                MODE                REPLICAS    IMAGE                PORTS
ttz1lxq9fd0g      wpstack_db          replicated          1/1          mysql:5.7
wudfpci7alzz      wpstack_wordpress   replicated          1/1          wordpress:latest     *:8080->80/tcp
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker ps
CONTAINER ID   IMAGE                COMMAND              CREATED        STATUS              PORTS
63a12c8e6737   wordpress:latest     "docker-entrypoint.s...  36 seconds ago Up 35 seconds       80/tcp
wpstack_wordpress.1.qitszzkmt1gx2q90bz6mn4
b8o
9bb7dba7ac1e   mysql:5.7           "docker-entrypoint.s...  40 seconds ago Up 38 seconds       3306/tcp, 33060/t
wpstack_db.1.hc19cnkf7zszuarailiw26gi0
cp
86b03f4c50e5   89a0b59e6d72        "httpd-foreground"     About a minute ago Up About a minute   k8s_httpd_apache-7b8c8c4b44-r7ktk_default_
6ded42f7-3d35-4d41-ac75-ef8ec47d400d_4
0b67edee3f42   89a0b59e6d72        "httpd-foreground"     About a minute ago Up About a minute   k8s_httpd_apache-7b8c8c4b44-rtrdb_default_
7dbd218c-afda-48e8-a630-fefa6e976b02_4
58c85f3203ad   jenkins/jenkins:lts "/usr/bin/tini -- /u...  7 days ago     Up 2 minutes       0.0.0.0:50000->50
000/tcp, [::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp
jenkins
70dee297b4cf   wordpress:latest     "docker-entrypoint.s...  7 days ago     Up 2 minutes       0.0.0.0:8082->80/
tcp, [::]:8082->80/tcp
wordpress_app
f529d17bd67b   mysql:5.7           "docker-entrypoint.s...  7 days ago     Up 2 minutes       3306/tcp, 33060/t
wordpress_db
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11>

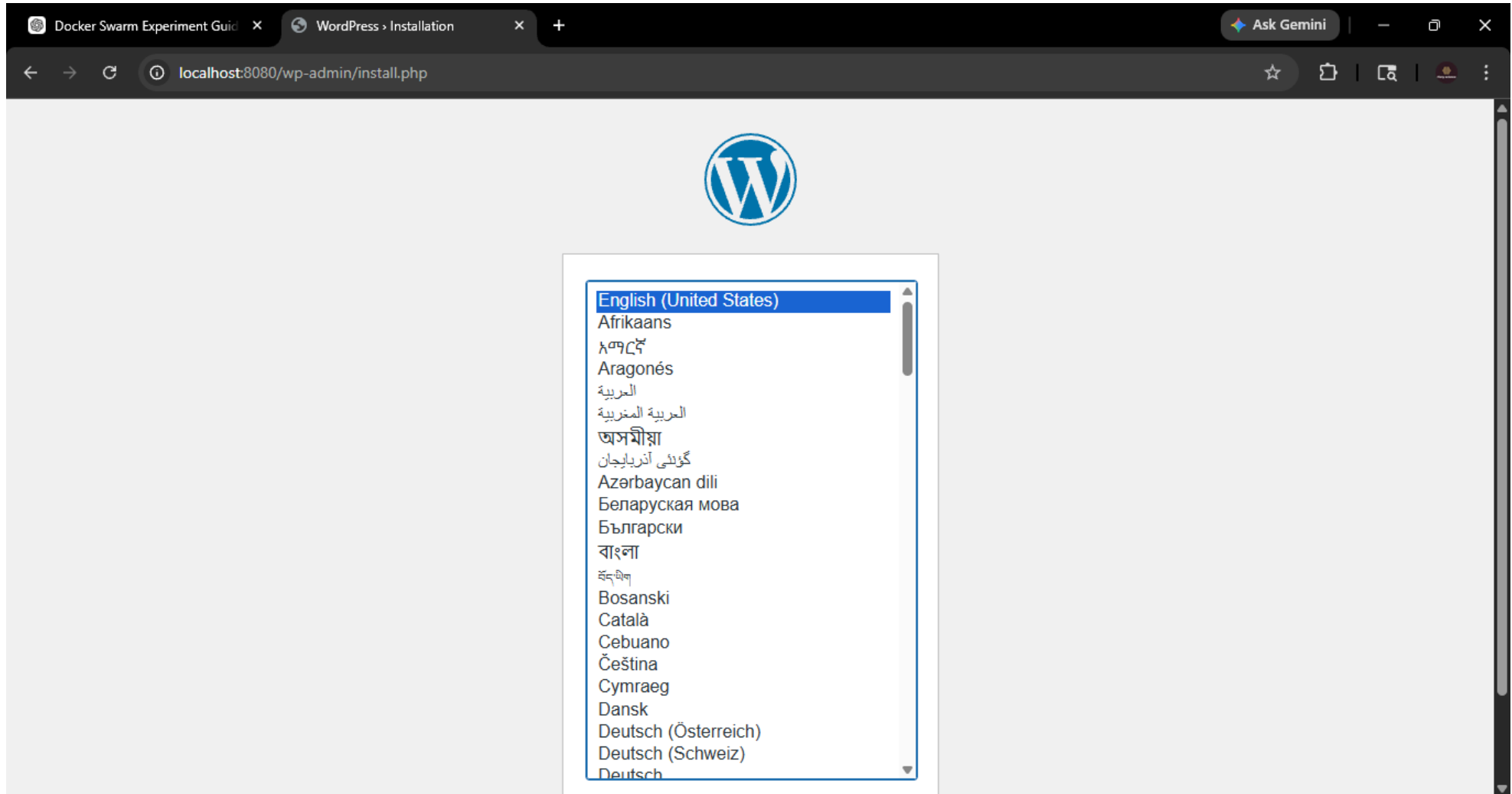
```

Step 6 – See Individual Tasks (Containers) for a Service

```
docker service ps wpstack_wordpress
```



Shows the actual container name, node, and current state.



Step 7 – Check Running Containers

Swarm-managed containers have names like `wpstack_wordpress.1.xxxxx`.

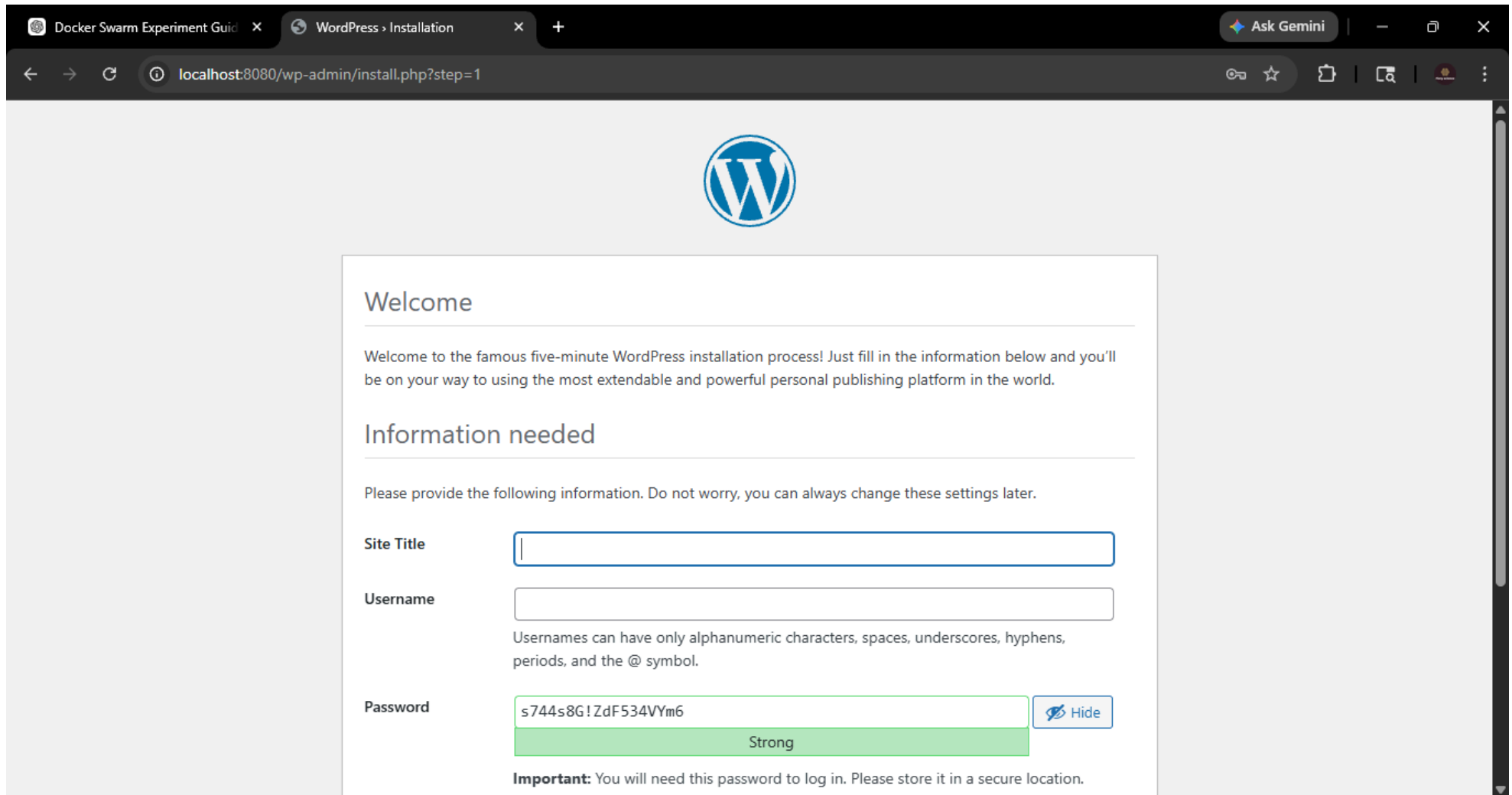
`docker ps`



```
Windows PowerShell
ttz1lxq9fd0g  wpstack_db      replicated  1/1      mysql:5.7
wudfpci7alzz  wpstack_wordpress replicated  1/1      wordpress:latest *:8080->80/tcp
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
63a12c8e6737   wordpress:latest                    "docker-entrypoint.s..." 36 seconds ago Up 35 seconds 80/tcp
wpstack_wordpress.1.qitszzkmtlgx2q90bz6mn4
b8o
9bb7dba7ac1e   mysql:5.7                          "docker-entrypoint.s..." 40 seconds ago Up 38 seconds 3306/tcp, 33060/t
cp
86b03f4c50e5   89a0b59e6d72                      "httpd-foreground"       About a minute ago Up About a minute
k8s_httpd_apache-7b8c8c4b44-r7ktk_default_
6ded42f7-3d35-4d41-ac75-ef8ec47d400d_4
0b67edee3f42   89a0b59e6d72                      "httpd-foreground"       About a minute ago Up About a minute
k8s_httpd_apache-7b8c8c4b44-rtrdb_default_
7dbd218c-afda-48e8-a630-fef6e976b02_4
58c85f3203ad   jenkins/jenkins:lts               "/usr/bin/tini -- /u..." 7 days ago     Up 2 minutes   0.0.0.0:50000->50
000/tcp, [::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp jenkins
70dee297b4cf   wordpress:latest                    "docker-entrypoint.s..." 7 days ago     Up 2 minutes   0.0.0.0:8082->80/
tcp, [::]:8082->80/tcp wordpress_app
f529d17bd67b   mysql:5.7                          "docker-entrypoint.s..." 7 days ago     Up 2 minutes   3306/tcp, 33060/t
cp wordpress_db
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker service scale wps
tack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 1 out of 3 tasks
1/3: starting [=====>]
2/3: running  [=====>]
3/3: starting [=====>]
```


Step 8 – Access WordPress

Open `http://localhost:8080` – the WordPress setup screen appears, exactly as in Experiment 6.



The screenshot shows a web browser window with the address bar displaying `localhost:8080/wp-admin/install.php?step=1`. The page features the WordPress logo at the top center. Below the logo, the heading "Welcome" is followed by a message: "Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world." The section "Information needed" follows, with a note: "Please provide the following information. Do not worry, you can always change these settings later." There are three input fields: "Site Title" (empty), "Username" (empty), and "Password" (containing the text "s744s8G!ZdF534VYm6"). Below the password field, a green bar indicates the password strength as "Strong". A "Hide" button is located to the right of the password field. At the bottom, an "Important" note states: "You will need this password to log in. Please store it in a secure location."

Step 9 – Scale the WordPress Service

Scale from 1 replica to 3 replicas with a single command.

```
docker service scale wpstack_wordpress=3
```



Output: wpstack_wordpress scaled to 3, overall progress: 3 out of 3 tasks.

```
Windows PowerShell
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running [=====>]
2/3: running [=====>]
3/3: running [=====>]
verify: Service wpstack_wordpress converged
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker ps
```

| CONTAINER ID | IMAGE | COMMAND | CREATED | STATUS | PORTS | NAMES |
|--------------------------------|------------------|--------------------------|--------------------|---------------|--|--|
| 208b062bc5da | wordpress:latest | "docker-entrypoint.s..." | 47 seconds ago | Up 46 seconds | 80/tcp | wpstack_wordpress.2.1lqzpu4os3cpuxzbuuugmot0i |
| eb542af81e77 | wordpress:latest | "docker-entrypoint.s..." | 47 seconds ago | Up 46 seconds | 80/tcp | wpstack_wordpress.3.wff7ot1sw2zgmbu9kwvmxm74v |
| 63a12c8e6737 | wordpress:latest | "docker-entrypoint.s..." | 2 minutes ago | Up 2 minutes | 80/tcp | wpstack_wordpress.1.qitszzkmt1gx2q90bz6mn4b8o |
| 9bb7dba7ac1e | mysql:5.7 | "docker-entrypoint.s..." | 2 minutes ago | Up 2 minutes | 3306/tcp, 33060/tcp | wpstack_db.1.hc19cnkf7zszuarailiw26gi0 |
| 86b03f4c50e5 | 89a0b59e6d72 | "httpd-foreground" | 3 minutes ago | Up 3 minutes | | k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7 |
| -3d35-4d41-ac75-ef8ec47d400d_4 | 0b67edee3f42 | 89a0b59e6d72 | "httpd-foreground" | 3 minutes ago | Up 3 minutes | k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c |
| -afda-48e8-a630-fefa6e976b02_4 | 58c85f3203ad | jenkins/jenkins:lts | 7 days ago | Up 4 minutes | 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp | jenkins |
| 70dee297b4cf | wordpress:latest | "docker-entrypoint.s..." | 7 days ago | Up 4 minutes | 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp | wordpress_app |
| f529d17bd67b | mysql:5.7 | "docker-entrypoint.s..." | 7 days ago | Up 4 minutes | 3306/tcp, 33060/tcp | wordpress_db |

```
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> |
```

Step 10 – Verify Scaling

```
docker service ls  
docker service ps wpstack_wordpress
```



The REPLICAS column shows 3/3; three separate tasks are now running.

```

Windows PowerShell
2/3: running [=====>]
3/3: running [=====>]
verify: Service wpstack_wordpress converged
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
208b062bc5da   wordpress:latest "docker-entrypoint.s..." 47 seconds ago Up 46 seconds 80/tcp
wpstack_wordpress.2.llqzpu4os3cpuxzbuuugmot0i
eb542af81e77   wordpress:latest "docker-entrypoint.s..." 47 seconds ago Up 46 seconds 80/tcp
wpstack_wordpress.3.wff7ot1sw2zgmbu9kwvmxm74v
63a12c8e6737   wordpress:latest "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  80/tcp
wpstack_wordpress.1.qitszzkmt1gx2q90bz6mn4b8o
9bb7dba7ac1e   mysql:5.7       "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  3306/tcp, 33060/tcp
wpstack_db.1.hc19cnkf7zszuarailw26gi0
86b03f4c50e5   89a0b59e6d72    "httpd-foreground"       3 minutes ago  Up 3 minutes
k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7
-3d35-4d41-ac75-ef8ec47d400d_4
0b67edee3f42   89a0b59e6d72    "httpd-foreground"       3 minutes ago  Up 3 minutes
k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c
-afda-48e8-a630-fefa6e976b02_4
58c85f3203ad   jenkins/jenkins:lts "/usr/bin/tini -- /u..." 7 days ago    Up 4 minutes  0.0.0.0:50000->50000/tcp,
[::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp jenkins
70dee297b4cf   wordpress:latest "docker-entrypoint.s..." 7 days ago    Up 4 minutes  0.0.0.0:8082->80/tcp, [::
]:8082->80/tcp wordpress_app
f529d17bd67b   mysql:5.7       "docker-entrypoint.s..." 7 days ago    Up 4 minutes  3306/tcp, 33060/tcp
wordpress_db

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker kill 208b062bc5da
208b062bc5da
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> |

```

Step 11 – Load Balancing in Action

Even though there are three containers, you still access the application on `http://localhost:8080`. Swarm's internal load balancer distributes requests across all replicas.

```

Windows PowerShell
9bb7dba7ac1e  mysql:5.7      "docker-entrypoint.s..." 2 minutes ago Up 2 minutes 3306/tcp, 33060/tcp
86b03f4c50e5  89a0b59e6d72   "httpd-foreground"      3 minutes ago Up 3 minutes
-3d35-4d41-ac75-ef8ec47d400d_4 0b67edee3f42  89a0b59e6d72   "httpd-foreground"      3 minutes ago Up 3 minutes
-afda-48e8-a630-fefa6e976b02_4 58c85f3203ad jenkins/jenkins:lts "/usr/bin/tini -- /u..." 7 days ago Up 4 minutes 0.0.0.0:50000->50000/tcp,
[::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp jenkins
70dee297b4cf  wordpress:latest "docker-entrypoint.s..." 7 days ago Up 4 minutes 0.0.0.0:8082->80/tcp, [::
]:8082->80/tcp wordpress_app
f529d17bd67b  mysql:5.7      "docker-entrypoint.s..." 7 days ago Up 4 minutes 3306/tcp, 33060/tcp
wordpress_db

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker kill 208b062bc5da
208b062bc5da
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker service ps wpstac
k_wordpress
ID NAME PORTS IMAGE NODE DESIRED STATE CURRENT STATE ERR
OR
qitszzkmtlgx wpstack_wordpress.1 wordpress:latest docker-desktop Running Running 3 minutes ago
hzjhl183xmqz wpstack_wordpress.2 wordpress:latest docker-desktop Running Running 5 seconds ago
llqzpu4os3cp \_ wpstack_wordpress.2 wordpress:latest docker-desktop Shutdown Failed 12 seconds ago "ta
sk: non-zero exit (137)"
wff7ot1sw2zg wpstack_wordpress.3 wordpress:latest docker-desktop Running Running 2 minutes ago

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11>

```

Step 12 – Test Self-Healing (Kill a Container)

1. Find a WordPress container ID:

```
docker ps | grep wordpress
```



2. Kill it:

```
docker kill <container-id>
```



3. Watch Swarm automatically replace it:

```
docker service ps wpstack_wordpress
```



The killed container shows `Shutdown / Failed` , and a new one is created to maintain the desired 3 replicas.

```

Windows PowerShell
hzhhl183xmq  wpstack_wordpress.2  wordpress:latest  docker-desktop  Running  Running 5 seconds ago
llqzpu4os3cp  \_ wpstack_wordpress.2  wordpress:latest  docker-desktop  Shutdown  Failed 12 seconds ago  "ta
sk: non-zero exit (137)"
wff7ot1sw2zg  wpstack_wordpress.3  wordpress:latest  docker-desktop  Running  Running 2 minutes ago

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
dcde8c2a3e36   wordpress:latest  "docker-entrypoint.s..."  26 seconds ago  Up 20 seconds  80/tcp
wpstack_wordpress.2.hzhhl183xmq2u5jqf1ckleqv
eb542af81e77   wordpress:latest  "docker-entrypoint.s..."  2 minutes ago  Up 2 minutes  80/tcp
wpstack_wordpress.3.wff7ot1sw2zgmbu9kwvmxm74v
63a12c8e6737   wordpress:latest  "docker-entrypoint.s..."  4 minutes ago  Up 4 minutes  80/tcp
wpstack_wordpress.1.qitszzkmtlgx2q90bz6mn4b8o
9bb7dba7ac1e   mysql:5.7        "docker-entrypoint.s..."  4 minutes ago  Up 4 minutes  3306/tcp, 33060/tcp
wpstack_db.1.hc19cnkf7zszuarailiw26gi0
86b03f4c50e5   89a0b59e6d72     "httpd-foreground"        4 minutes ago  Up 4 minutes
k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7
-3d35-4d41-ac75-ef8ec47d400d_4
0b67edee3f42   89a0b59e6d72     "httpd-foreground"        4 minutes ago  Up 4 minutes
k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c
-afda-48e8-a630-fefa6e976b02_4
58c85f3203ad   jenkins/jenkins:lts  "/usr/bin/tini -- /u..."  7 days ago  Up 6 minutes  0.0.0.0:50000->50000/tcp,
[::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp  jenkins
70dee297b4cf   wordpress:latest  "docker-entrypoint.s..."  7 days ago  Up 6 minutes  0.0.0.0:8082->80/tcp, [::
]:8082->80/tcp  wordpress_app
f529d17bd67b   mysql:5.7        "docker-entrypoint.s..."  7 days ago  Up 6 minutes  3306/tcp, 33060/tcp
wordpress_db

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11>

```

Step 13 – Confirm Self-Healing

docker ps | grep wordpress



Three containers are still running – Swarm healed the service automatically.

```

Windows PowerShell
wff7ot1sw2zg  wpstack_wordpress.3  wordpress:latest  docker-desktop  Running  Running 2 minutes ago

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
dcde8c2a3e36   wordpress:late "docker-entrypoint.s..." 26 seconds ago Up 20 seconds 80/tcp
wpstack_wordpress.2.hzjhl183xmq2u5jqf1ckleqv
eb542af81e77   wordpress:late "docker-entrypoint.s..." 2 minutes ago  Up 2 minutes  80/tcp
wpstack_wordpress.3.wff7ot1sw2zgmbu9kwvmxm74v
63a12c8e6737   wordpress:late "docker-entrypoint.s..." 4 minutes ago  Up 4 minutes  80/tcp
wpstack_wordpress.1.qitszzkmt1gx2q90bz6mn4b8o
9bb7dba7ac1e   mysql:5.7      "docker-entrypoint.s..." 4 minutes ago  Up 4 minutes  3306/tcp, 33060/tcp
wpstack_db.1.hc19cnkf7zszuarailiw26gi0
86b03f4c50e5   89a0b59e6d72   "httpd-foreground"       4 minutes ago  Up 4 minutes
k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7
-3d35-4d41-ac75-ef8ec47d400d_4
0b67edee3f42   89a0b59e6d72   "httpd-foreground"       4 minutes ago  Up 4 minutes
k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c
-afda-48e8-a630-fefa6e976b02_4
58c85f3203ad   jenkins/jenkins:lts "/usr/bin/tini -- /u..." 7 days ago     Up 6 minutes  0.0.0.0:50000->50000/tcp,
[:]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [:]:9090->8080/tcp jenkins
70dee297b4cf   wordpress:late "docker-entrypoint.s..." 7 days ago     Up 6 minutes  0.0.0.0:8082->80/tcp, [:
]:8082->80/tcp wordpress_app
f529d17bd67b   mysql:5.7      "docker-entrypoint.s..." 7 days ago     Up 6 minutes  3306/tcp, 33060/tcp
wordpress_db

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11>

```

Step 14 – Remove the Stack


```
docker stack rm wpstack
```



This removes the services and network. Volumes remain unless deleted separately.

```
Windows PowerShell
[::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp jenkins
70dee297b4cf wordpress:latest "docker-entrypoint.s..." 7 days ago Up 6 minutes 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp wordpress_app
f529d17bd67b mysql:5.7 "docker-entrypoint.s..." 7 days ago Up 6 minutes 3306/tcp, 33060/tcp wordpress_db

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker service ls
ID NAME MODE REPLICAS IMAGE PORTS
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
86b03f4c50e5 89a0b59e6d72 "httpd-foreground" 6 minutes ago Up 6 minutes k8s_httpd_apache-7b8c8c4b44-r7ktk_default_6ded42f7-3
d35-4d41-ac75-ef8ec47d400d_4
0b67edee3f42 89a0b59e6d72 "httpd-foreground" 6 minutes ago Up 6 minutes k8s_httpd_apache-7b8c8c4b44-rtrdb_default_7dbd218c-a
fda-48e8-a630-fefa6e976b02_4
58c85f3203ad jenkins/jenkins:lts "/usr/bin/tini -- /u..." 7 days ago Up 7 minutes 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp, 0.0.0.0:9090->8080/tcp, [::]:9090->8080/tcp jenkins
70dee297b4cf wordpress:latest "docker-entrypoint.s..." 7 days ago Up 7 minutes 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp wordpress_app
f529d17bd67b mysql:5.7 "docker-entrypoint.s..." 7 days ago Up 7 minutes 3306/tcp, 33060/tcp wordpress_db

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> docker swarm leave --force
Node left the swarm.
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-11> |
```

Step 15 – Verify Cleanup

```
docker service ls
docker ps
```



No services, no WordPress/MySQL containers.

 [clean after stack remove](#)

Key Concepts

| Feature | Docker Compose | Docker Swarm |
|-------------------|---|---|
| Scope | Single host | Multi-node cluster |
| Scaling | <code>--scale</code> flag, no balancing | <code>docker service scale</code> with built-in load balancer |
| Self-healing | Manual restart required | Automatic container replacement |
| Service Discovery | Container names | DNS + virtual IP |

Observations

- The **same** `docker-compose.yml` was used for both Compose and Swarm.
- Swarm replaces direct container management with **services**.
- Scaling a service did **not** cause port conflicts – Swarm's load balancer handles it.
- Killing a container triggered automatic recreation – no manual intervention needed.

Cleanup (Optional)

```
docker swarm leave --force    # if you want to exit Swarm mode
docker volume prune           # to remove unused volumes
```







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Containerization-and-Devops / lab / Experiment-12 / README.md



varun124477 Add Experiments 11 & 12, update Experiments 6 & 7, and update homepage ✓

abb8bef · 22 minutes ago

History

Preview

Code

Blame

Raw



Below are the **README.md** and **index.html** for Experiment 12.

Place both files inside `lab/Experiment-12/` (next to your `Images` folder and the YAML files).

The `index.html` will automatically load and beautifully render the README – no changes to the parent homepage needed.

lab/Experiment-12/README.md

Experiment 12: Study and Analyse Container Orchestration using Kubernetes



Aim

To understand core Kubernetes concepts by deploying a WordPress application, exposing it with a Service, scaling the deployment.

Pre-requisites

- Docker Desktop with Kubernetes enabled (or any running Kubernetes cluster)
- `kubect1` configured and working
- Basic YAML knowledge

Step-by-Step Procedure

Step 1 – Verify kubectl and Cluster Nodes

Check that `kubectl` is available and that the local Docker Desktop node is ready.

```
```powershell
kubectl version
kubectl get nodes
```

*The node `docker-desktop` appears as **Ready**, control-plane.*

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl version --client

Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> minikube start
minikube : The term 'minikube' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ minikube start
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (minikube:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get nodes
NAME STATUS ROLES AGE VERSION
docker-desktop Ready control-plane 37d v1.34.1
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> minikube start
minikube : The term 'minikube' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ minikube start
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (minikube:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12>
```

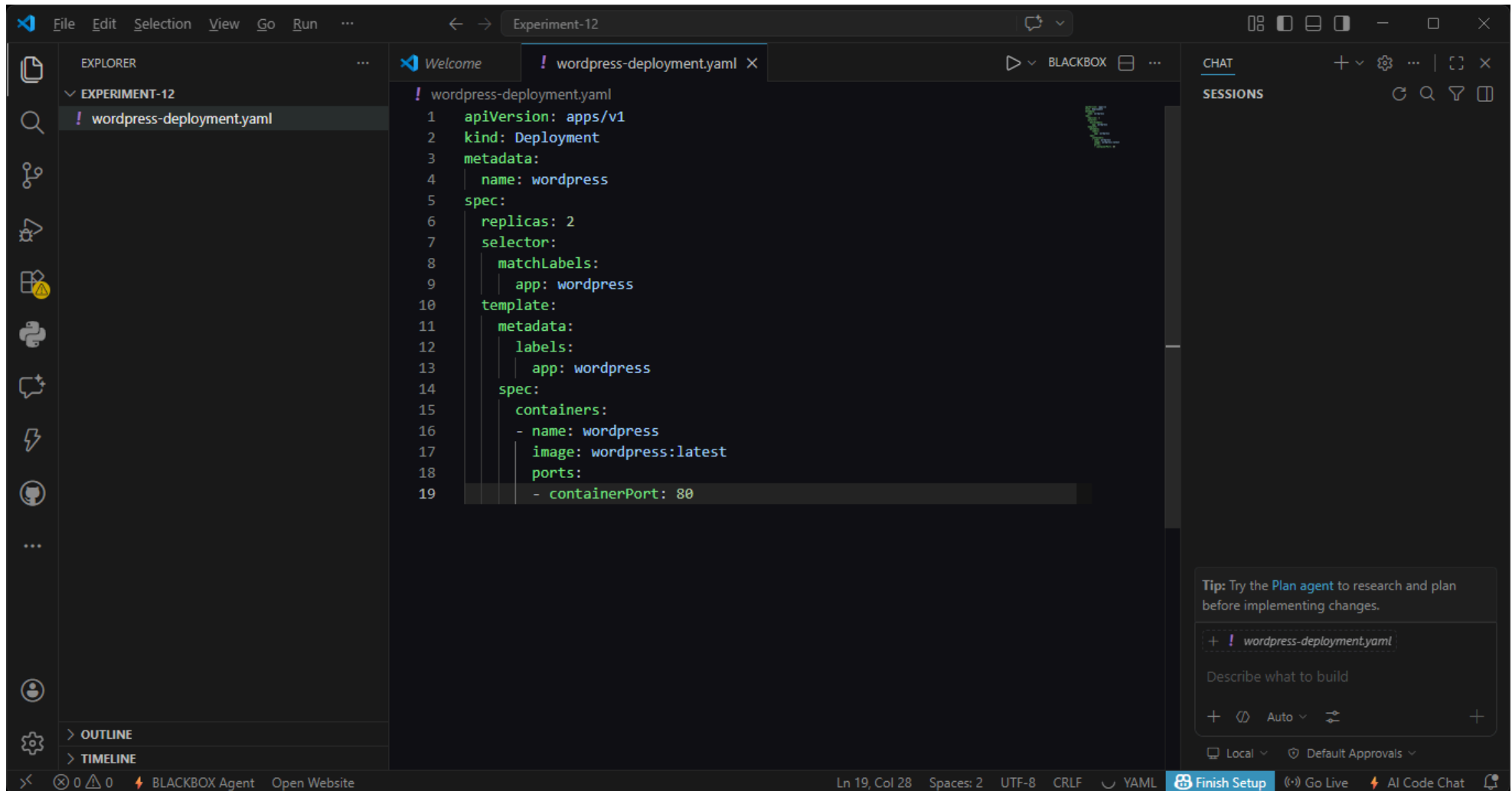
## Step 2 – Create the WordPress Deployment YAML

Write a deployment manifest that requests 2 replicas of the WordPress container.

wordpress-deployment.yaml :



```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: wordpress
spec:
 replicas: 2
 selector:
 matchLabels:
 app: wordpress
 template:
 metadata:
 labels:
 app: wordpress
 spec:
 containers:
 - name: wordpress
 image: wordpress:latest
 ports:
 - containerPort: 80
```



The screenshot shows the Visual Studio Code editor interface. The Explorer panel on the left shows a project named 'EXPERIMENT-12' with a file named 'wordpress-deployment.yaml'. The main editor area displays the content of this file, which is a Kubernetes Deployment manifest for WordPress. The manifest includes metadata, replica count (2), selector, and a template with labels and a container specification. The status bar at the bottom indicates the file is at line 19, column 28, with 2 spaces, UTF-8 encoding, CRLF line endings, and a YAML file type. The right sidebar shows the 'CHAT' panel with a 'SESSIONS' list and a 'Tip' box suggesting the use of the 'Plan agent'.

```
1 apiVersion: apps/v1
2 kind: Deployment
3 metadata:
4 name: wordpress
5 spec:
6 replicas: 2
7 selector:
8 matchLabels:
9 app: wordpress
10 template:
11 metadata:
12 labels:
13 app: wordpress
14 spec:
15 containers:
16 - name: wordpress
17 image: wordpress:latest
18 ports:
19 - containerPort: 80
```

### Step 3 – Apply the Deployment

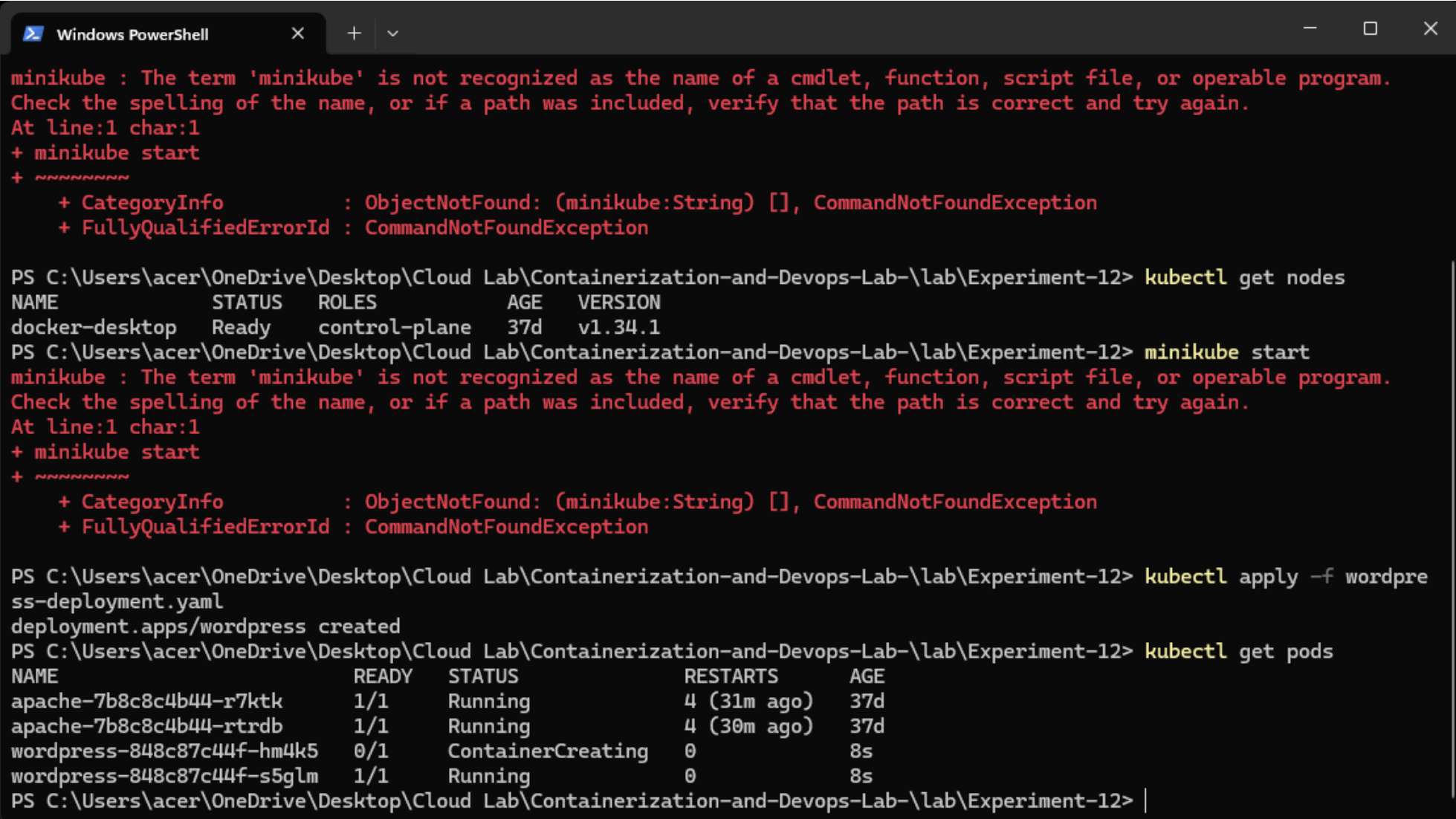
Submit the deployment to Kubernetes and list the resulting pods.

```
kubectl apply -f wordpress-deployment.yaml
kubectl get pods
```





Two WordPress pods are created (one may show *ContainerCreating* briefly).



```

Windows PowerShell
minikube : The term 'minikube' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ minikube start
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (minikube:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get nodes
NAME STATUS ROLES AGE VERSION
docker-desktop Ready control-plane 37d v1.34.1
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> minikube start
minikube : The term 'minikube' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ minikube start
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (minikube:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7k7k 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (30m ago) 37d
wordpress-848c87c44f-hm4k5 0/1 ContainerCreating 0 8s
wordpress-848c87c44f-s5glm 1/1 Running 0 8s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> |

```

#### Step 4 – Wait for All Pods to Become Ready

After a few seconds, both pods should be in **Running** state.

kubectl get pods



*Both WordPress pods are 1/1 Running.*

```

Windows PowerShell
NAME STATUS ROLES AGE VERSION
docker-desktop Ready control-plane 37d v1.34.1
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> minikube start
minikube : The term 'minikube' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ minikube start
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (minikube:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7k7k 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (30m ago) 37d
wordpress-848c87c44f-hm4k5 0/1 ContainerCreating 0 8s
wordpress-848c87c44f-s5glm 1/1 Running 0 8s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress unchanged
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7k7k 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (31m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 32s
wordpress-848c87c44f-s5glm 1/1 Running 0 32s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> |

```

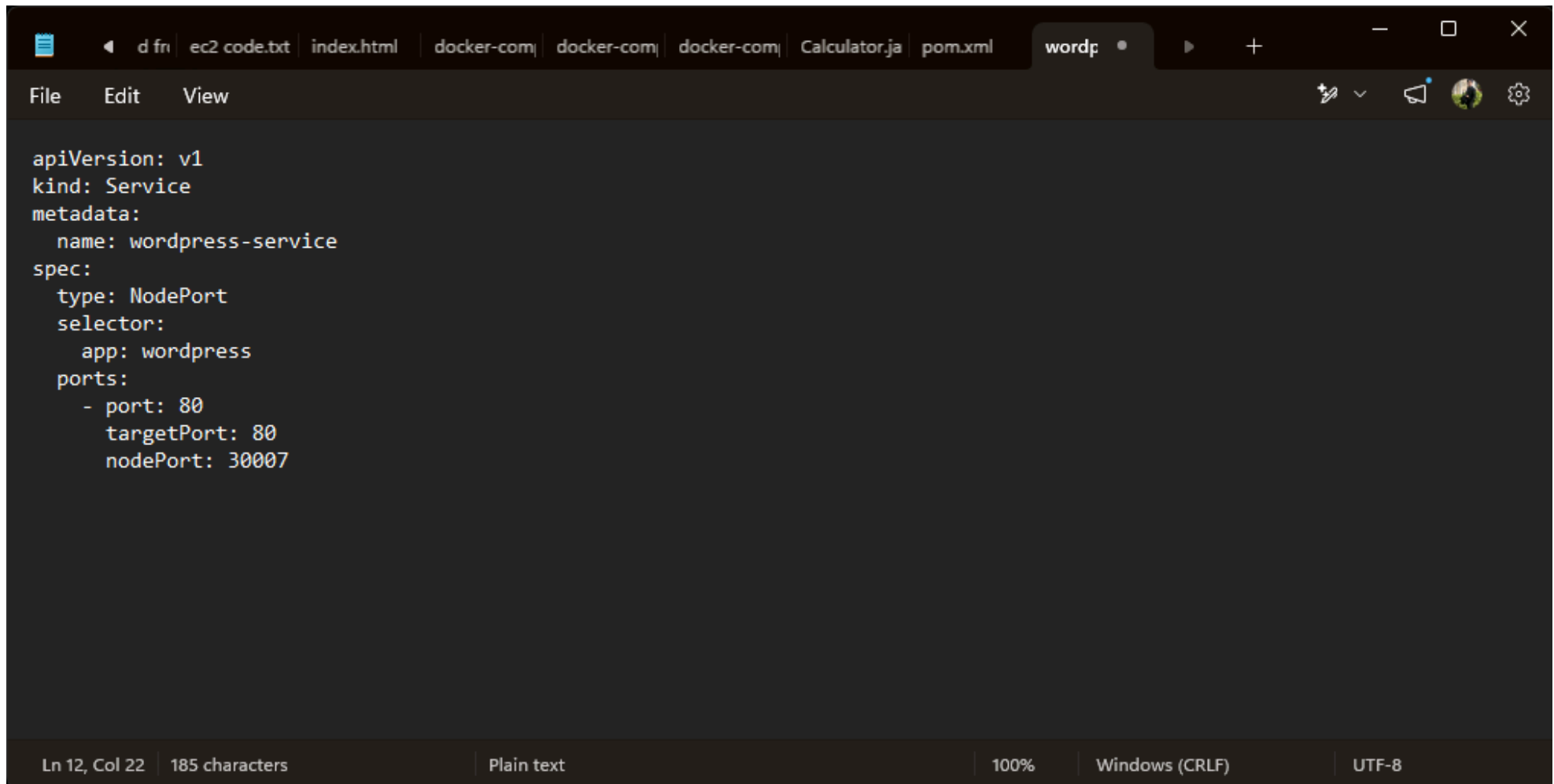
## Step 5 – Create the WordPress Service YAML

Define a `NodePort` Service to expose the deployment externally on port **30007**.

wordpress-service.yaml :

```
apiVersion: v1
kind: Service
metadata:
 name: wordpress-service
spec:
 type: NodePort
 selector:
 app: wordpress
 ports:
 - port: 80
 targetPort: 80
 nodePort: 30007
```





The screenshot shows a code editor window with a dark theme. The title bar at the top displays several open files: 'd fn', 'ec2 code.txt', 'index.html', 'docker-com', 'docker-com', 'docker-com', 'Calculator.js', 'pom.xml', and 'wordp'. The 'wordp' file is the active tab. The editor's menu bar includes 'File', 'Edit', and 'View'. On the right side of the menu bar, there are icons for search, a dropdown arrow, a chat window, a user profile, and settings. The main text area contains a YAML manifest for a Kubernetes Service. The status bar at the bottom indicates the cursor is at 'Ln 12, Col 22', the file contains '185 characters', it is 'Plain text', the zoom is '100%', the line endings are 'Windows (CRLF)', and the encoding is 'UTF-8'.

```
apiVersion: v1
kind: Service
metadata:
 name: wordpress-service
spec:
 type: NodePort
 selector:
 app: wordpress
 ports:
 - port: 80
 targetPort: 80
 nodePort: 30007
```

## Step 6 – Apply the Service

Create the service and then list all services.

```
kubectl apply -f wordpress-service.yaml
kubectl get svc
```



*The new **wordpress-service** appears with type **NodePort** and port mapping **80:30007/TCP**.*

```
Windows PowerShell
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> minikube start
minikube : The term 'minikube' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ minikube start
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (minikube:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (30m ago) 37d
wordpress-848c87c44f-hm4k5 0/1 ContainerCreating 0 8s
wordpress-848c87c44f-s5glm 1/1 Running 0 8s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress unchanged
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (31m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 32s
wordpress-848c87c44f-s5glm 1/1 Running 0 32s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> notepad wordpress-servic
e.yaml
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> |
```

```

Windows PowerShell
At line:1 char:1
+ minikube start
+ ~~~~~
+ CategoryInfo : ObjectNotFound: (minikube:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

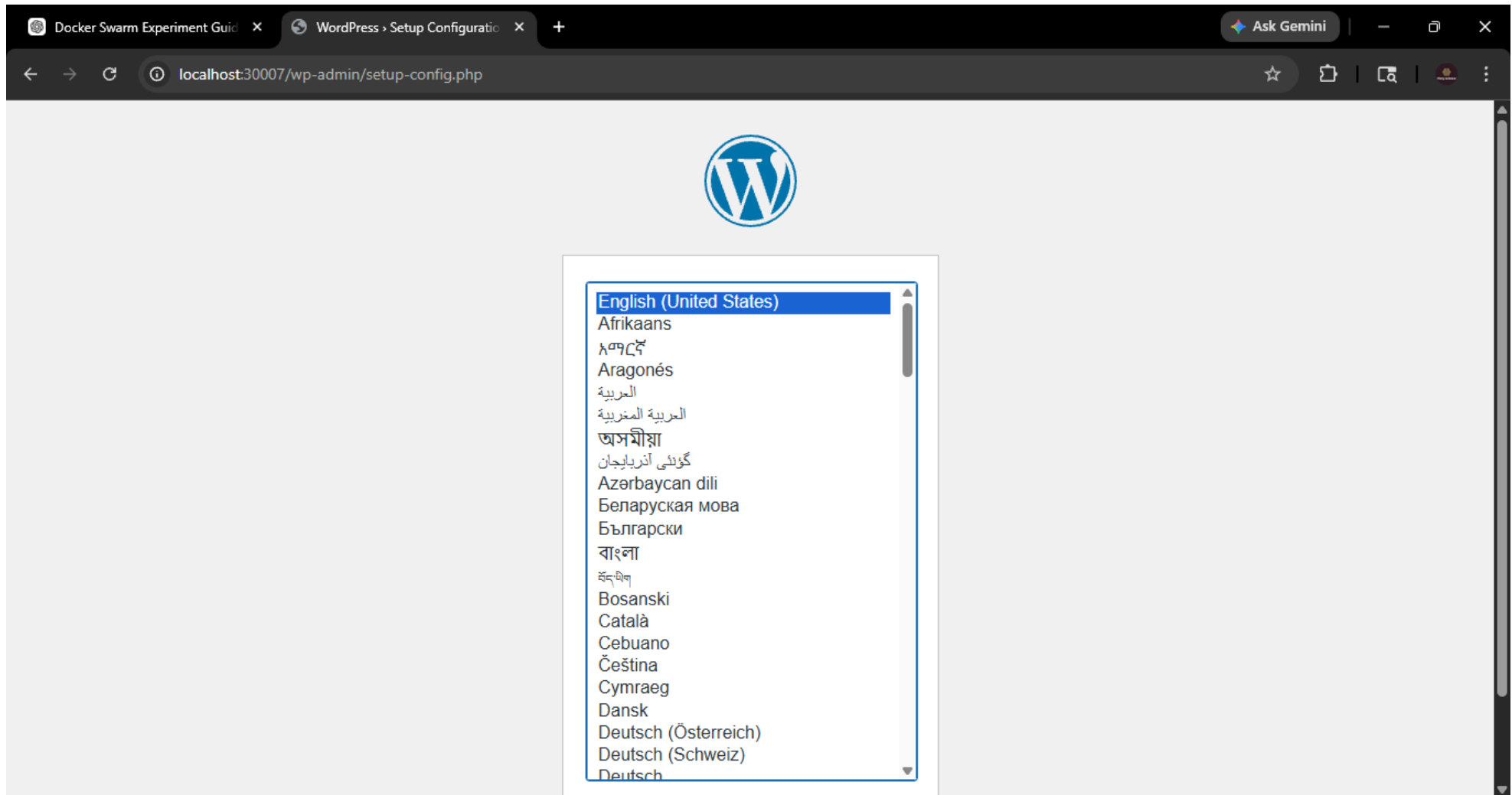
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (30m ago) 37d
wordpress-848c87c44f-hm4k5 0/1 ContainerCreating 0 8s
wordpress-848c87c44f-s5glm 1/1 Running 0 8s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress unchanged
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (31m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 32s
wordpress-848c87c44f-s5glm 1/1 Running 0 32s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> notepad wordpress-servic
e.yaml
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-service.yaml
service/wordpress-service created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> |

```

## Step 7 – Access WordPress in Browser

Open a browser and navigate to <http://localhost:30007> .

*The WordPress language selection screen appears, confirming the application is reachable.*



## Step 8 – Scale the Deployment

Increase the number of replicas from 2 to 4.

```
kubectl scale deployment wordpress --replicas=4
kubectl get pods
```





Four pods are now listed – some may still be starting.

```

Windows PowerShell
deployment.apps/wordpress created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7k7k 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (30m ago) 37d
wordpress-848c87c44f-hm4k5 0/1 ContainerCreating 0 8s
wordpress-848c87c44f-s5glm 1/1 Running 0 8s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-deployment.yaml
deployment.apps/wordpress unchanged
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7k7k 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (31m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 32s
wordpress-848c87c44f-s5glm 1/1 Running 0 32s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> notepad wordpress-servic
e.yaml
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpre
ss-service.yaml
service/wordpress-service created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
apache NodePort 10.101.193.126 <none> 80:32253/TCP 37d
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 37d
wordpress-service NodePort 10.97.35.246 <none> 80:30007/TCP 9s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl scale deployment
wordpress --replicas=4
deployment.apps/wordpress scaled
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> |

```

## Step 9 – Verify Scaling Completes

After a few moments, all four pods should be Running.

```
kubectl get pods
```



*All replicas are 1/1 Running.*

```

ss-deployment.yaml
deployment.apps/wordpress unchanged
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (31m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (31m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 32s
wordpress-848c87c44f-s5glm 1/1 Running 0 32s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> notepad wordpress-service.yaml
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl apply -f wordpress-service.yaml
service/wordpress-service created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
apache NodePort 10.101.193.126 <none> 80:32253/TCP 37d
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 37d
wordpress-service NodePort 10.97.35.246 <none> 80:30007/TCP 9s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl scale deployment wordpress --replicas=4
deployment.apps/wordpress scaled
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (34m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (34m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 3m50s
wordpress-848c87c44f-nm78f 1/1 Running 0 10s
wordpress-848c87c44f-pptqx 0/1 ContainerCreating 0 10s
wordpress-848c87c44f-s5glm 1/1 Running 0 3m50s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12>

```

## Step 10 – Demonstrate Self-Healing

Delete one pod manually and watch Kubernetes immediately recreate it.

```
kubectl delete pod wordpress-84b8c87c44f-hm4k5
kubectl get pods
```



*A new pod (with a different name) is automatically started, keeping the total at 4.*

```

Windows PowerShell
ss-service.yaml
service/wordpress-service created
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
apache NodePort 10.101.193.126 <none> 80:32253/TCP 37d
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 37d
wordpress-service NodePort 10.97.35.246 <none> 80:30007/TCP 9s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl scale deployment
wordpress --replicas=4
deployment.apps/wordpress scaled
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (34m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (34m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 3m50s
wordpress-848c87c44f-nm78f 1/1 Running 0 10s
wordpress-848c87c44f-pptqx 0/1 ContainerCreating 0 10s
wordpress-848c87c44f-s5glm 1/1 Running 0 3m50s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (35m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (35m ago) 37d
wordpress-848c87c44f-hm4k5 1/1 Running 0 4m43s
wordpress-848c87c44f-nm78f 1/1 Running 0 63s
wordpress-848c87c44f-pptqx 1/1 Running 0 63s
wordpress-848c87c44f-s5glm 1/1 Running 0 4m43s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl delete pod wordp
ress-848c87c44f-hm4k5
pod "wordpress-848c87c44f-hm4k5" deleted from default namespace
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> |

```

```

Windows PowerShell
wordpress-848c87c44f-hm4k5 1/1 Running 0 4m43s
wordpress-848c87c44f-nm78f 1/1 Running 0 63s
wordpress-848c87c44f-pptqx 1/1 Running 0 63s
wordpress-848c87c44f-s5glm 1/1 Running 0 4m43s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl delete pod wordp
ress-848c87c44f-hm4k5
pod "wordpress-848c87c44f-hm4k5" deleted from default namespace
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (36m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (36m ago) 37d
wordpress-848c87c44f-8t9mf 1/1 Running 0 11s
wordpress-848c87c44f-nm78f 1/1 Running 0 115s
wordpress-848c87c44f-pptqx 1/1 Running 0 115s
wordpress-848c87c44f-s5glm 1/1 Running 0 5m35s
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl delete deploymen
t wordpress
deployment.apps "wordpress" deleted from default namespace
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl delete service w
ordpress-service
service "wordpress-service" deleted from default namespace
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get pods
NAME READY STATUS RESTARTS AGE
apache-7b8c8c4b44-r7ktk 1/1 Running 4 (37m ago) 37d
apache-7b8c8c4b44-rtrdb 1/1 Running 4 (37m ago) 37d
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> kubectl get svc
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
apache NodePort 10.101.193.126 <none> 80:32253/TCP 37d
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 37d
PS C:\Users\acer\OneDrive\Desktop\Cloud Lab\Containerization-and-Devops-Lab-\lab\Experiment-12> |

```

## Step 11 – Clean Up Resources

Remove the deployment and service.



```
kubectl delete deployment wordpress
kubectl delete service wordpress-service
kubectl get pods
kubectl get svc
```

*Only the pre-existing apache pods remain; WordPress resources are gone.*

 [Cleanup and final state](#)

## Key Concepts

| Kubernetes Object | Role                                                                |
|-------------------|---------------------------------------------------------------------|
| Pod               | The smallest deployable unit – one or more containers               |
| Deployment        | Manages replicas and rolling updates                                |
| Service           | Provides a stable network identity and load balancing               |
| ReplicaSet        | Ensures a specified number of pod replicas are running at all times |
| NodePort          | Exposes a service on a static port of each node's IP                |

## Observations

- **Minikube** was not installed; Docker Desktop's built-in Kubernetes was used instead.
- Scaling from 2 to 4 replicas took only one command.
- After deleting a pod, Kubernetes automatically recreated it – no manual restart required.
- A `NodePort` service made the application accessible via `localhost:30007` without port-forwarding.

- The deployment ensured the desired state was always maintained.

## Cleanup (Optional)

---

If you wish to remove the previously deployed Apache resources:

```
kubectl delete deployment apache
kubectl delete service apache
```



---

